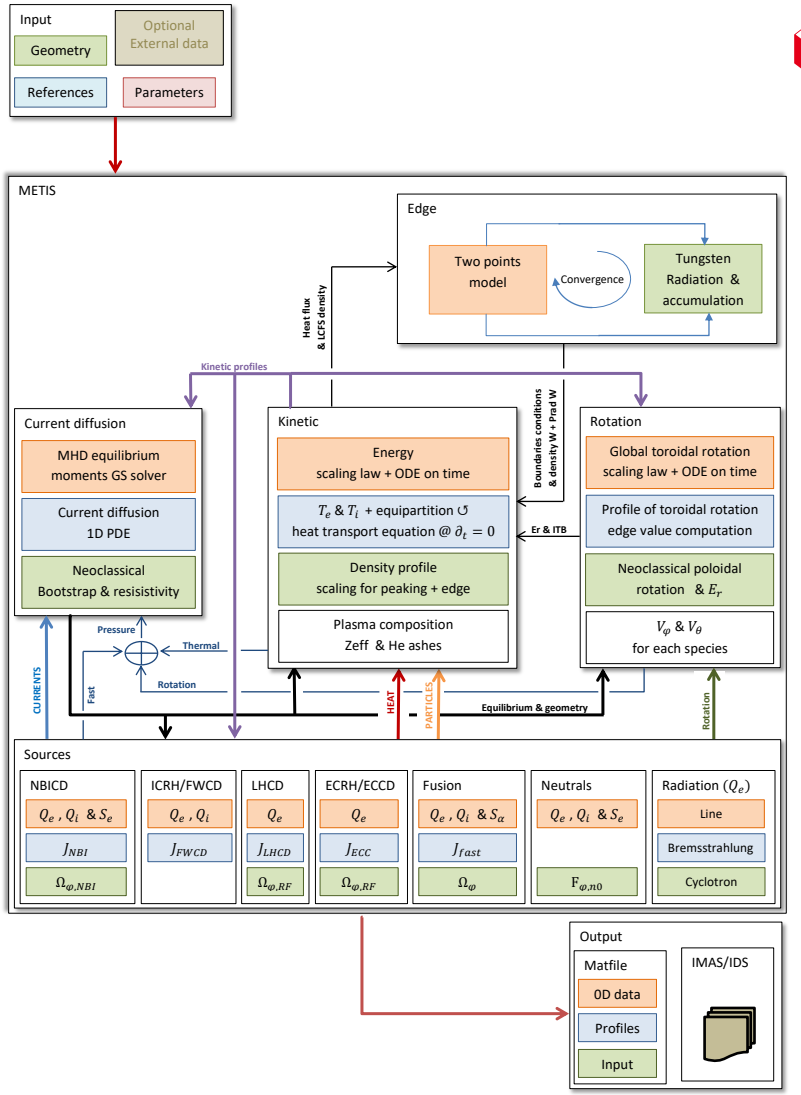
HOWTO METIS

Jean-François Artaud

Pierre Manas

2/18/2026



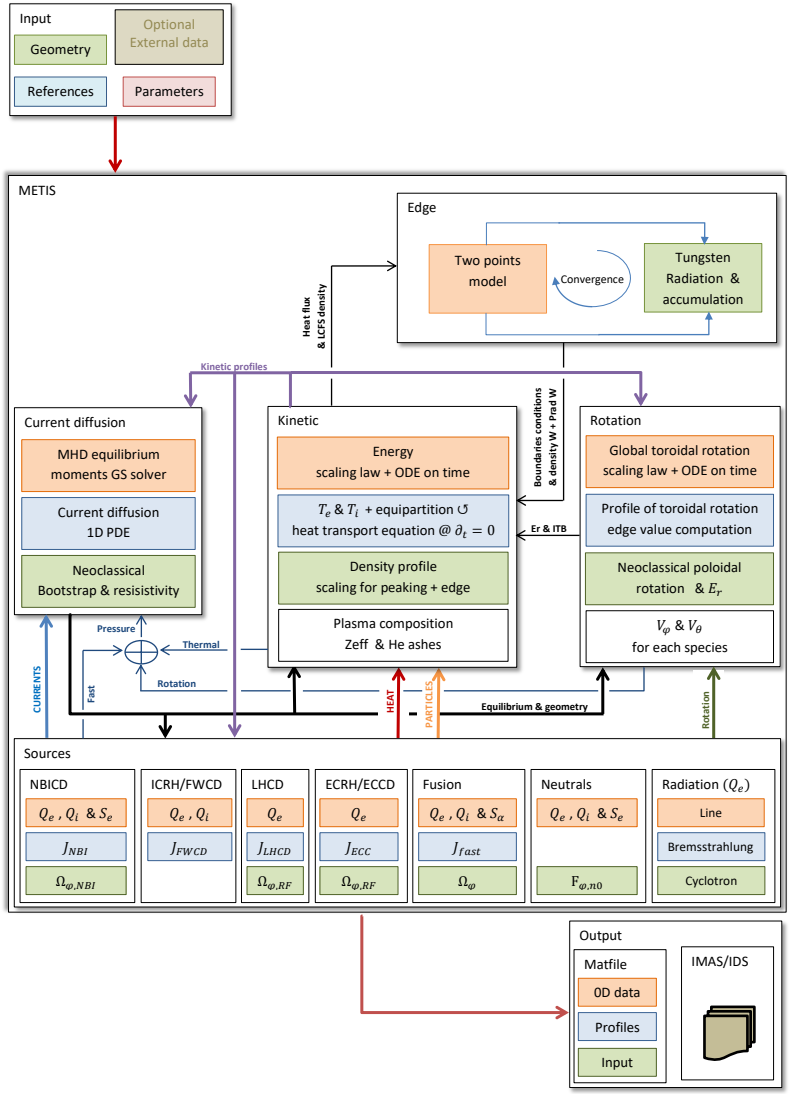
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1. METIS OVERVIEW

Main METIS features

- ✓ **Mixed 1D and 0D equations**
  - ✓ **Current diffusion 1.5D with moment equilibrium**
  - ✓ **Source profiles deduced from simple models**
  - ✓ **Global energy content from 0D ODE (scaling, transients)**
  - ✓ **Temperature profiles : stationary 1D solution, χ scaled to W_{th}**
  - ✓ **All non-linearities solved (dependence of sources on profiles, fusion power, He ash transport)**
  - ✓ **Bootstrap and resistivity :Sauter O. et al 1999 Phys. Plasmas 6, 2834; ibid. 9 (2002) & Redl update**
- ✓ **Input : Power references, I_P, plasma density, Z_{eff} , LCMS geometry**
- ➢ **Output : all standard 1D and 0D data that you would expect from a transport code**
- ➢ **Global time convergence (wave form relaxation)**



METIS databases Connections

- Direct connection:
  - Tore Supra / WEST: pre and post shot access (Tslib/IMAS)
  - JET: SAL access to JET data
  - Compass: dedicated library
  - IMAS : UAL V5 and DD v3.x and 4.0.0
- Indirect connection:
  - DIII-D: dedicated tool (to be run on DIII-D computers)
  - EAST: dedicated tool (to be run on EAST computers)
  - Asdex-U: still incoming
  - TCV: MDS + access from SPC

➢ Allows comparison to experimental data.

METIS SCENARIO GENERATORS

- Tools to prepare a complete scenario ab-initio:
  - Set waveforms and parameters
  - Including ramp-up, flattop and ramp-down
  - Take into account specificity of each device
  - Can be fine tuned from METIS GUI

- Available generators:
  - WEST
  - JT-60SA
  - Reactors (ITER, DEMO, …)

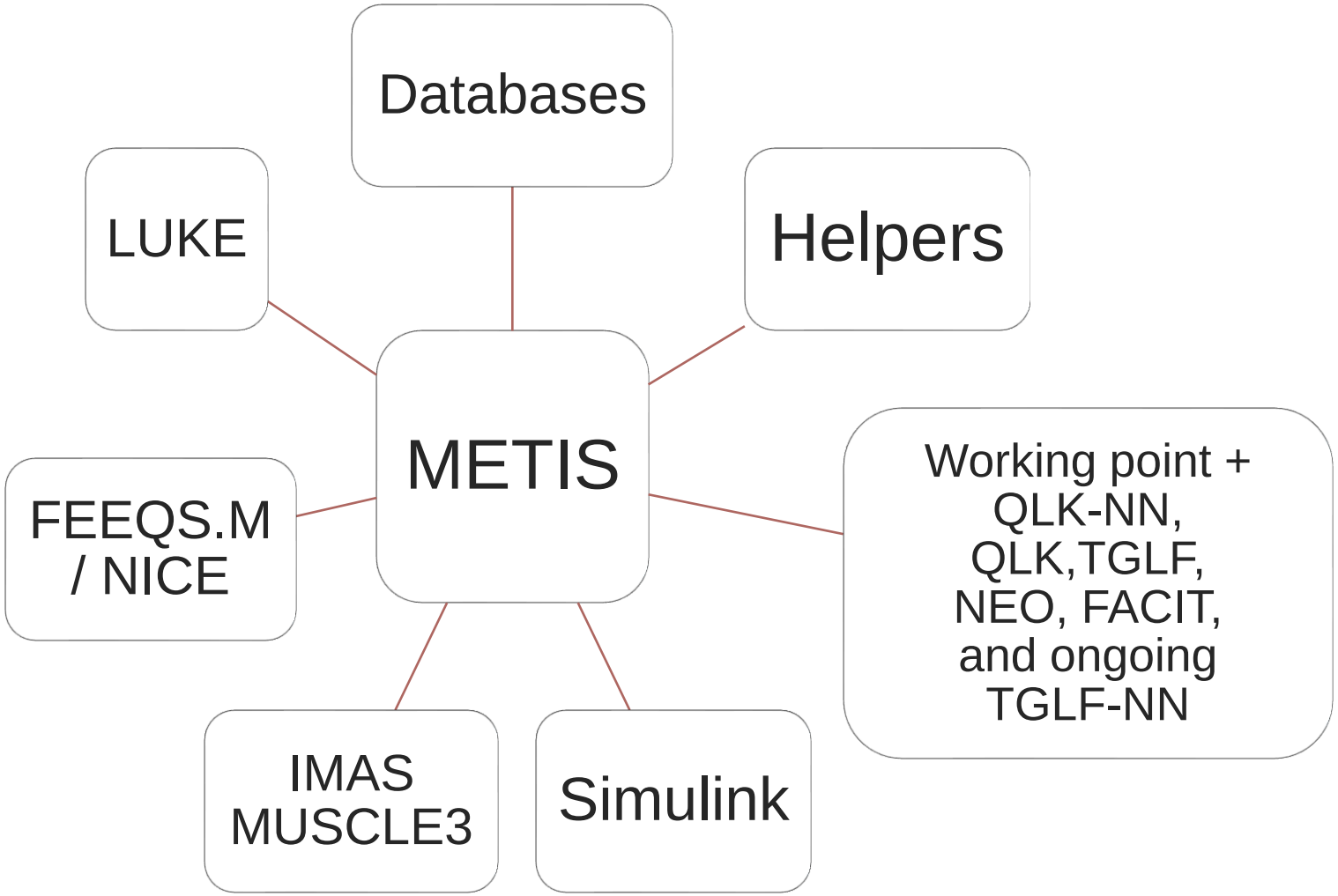
METIS versions

- **METIS for IMAS :**
  - Access through AL
  - METIS actor for MUSCLE3
  - Coupling with Kepler and Python workflow

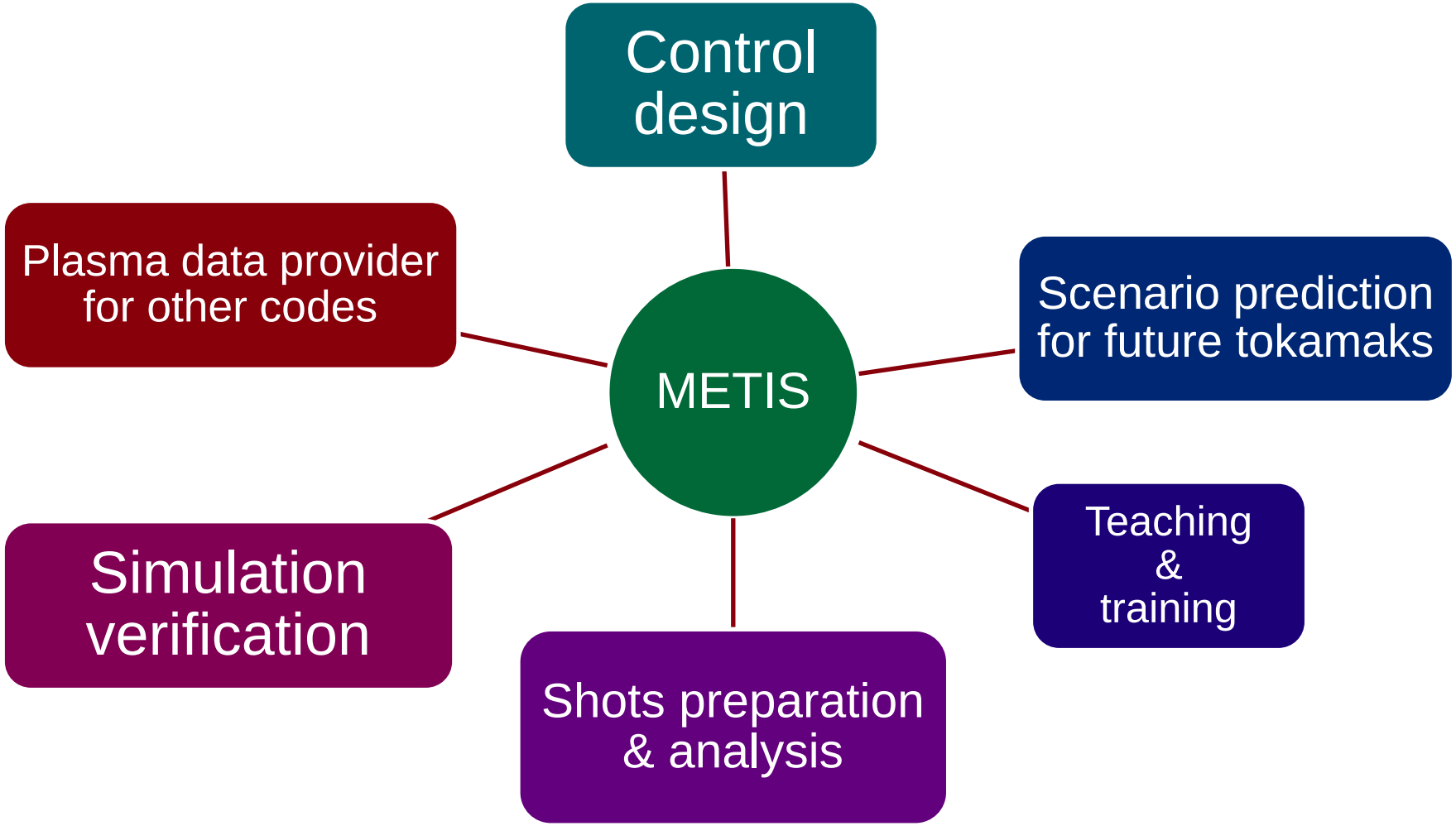
- **METIS standalone:**
  - Simplified installation
  - Can be called from Simulink

- **METIS runtime (compiled version):**
  - No Matlab license needed (= no Matlab workspace)
  - Work on many OS
    (available for Windows, on demand for Linux)

METIS ecosystem overview



METIS Applications



2. INSTALL & LAUNCH METIS

Where to find METIS ?

- **METIS source code can be cloned from:**
  - **Public repository (only main branch available):**

    https://github.com/IRFM/METIS

  - **On ITER Git server:**

    https://git.iter.org/projects/SCEN/repos/metis

  - **In EUROfusion Git server:**

    https://gitlab.eufus.psnc.pl/g2jfa/metis

Install & launch METIS

METIS can work out of the box but it requires MATLAB to run.

- To run METIS:
  - Load MATLAB module.
  - Launch the script **metis** in the root directory of METIS source code
    "./metis"
  - If not done, launch the compilation of mexfiles in Matlab command line window:
    make_metis_linux

- More information on METIS installation are available in:

  <METIS_root_directory>/doc/13. METIS installation guide CeCiLL C.pdf

- For Windows, the script make_metis_windows should be run separtely from METIS: the script should be edited and modified depending on the version of Matlab and Windows.

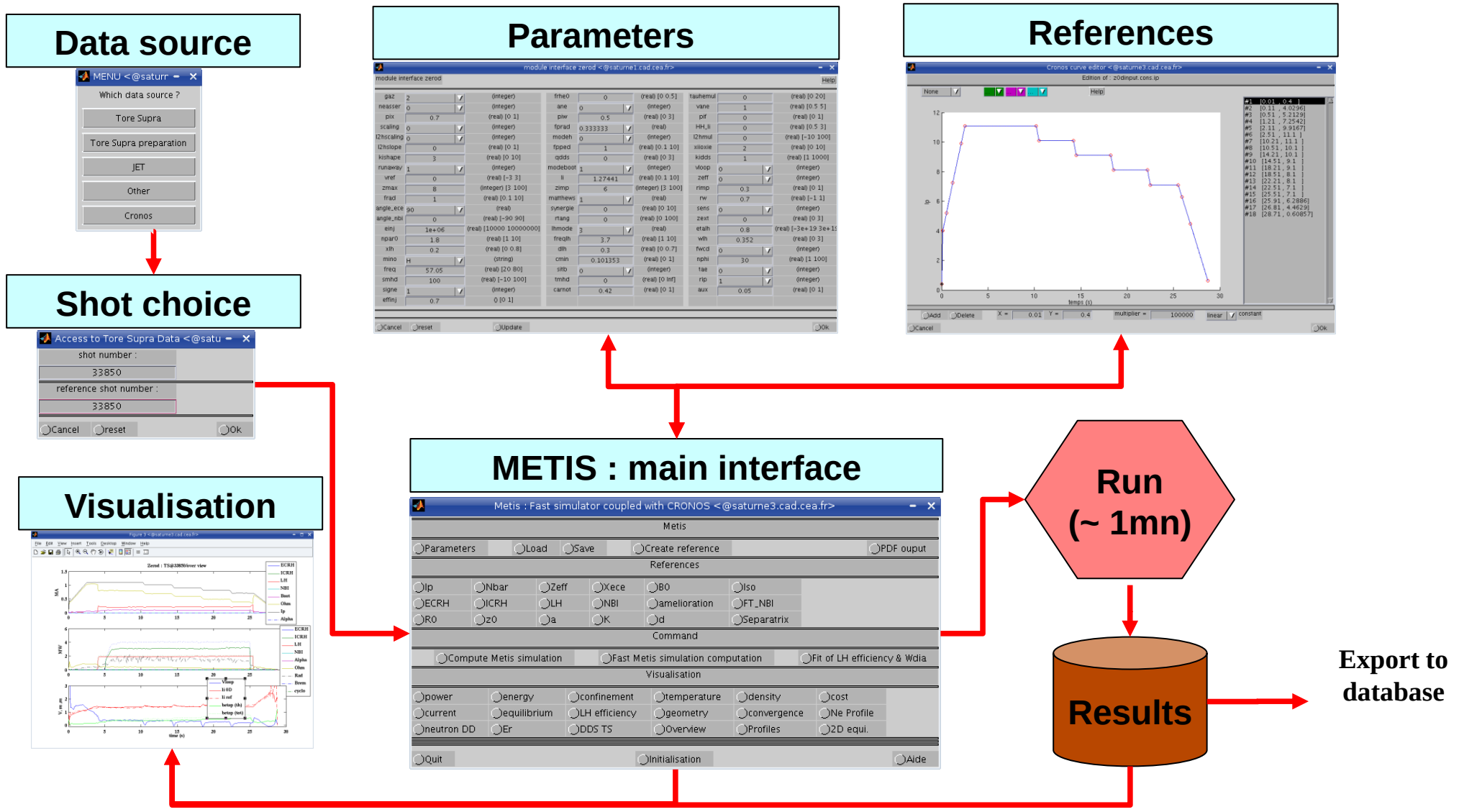
METIS documentation

METIS documentations are available in the directory “doc” under the root installation directory of METIS or here: https://github.com/IRFM/METIS/tree/main/doc.

- Main files:
  - Artaud_2018_Nucl._Fusion_58_105001.pdf
    → Reference paper
  - Howto_METIS_final.pdf
    → This is a “How to” introduction to METIS
  - METIS installation guide CeCiLL C.pdf
    → This is the METIS installation guide.
  - METIS actor for MUSCLE 3.pdf
    → This document describes the use of MUSCLE 3 actor for METIS.
  - Index_METIS_documentation.txt
    → List of available documents
  - ...

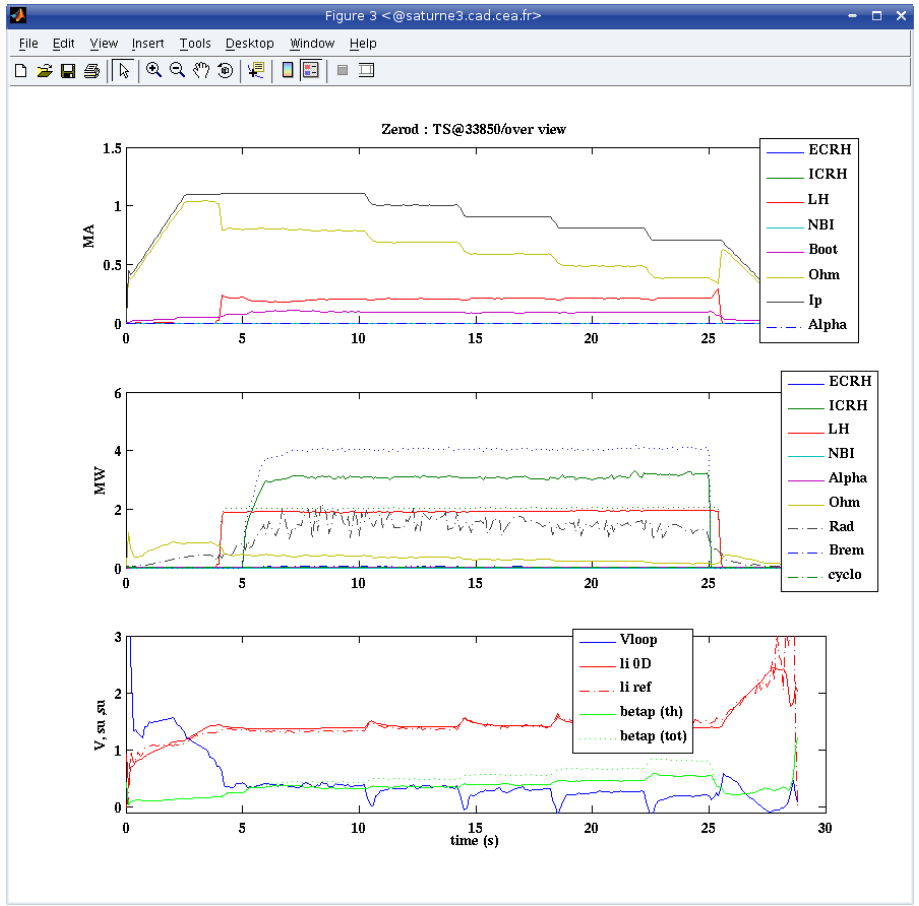
3. METIS GUI

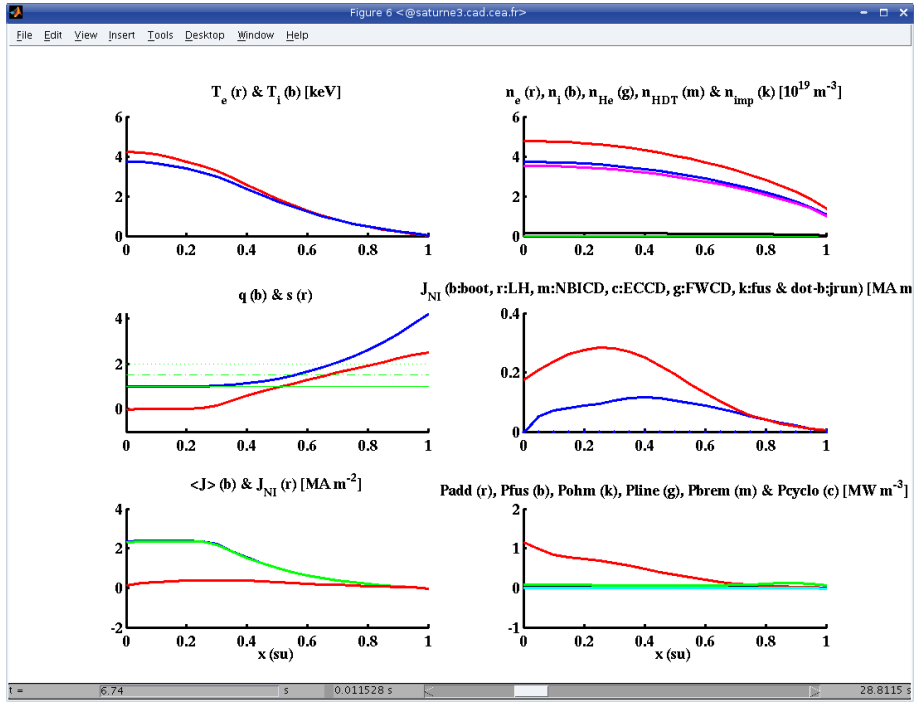
GUI overview



METIS output example (1/2)

Results: 0D data

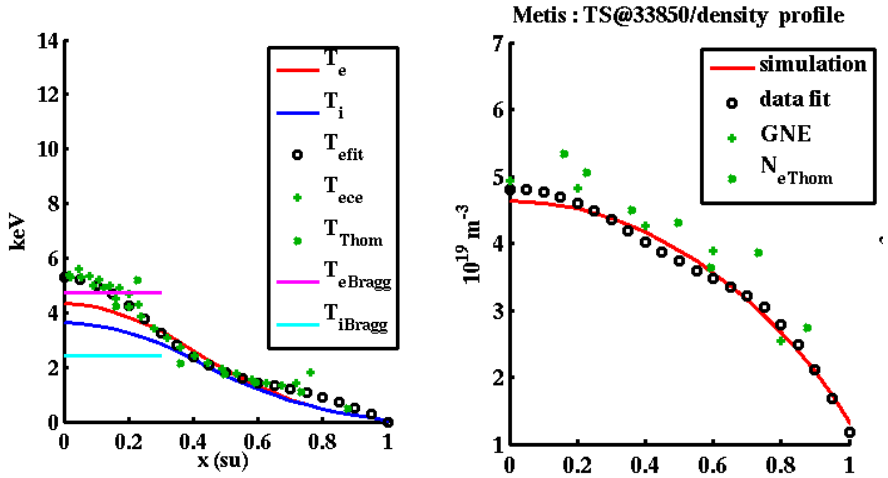


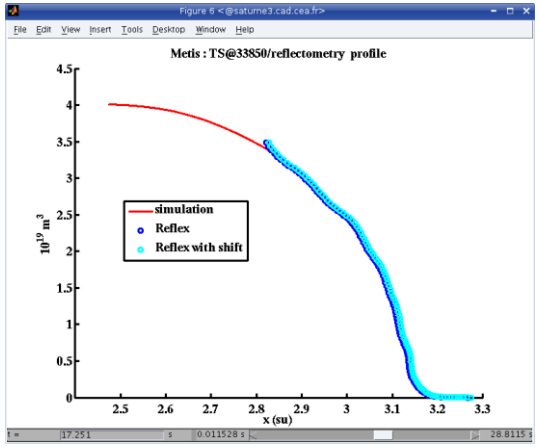


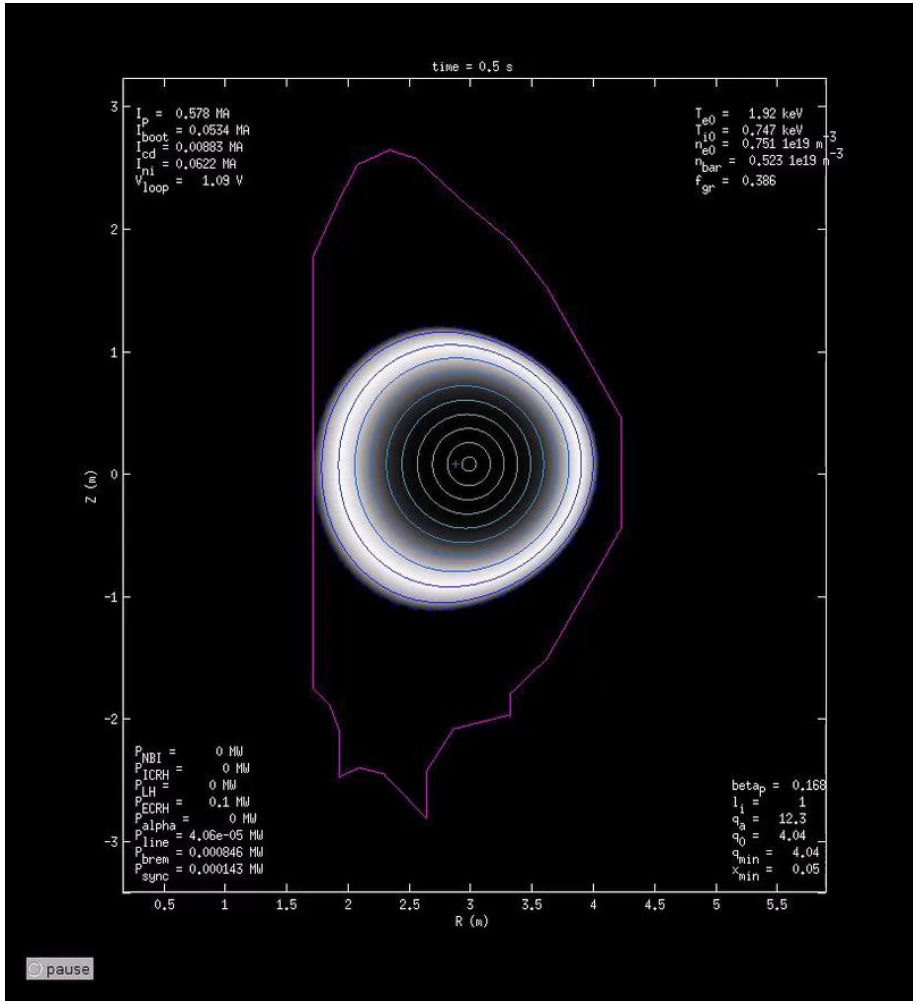
Results: profiles

METIS output example (2/2)

comparaison with diagnostics data







Internal 2D METIS equilibrium

Main GUI windows (1/2)

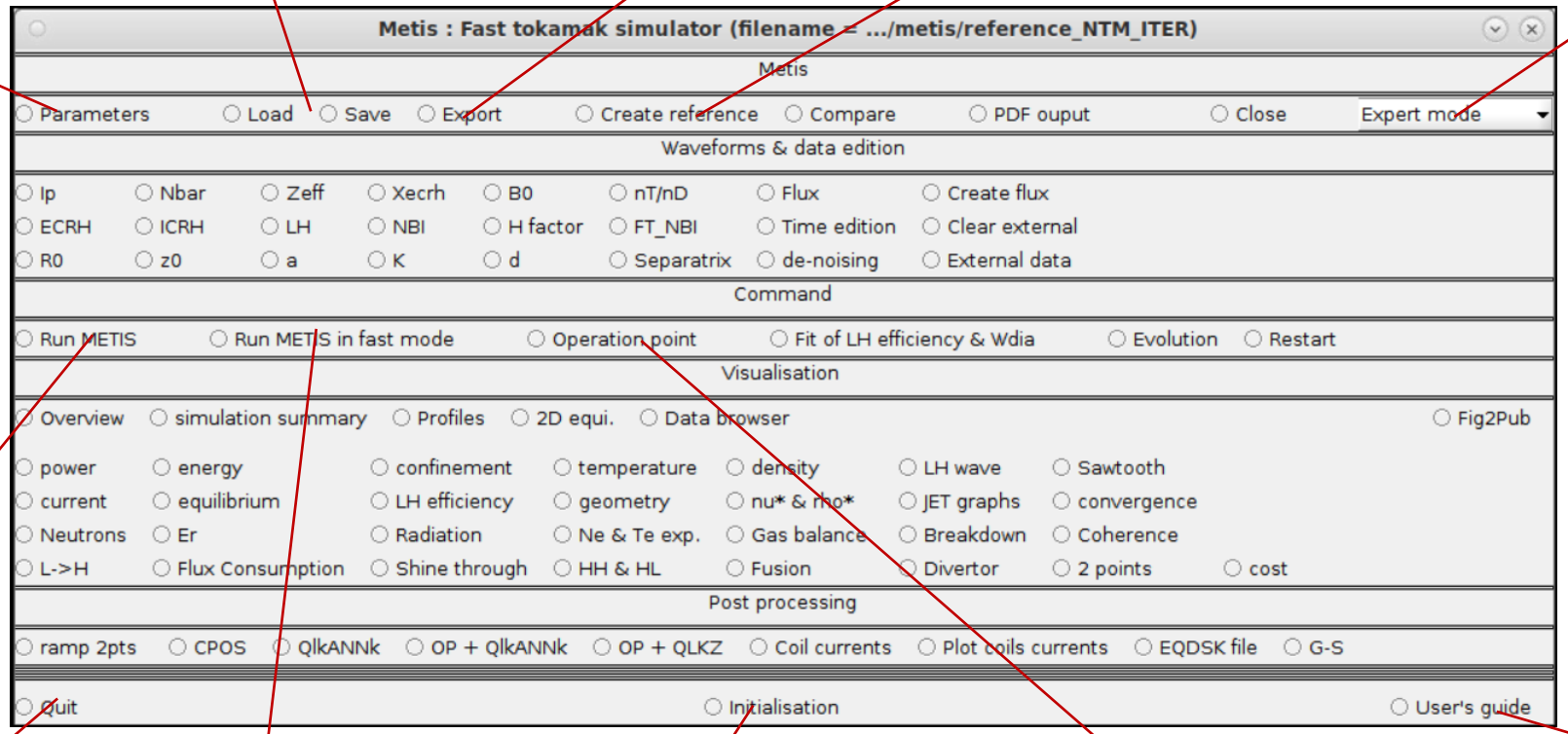
Edition of simulation parameters

Load & Save simulation

Export IDSs in matfile

Create reference data set for comparison

Mode switching



Edition references

Pre-defined graphs

Full simulation (all time slices, to be use for final result)

Quit METIS

Fast simulation (selected time slices for overview)

Initialization (create a new simulation)

Data browser (plot any METIS data)

Open METIS manual (need a pdf reader installed)

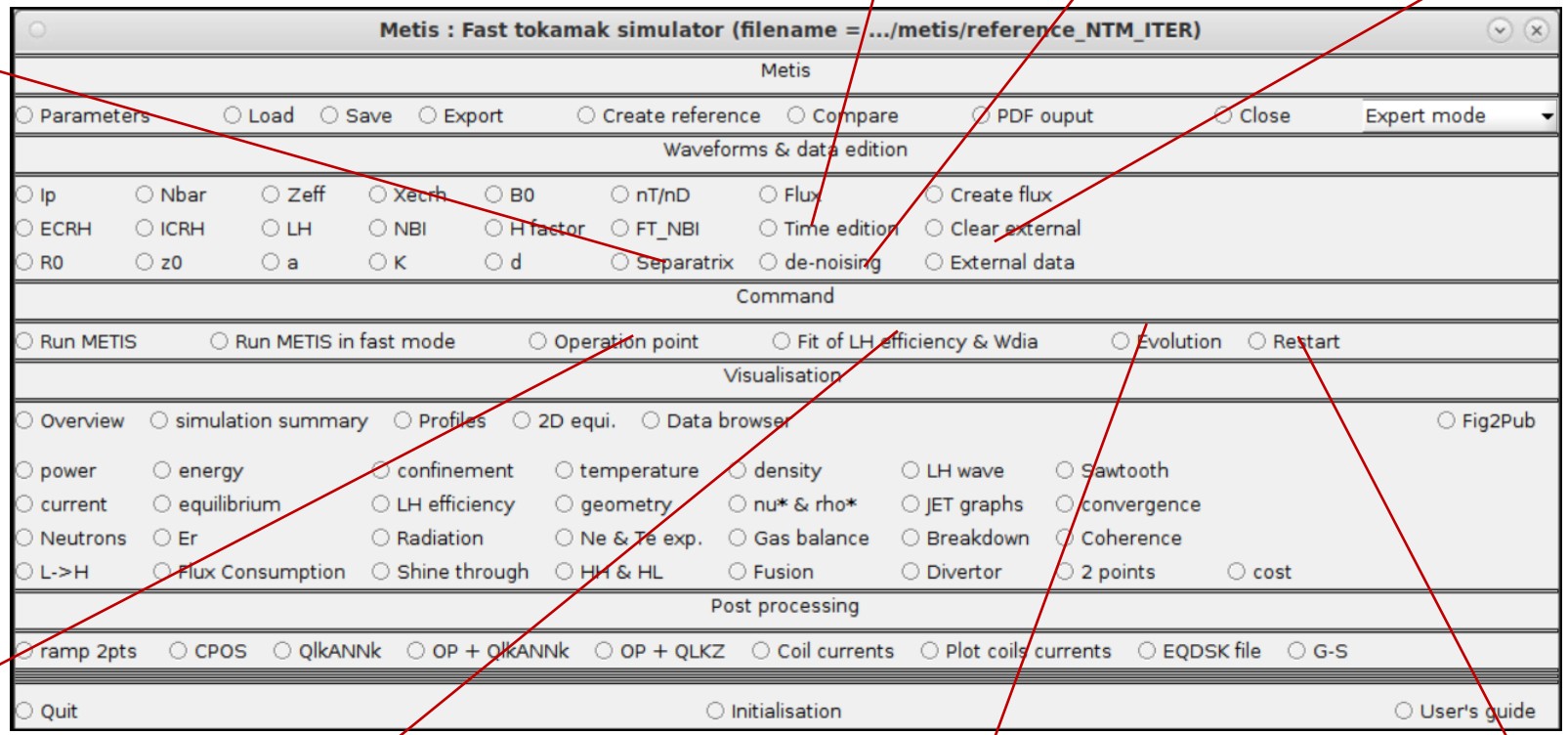
Main GUI windows (2/2)

Noise filter for references

Handle external data (measurement, …)

Time resampling (input)

Generate separatrix (LCFS) with X-point



Main graphs

Post processing

Steady-state fast computation

Fit experimental energy content and search LHCD efficiency (if present)

Run in evolution mode (for testing)

Restart evolution mode (to continue the simulation)

Simple Recipe for a METIS simulation

1. Create or load a data set
2. Edit and modify references
3. Optionally, edit the LCFS (if X-point)
4. Edit and modify parameters
5. Run the simulation in fast mode
6. Visualize results
7. Come back to 2, if changes are needed
8. Run complete simulation
9. Visualize and check results
10. Save results

Read data from databases



- From « initialization » :
  - Access to databases
  - Possibility to create a simulation ab-initio
  - Create simulation with scenario generator for WEST, JT-60SA, Reactor (ITER/DEMO)
  - Read METIS input data from IMAS IDSs

➢ Create a new simulation

✓ Data are only in memory : must be saved (after first run) !

➢ Menu change depending on context

Load data

- Button load open a file selector :
  - Just selected a file and open it

- METIS has unified file for input/output

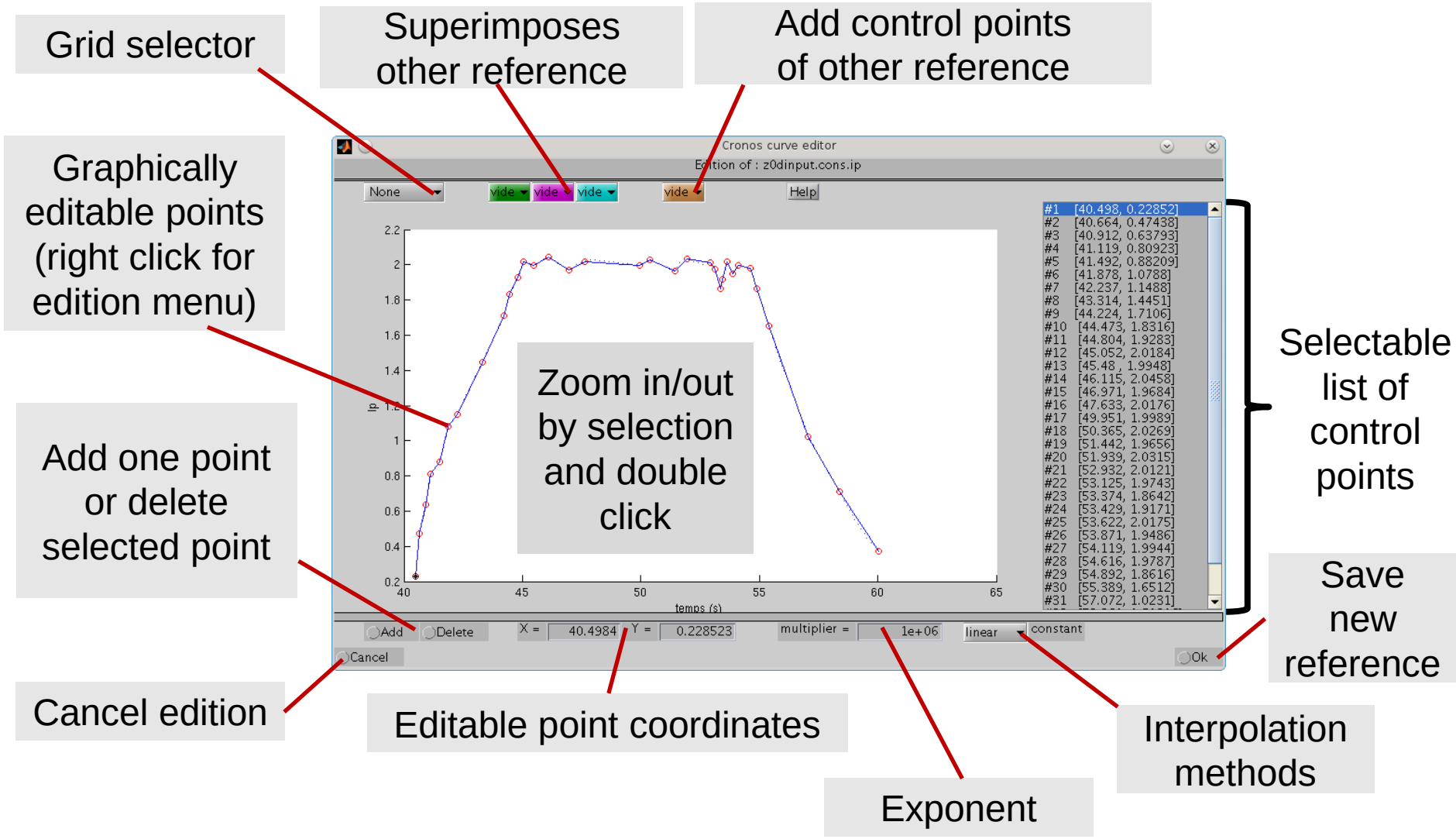
- All METIS files are compatible with new METIS versions (ascendant compatibility)

- Some sample files (used for non-regression tests) are in the directory certification/metis :

  <METIS_root>/certification/metis

Reference edition

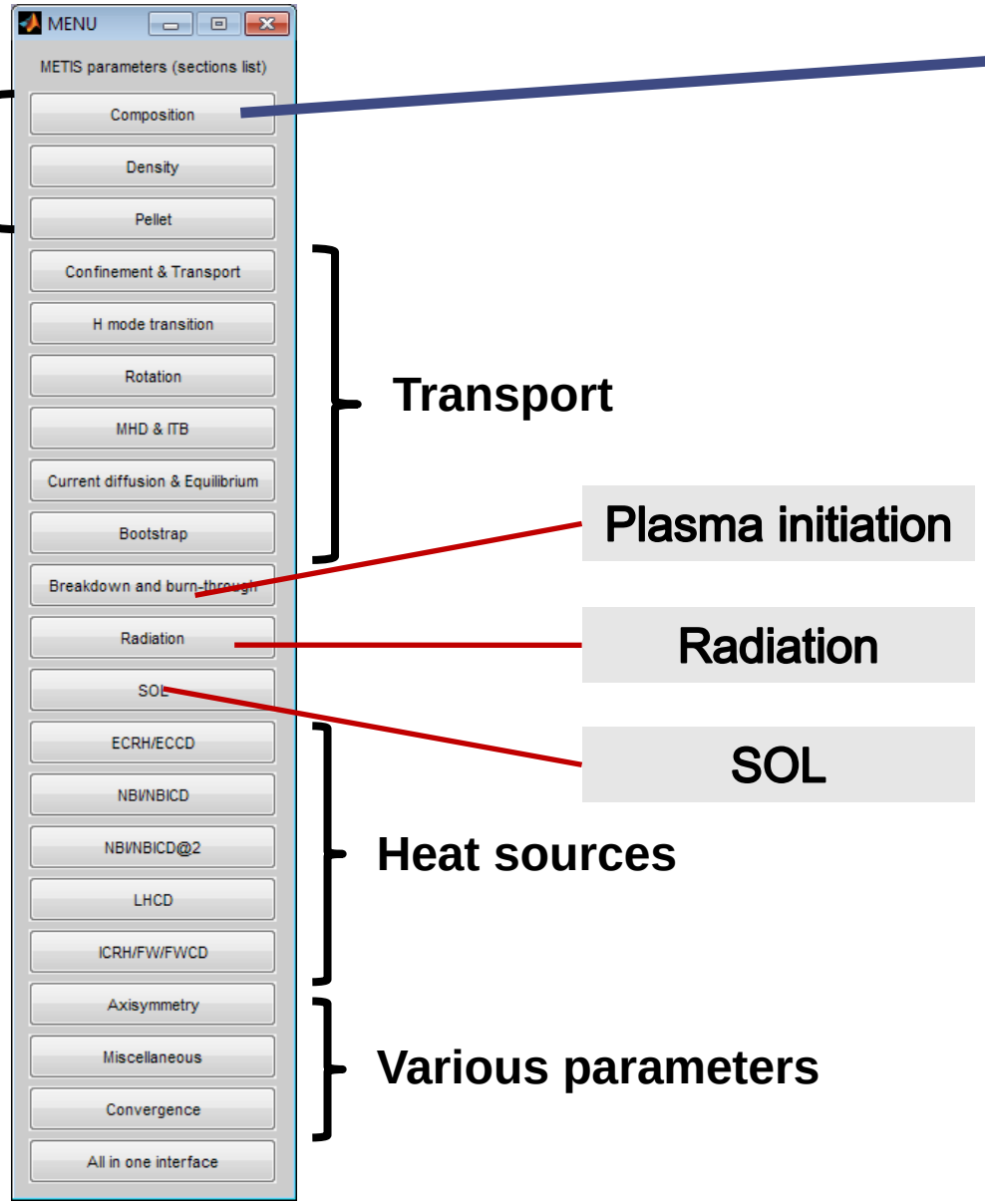
- Time dependent reference can be edited with the help of graphic editor

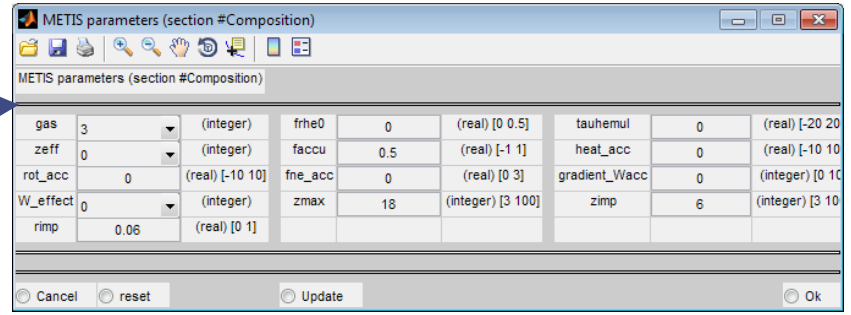


List of references

- Plasma parameters:
  - ip : plasma current
  - flux : edge poloidal flux
  - nbar : line averaged density
  - zeff : line averaged effective charge
  - iso : isotopic plasma composition (n_D/n_T)
  - picrh : ICRH/FW input power
  - plh : LFCD input power
  - pnbi : NBI input power (can open 2 reference editors)
  - pecrh : ECRH input power
  - hmore : enhancement factor
  - ftnbi : NBI composition
  - xece : ECRH maximum deposition position.
- Plasma geometry :
  - a : minor radius
  - R0 : major radius
  - z0 : vertical shift
  - K : elongation
  - d : averaged triangularity
  - b0 : toroidal magnetic field at the major radius
- LCFS generator:
  - More moments
  - Real X-point
  - For given interval of time

Modification of parameters





- Parameters grouped by topic
- Each button opens a form
- Change the input parameters

- If IMAS is available, additional entries allow to tune production of IMAS data

- In form, each button display an helper tooltip

METIS simulation modes

- Main modes:
  - Fast: just to have a look and tune parameters and references
  - Compute: Full computation for final results
  - Fit: Compute energy using measurement
    (and optionally search for LHCD current drive efficiency)
  - Operation point: One time slice, steady state

- Other modes (for testing):
  - Evolution: test the simulation in evolution mode
  - Restart: restart a simulation in evolution mode

Results visualizations

- At the end of the simulation overview graph is opened.
- Generic graph (0D) : just push the button
  - Power, energy, current, confinement, temperature density, equilibrium, geometry, convergence, …
- 2D plot :
  - equi 2D, Profiles, Er, …
- Specialize graphs :
  - L→ H, Consumed flux, LH wave , …
- Machine dependent graphs :
  - Neutron DD, DDS TS, NBI JET, Ne profile
- Generic data browser and plotter.

Data browser and plotter

Matlab figure menu

0D data

experimental data

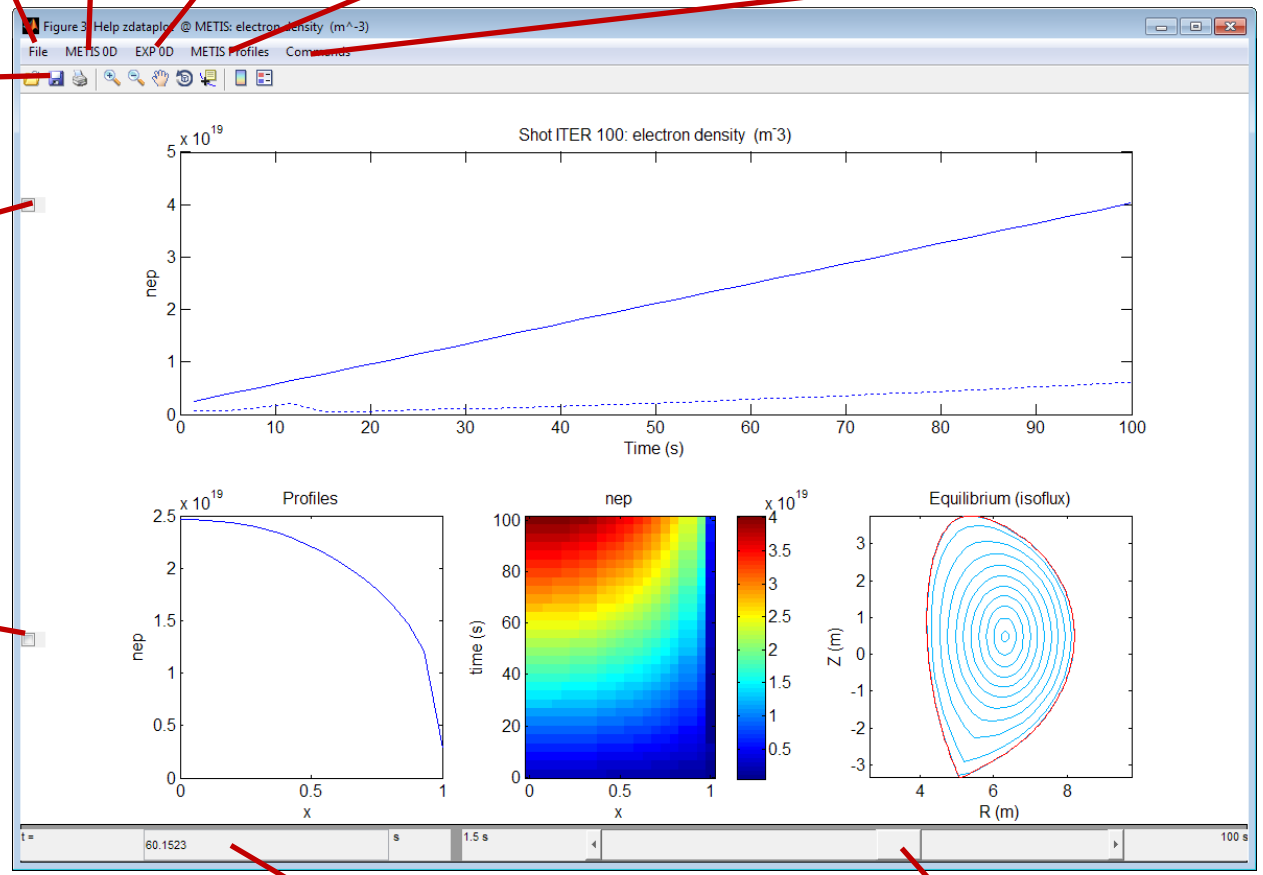
profile data

Graphs options (superimpose, …)

Matlab figure toolbar

time selection With cursor

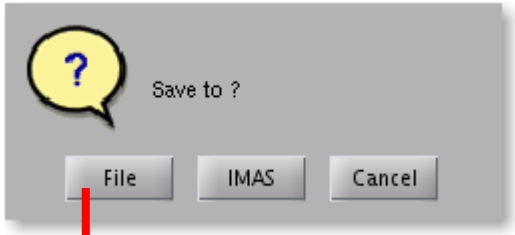
Memorize profile and superimpose

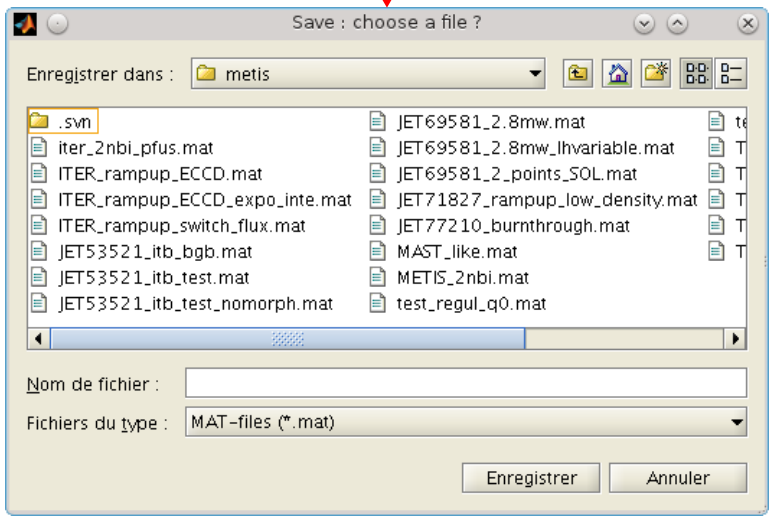


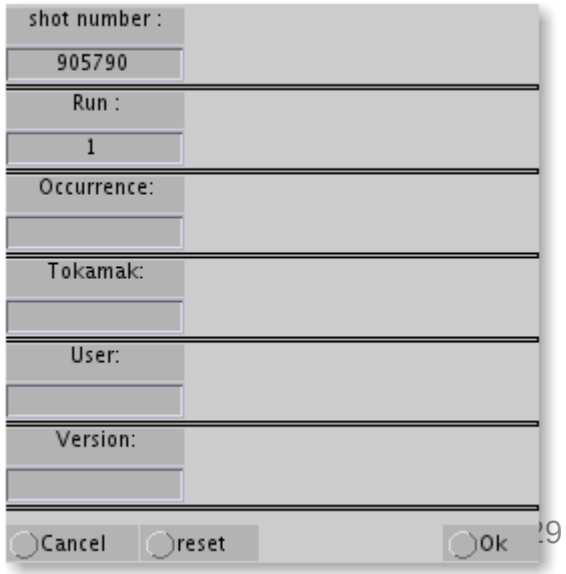
Time (edit field)

Time slider

Save results (matfile format or IMAS IDSs)

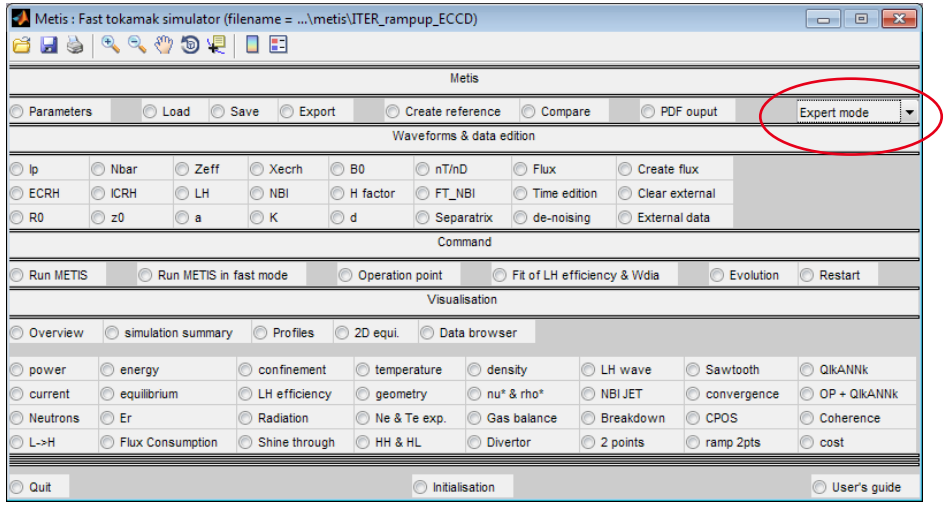






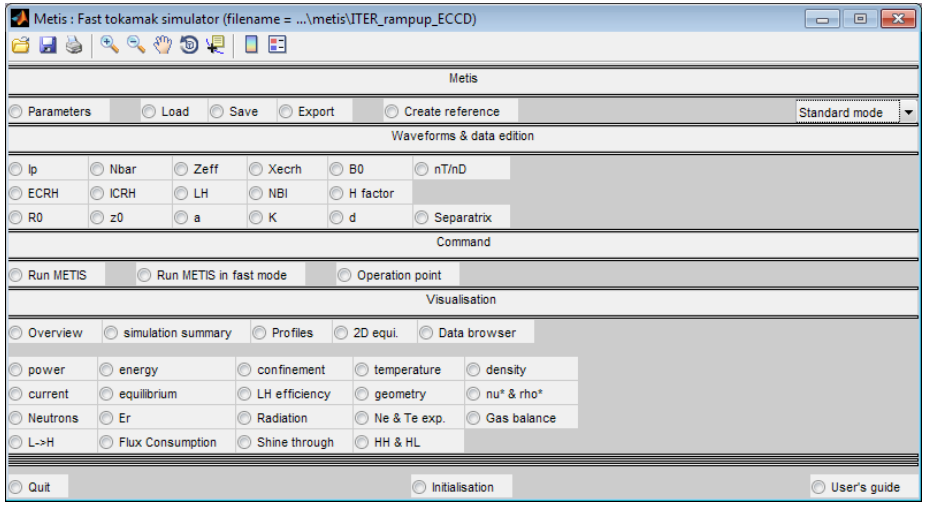
4. Expert & standard Mode

GUI has 2 modes: Standard and Expert (1/2)



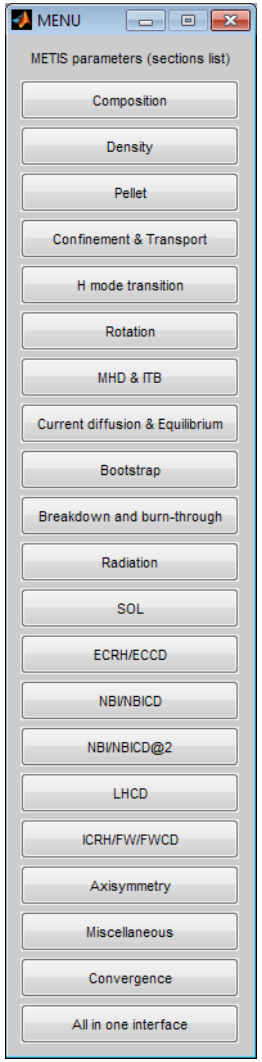
Expert

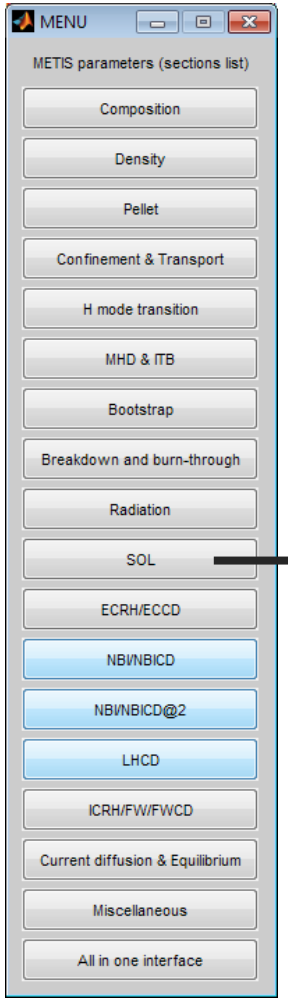
Standard



GUI has 2 modes: Standard and Expert (2/2)

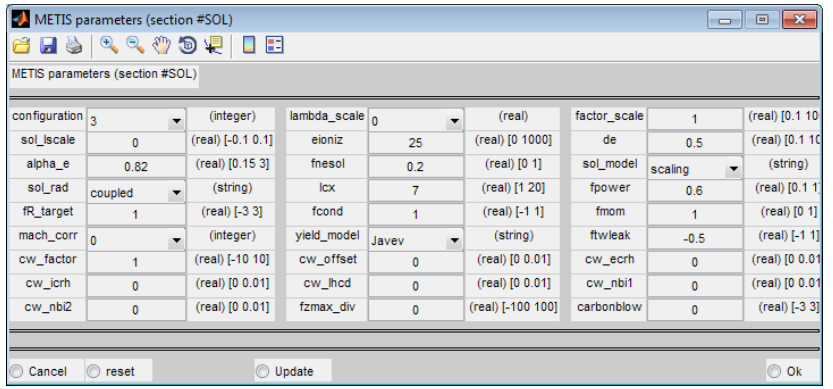
Expert





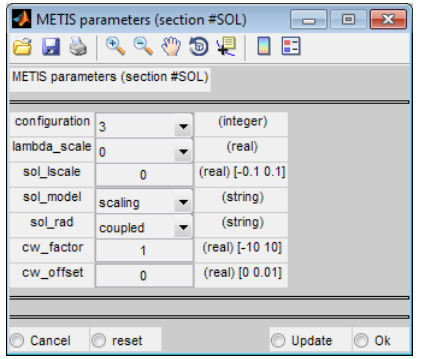
Standard

One example:
SOL Parameters



Expert

Standard



5. External data

METIS & external data (1/3)

- METIS can use external data for:
  - Heat and current drive sources :
    LHCD, ICRH, ECCD, NBICD, Pline, Runaway
  - Kinetic profiles: Ne, Te, Ti, Ω_φ

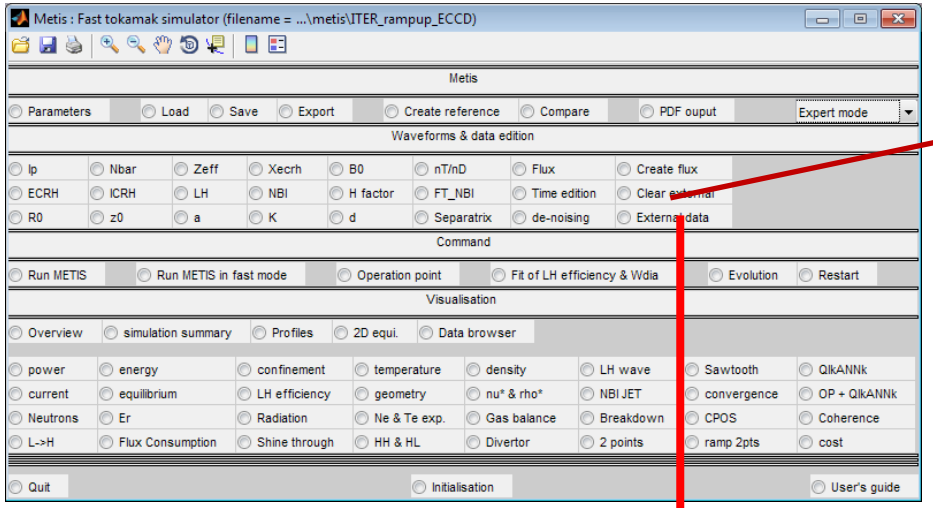
- External data override internal METIS computation
- Heat sources are renormalized on references.
- Density is renormalized on $\bar{n}$ reference

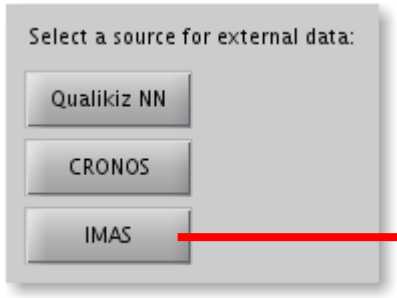
➢ Details available in file:
<METIS_root>/zerod/external_data_rule_for_METIS.m

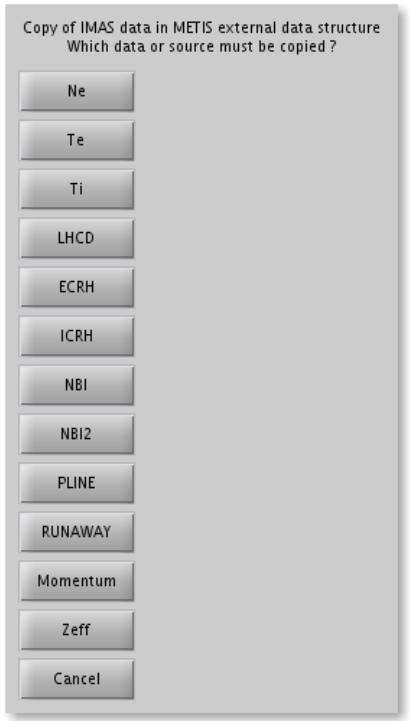
METIS & external data (2/3)

- METIS GUI for external:

Remove external data
(METIS stops to use external data)







Use of external data from CLI (3/3)

- External data can bet set from Matlab command line:
  1. Create the data structure in Matlab, generaly containing:
     - ✓ Time vector
     - ✓ Space vector
     - ✓ Data [time x space]
  2. Export it to global application data stack (this data is used by METIS as soon as it has been defined)
  3. During METIS run, statements begening with ”using external data in METIS for …” should be displayed in Matlab console

- Get details help from Matlab CLI:

  help external_data_rule_for_METIS

- Example:

```
NE_EXP.temps =  linspace(0,100,101)';
NE_EXP.x        =  linspace(0,1,21);
NE_EXP.ne      =  1e19 .* (1 + NE_EXP.temps / 10) * (1 -  NE_EXP.x .^ 2) .^ 2 + 1e19;
setappdata(0,'NE_EXP',NE_EXP)
```

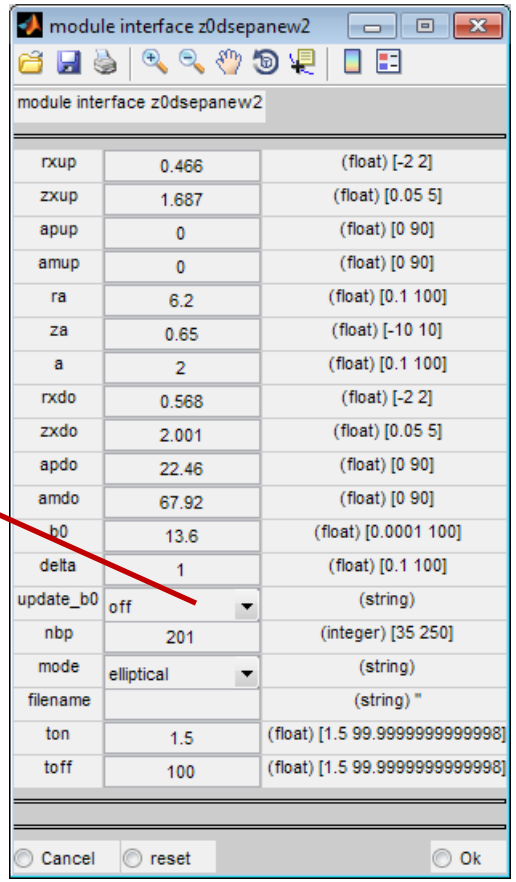
  - During METIS run, the statement ”using external data in METIS for NE” is now displayed

6. LCFS generator

LCFS generator (1/3)

- LCFS parameters:
  - rxup : upper triangularity (normalized)
  - zxup : upper elongation (normalized)
  - apup, amup : upper X-point angles
  - ra : major radius
  - za : vertical shift
  - a : minor raduis
  - rxdo : lower triangularity
  - zxdo : upper elongation
  - apdo, amdo : lower X-point angles
  - b0 : maximum magnetic field on TF conductor
  - delta : gap between TF and inner plasma point
  - nbp : number points in LCFS
  - mode : interpolation curve type
  - filename : filename of LCFS given by point
  - ton : first time for X-point LCFS
  - toff : last time for X-point LCFS

Update or not B_0



Before and after, use references R, a, K, d

LCFS generator (2/3)

- LCFS GENERATOR:
  - Override references (R,a,K,d) between ton and toff
  - May re-computes B0 (you must check or reset B0 after)

- If you edit again any R, a, K or d reference:
  - LCFS is reset to moments only
  - You must restart the LCFS design

- Shared by HELIOS/SYCOMORE and METIS:
  - Same set of parameters

[HELIOS: A zero-dimensional tool for next step and reactor studies, J. Johner, FUSION SCIENCE AND TECHNOLOGY VOL. 59 FEB. 2011]

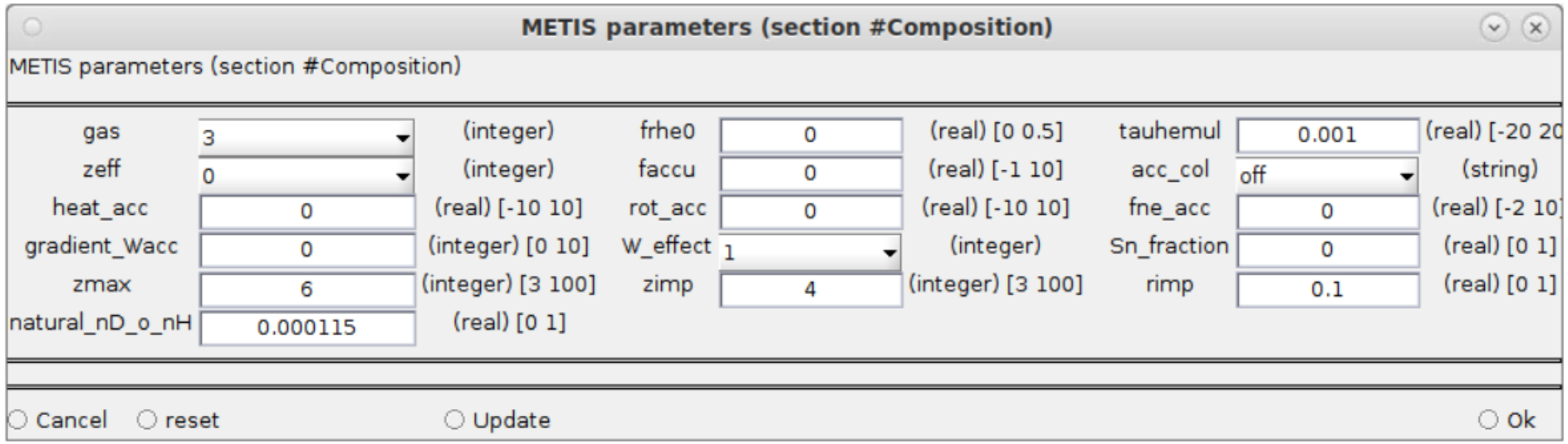
LCFS generator (3/3)

- Load a LCFS given by point:
  - Load file pointed by filename field, if field is not empty
  - If filename is invalid or “?”, open a file selector helping to choose the file
  - File must be a matfile containing R vector and Z vector of same length.
  - Override references (R,a,K,d) between ton and toff
  - Override LCFS given by parameters.

- If you edit again any R, a, K or d reference:
  - **LCFS is reset to moments only**
  - You must restart the LCFS design to get if back.

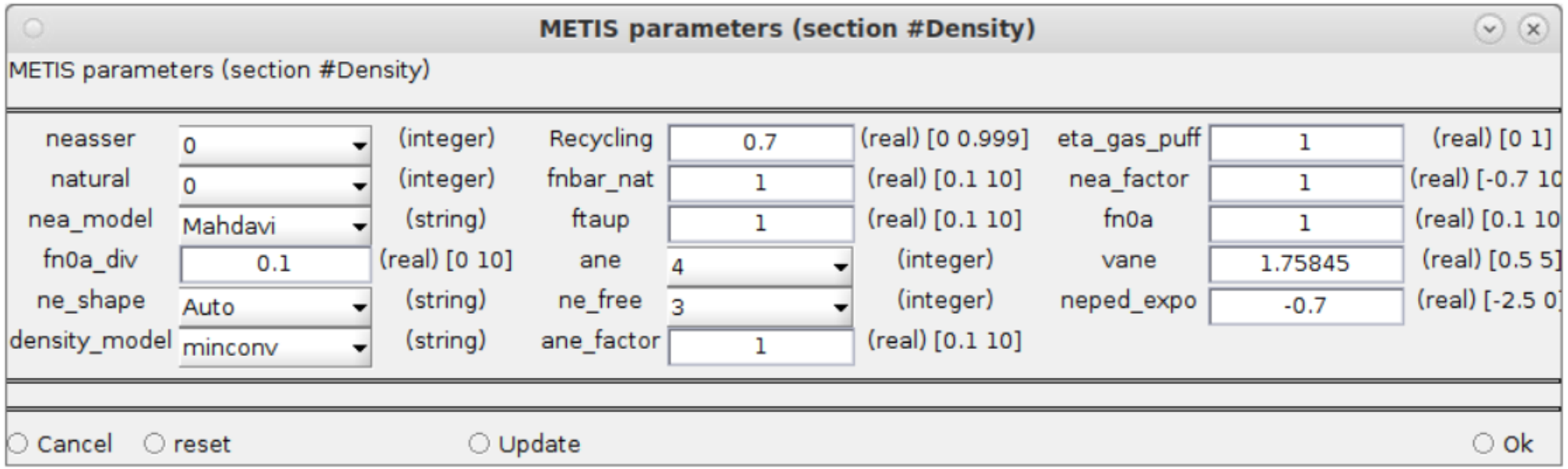
7. Main parameters for first use

METIS parameters GUI (Composition)



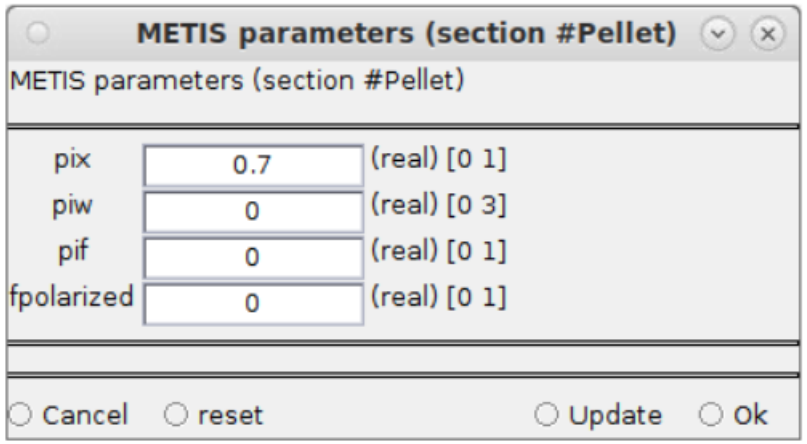
- Main Keys (for first use):
  - gas: main plasma ion species (H/D/He/DT)
  - zimp: charge of light/main impurity
  - zmax: charge of heavy/secondary impurity
  - rimp : density ratio of secondary over main impurity

METIS parameters GUI (Density)



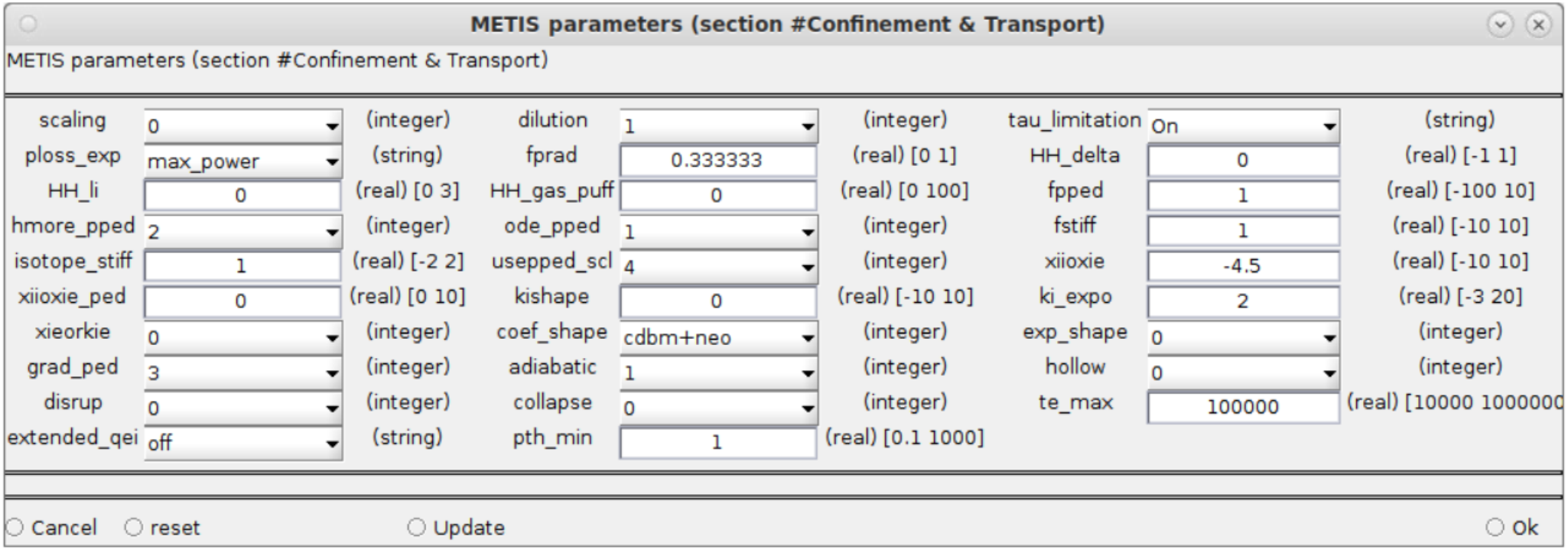
- Main Keys (for first use):
  - ane: model for density peaking factor
  - vane: value of density peaking factor if provided

METIS parameters GUI (Pellet)



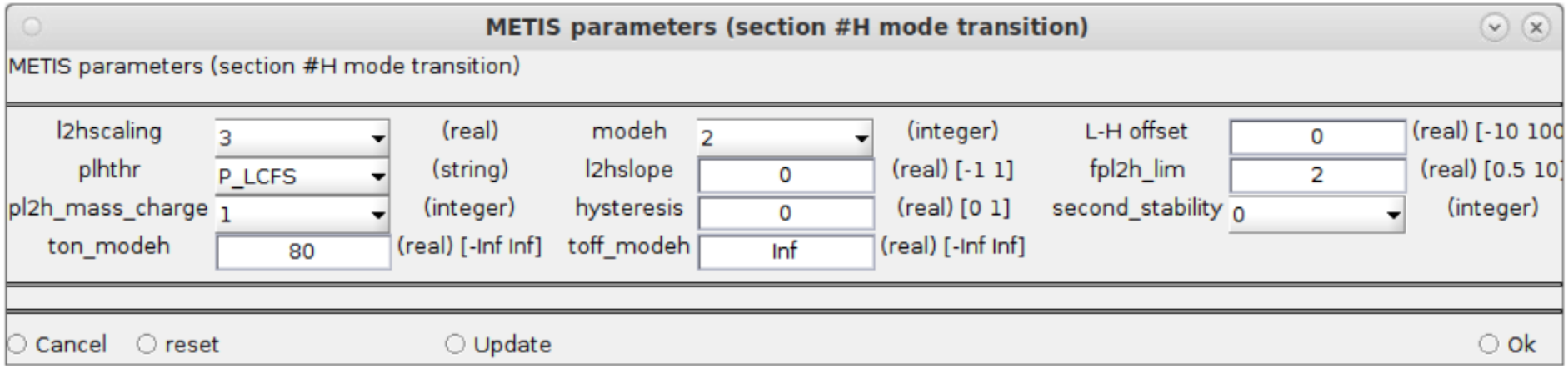
- Main Keys (for first use):
  - pif: set the fraction of fueling due to pellet

METIS parameters GUI (Confinement & transport)



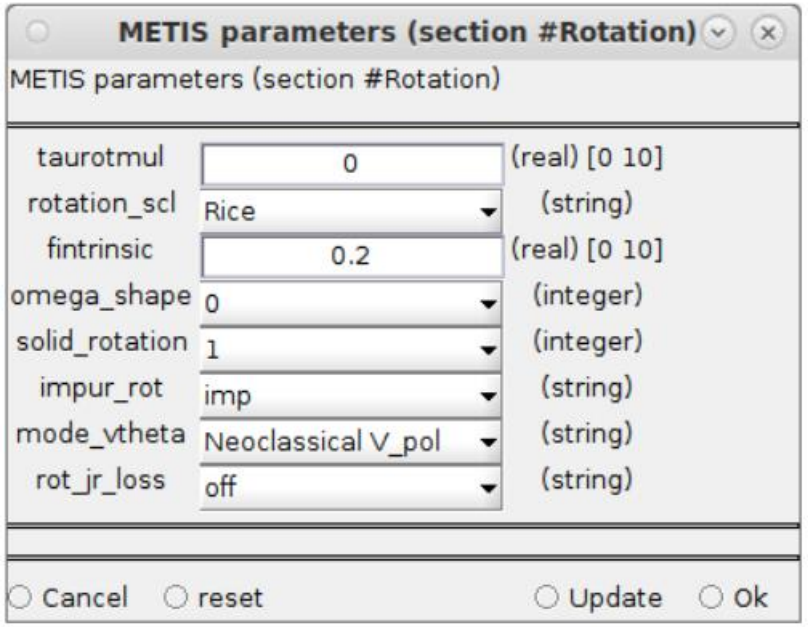
- Main Keys (for first use):
  - scaling: scaling law for energy content choice
  - fpped: pressure at the top of pedestal (factor)
  - usepped_scl: set on scaling for pedestal pressure

METIS parameters GUI (H mode transition)



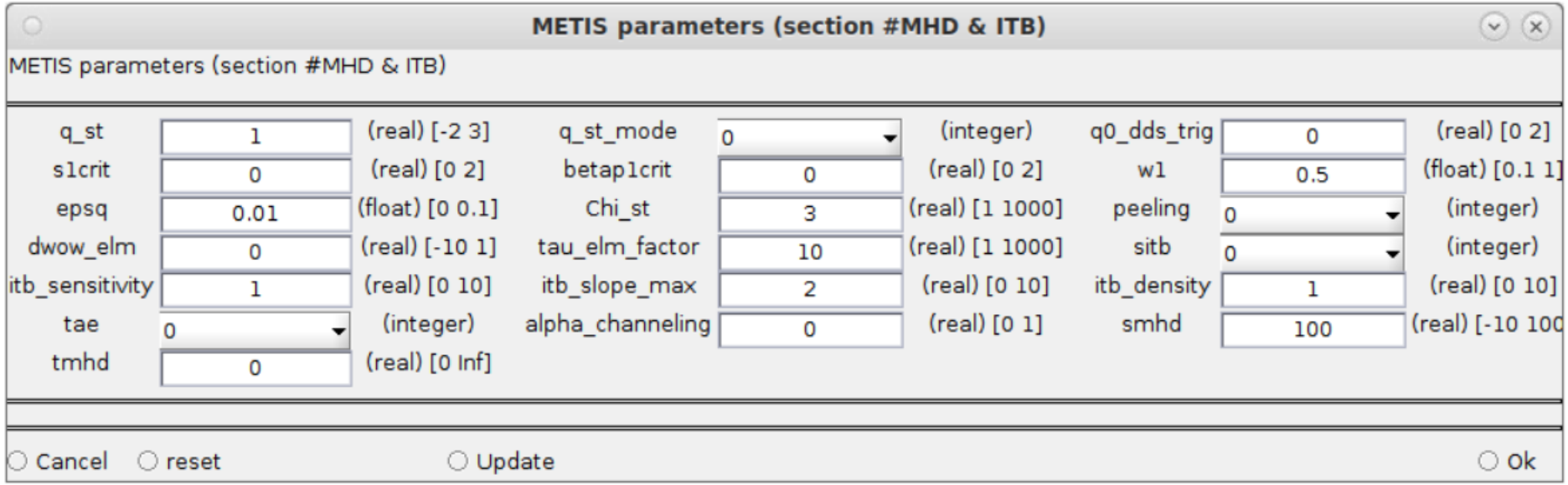
- Main Keys (for first use):
  - l2hscaling: scaling law for L->H transition
  - modeh: allow/force/prevent H-mode transition
  - l2hmul: offset for the transition

METIS parameters GUI (Rotation)



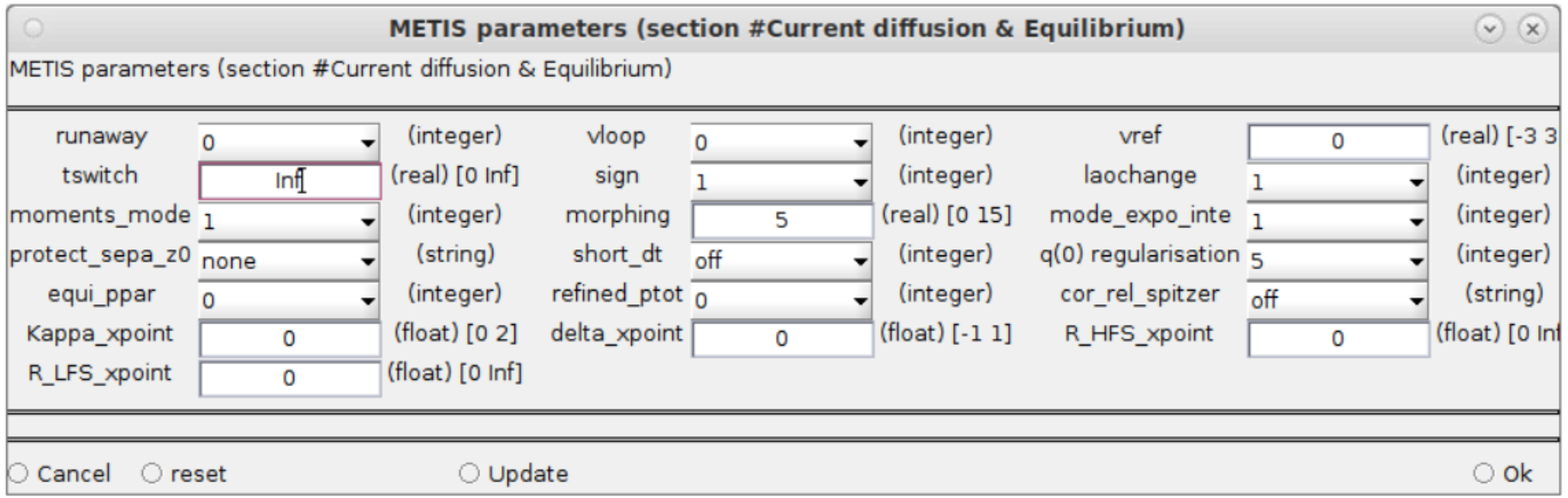
- Main Keys (for first use):
  - taurotmul: τ_φ / τ_E
  - fintrinsic: fraction of intrinsic rotation taking into account

METIS parameters GUI (MHD & ITB)



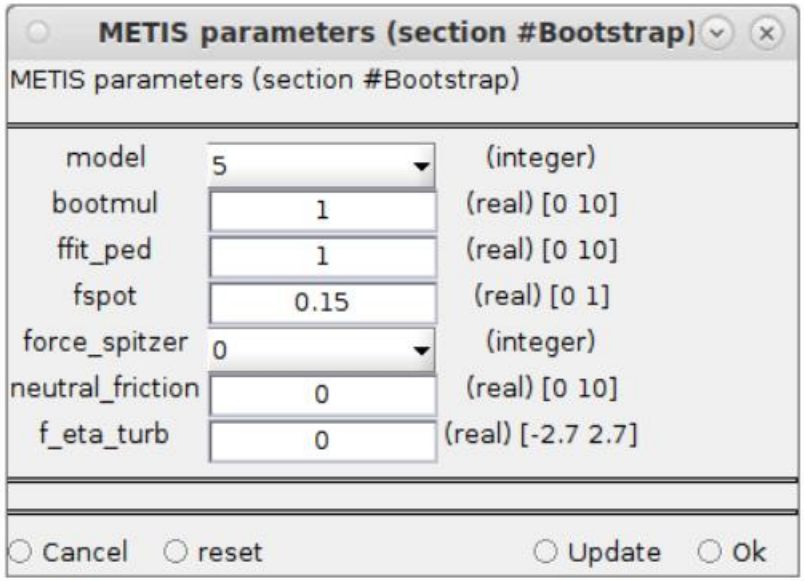
- Main Keys (for first use):
  - q_st: set on sawtooth model
  - Chi_st: effect of sawtoth on transport
  - smhd: limitation of β_N

METIS parameters GUI (Current diffusion & Equilibrium)



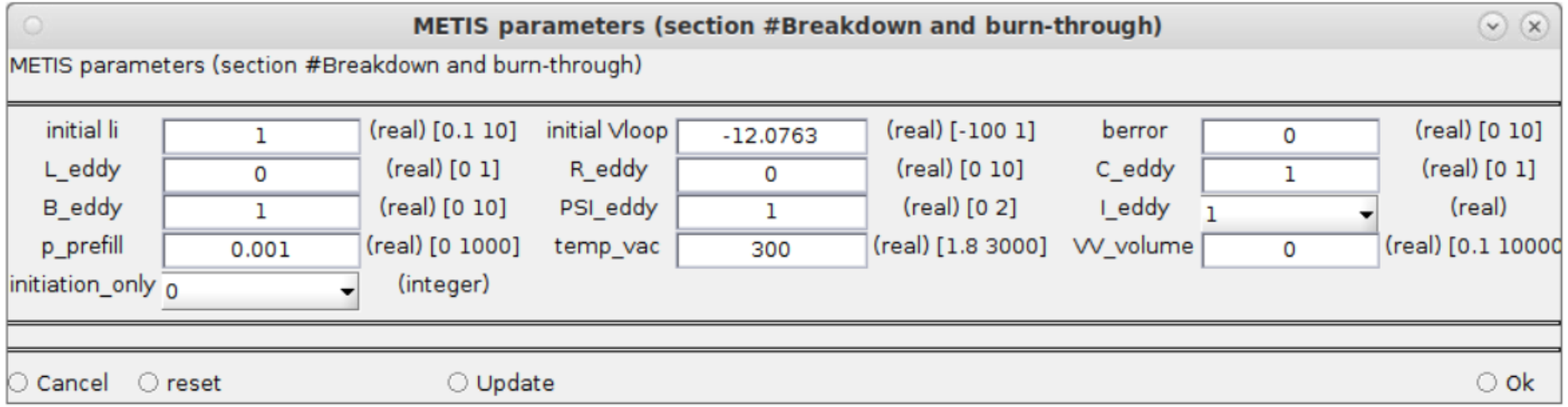
- Main Keys (for first use):
  - runaway: switch on/off runaway model (runaway can be a problem !)
  - signe: sign of Ip * Bphi
  - Cronos_regul: methode to compute safety factor @ magnetic axis

METIS parameters GUI (Bootstrap)



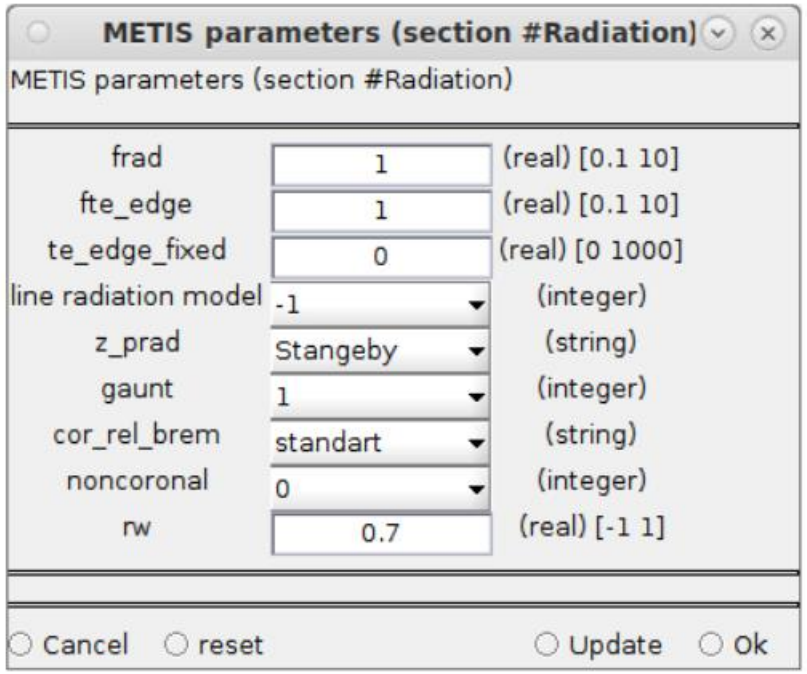
- Main Keys (for first use):
  - bootmul: factor applied to boostrap current

METIS parameters GUI (Breakdown and burn-through)



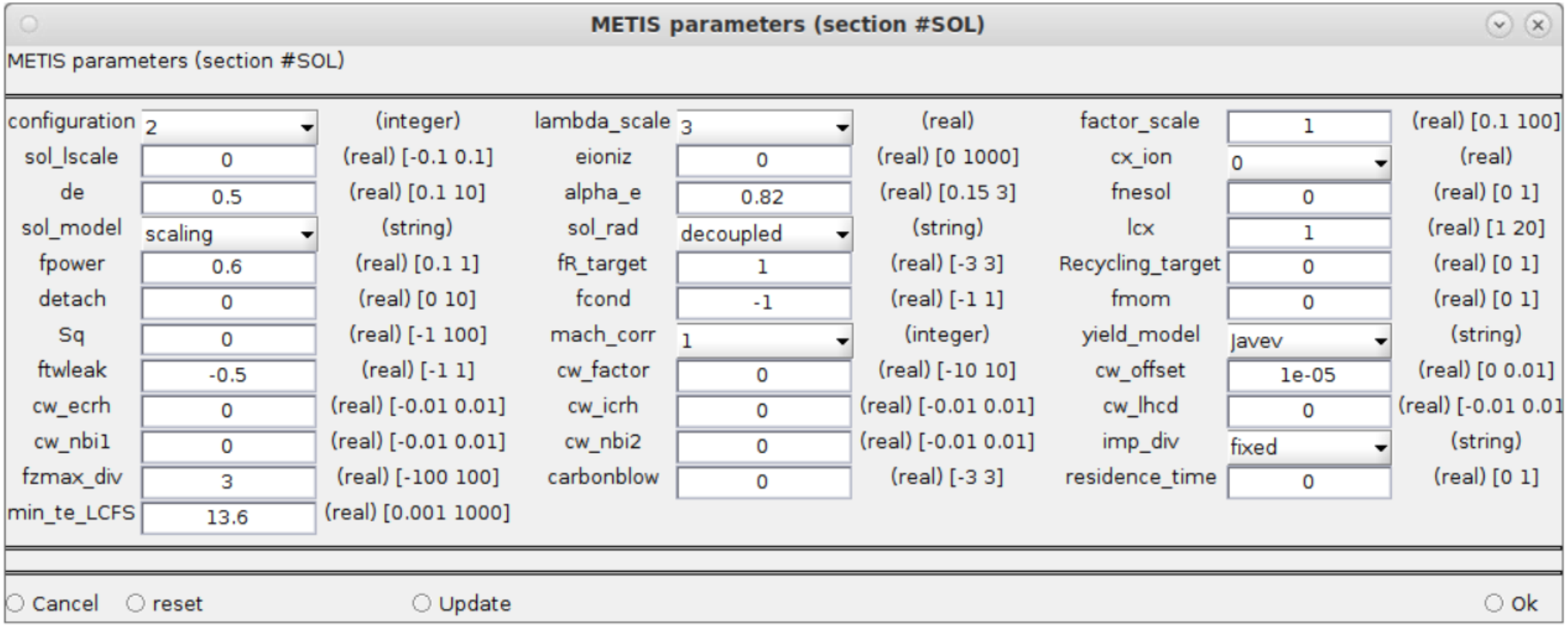
- Main Keys (for first use):
  - berror: should be set to 0 to keep the breakdown and burn-through model switched off

METIS parameters GUI (MHD & ITB)



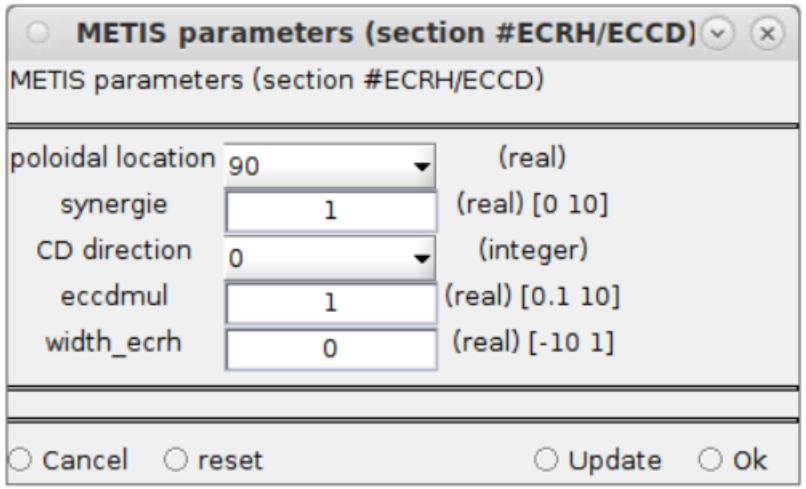
- Main Keys (for first use):
  - frad: factor applied to line radiative power
  - line radiation model: use Matthews scaling (for carbon machine) or only cooling rate
  - rw: effective wall cyclotron radiation reflection fraction

METIS parameters GUI (Scrape of layer)



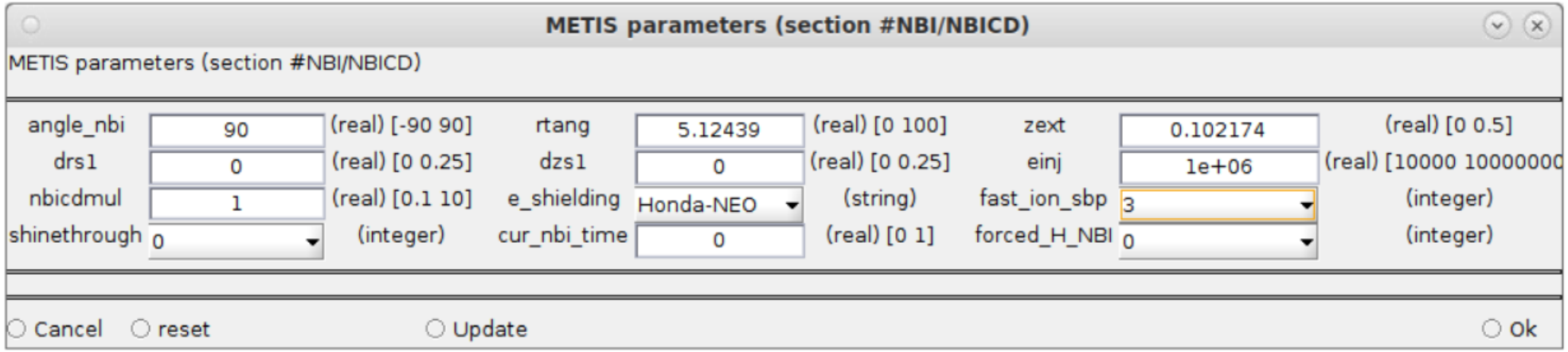
- Main Keys (for first use):
  - configuration: tokamak configuration (poloidal limiter, toroidal limiter, divertor, …)
  - sol_rad: should be set to decoupled for first use
  - cw_offset: set the basic plasma content in tungsten

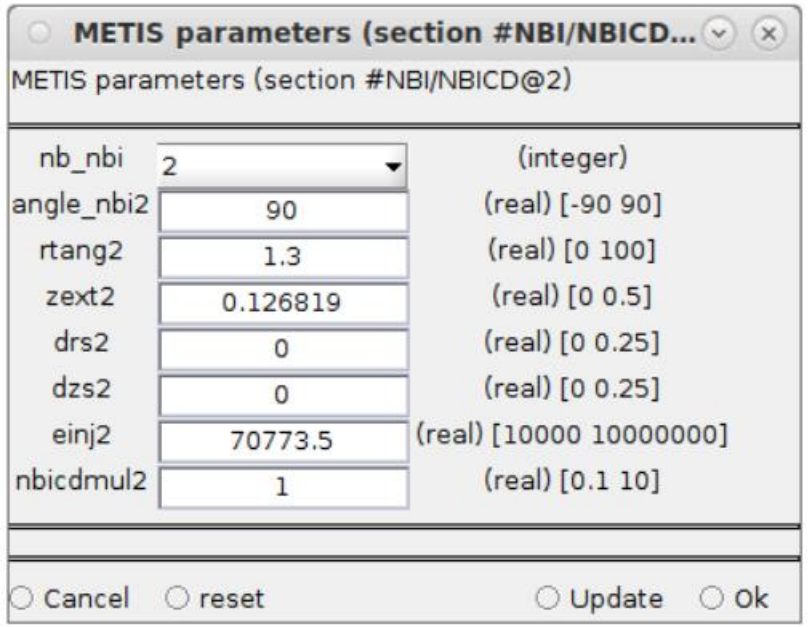
METIS parameters GUI (ECRH/ECCD)



- Main Keys (for first use): all
  - angle_ecrh: poloidal position of absorption
  - sens: co current, perpendicular or counter current

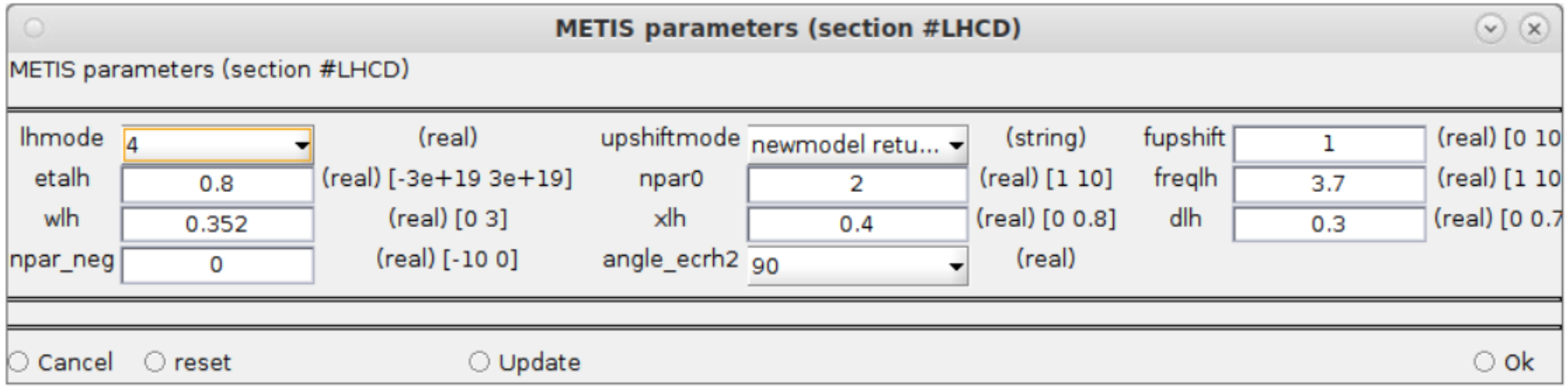
METIS parameters GUI (NBI/NBICD)





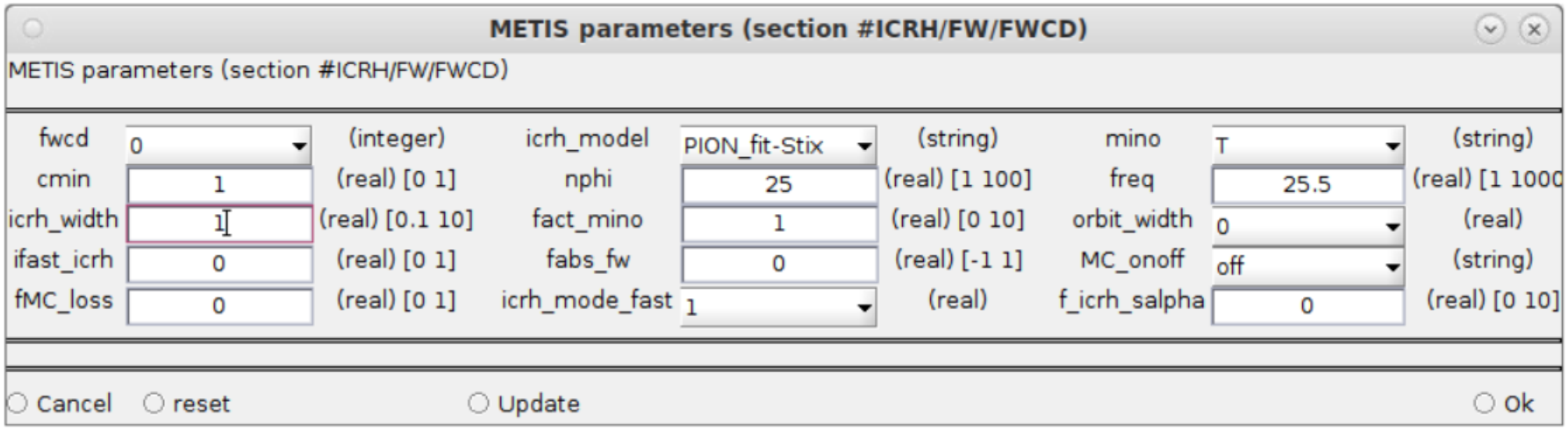
- Main Keys (for first use): all
  - Standard use : provided rtang & set current direction with angle_nbi
  - Set current direction with angle_nbi
    (-90= counter, 0 = perpendiculr, 90 = co)
  - Adjust deposition with zext
  - nb_nbi: to set the use of one or two NBI

METIS parameters GUI (LHCD)



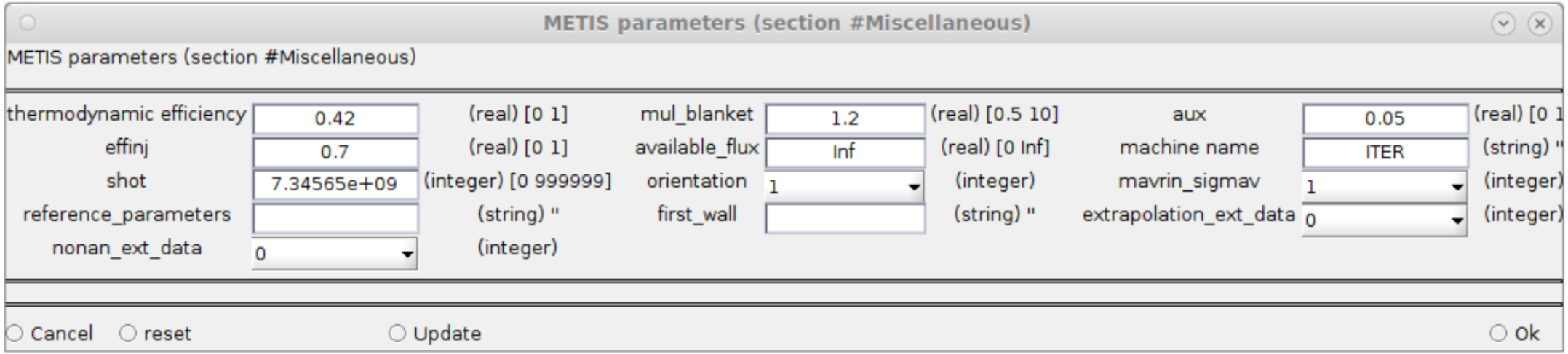
- Main Keys (for first use): all
  - Internal METIS model must be used cautiously
  - Recommended mode: newmodel retuned
  - For Tore Supra and WEST, can use hard X-rays bremsstrahlung data
  - For current drive efficiency: scaling (lhmode = 4) is recommended

METIS parameters GUI (ICRH)



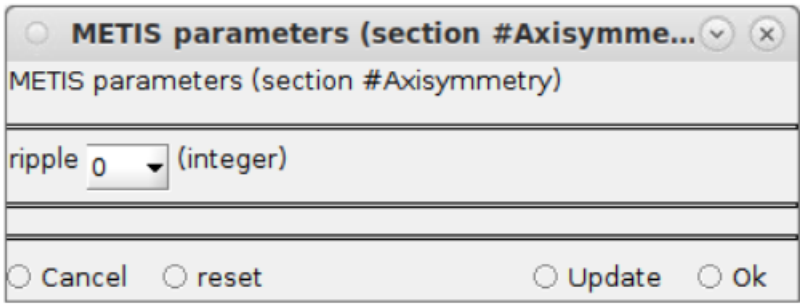
- Main Keys (for first use): all
  - fwcd: selection of heating scheme
  - mino & cmin : type and concentration of minority
  - nphi: toroidal wave number
  - freq : frequency of the injected wave (set to position of deposition)
  - Icrh_width: focalization of wave in plasma

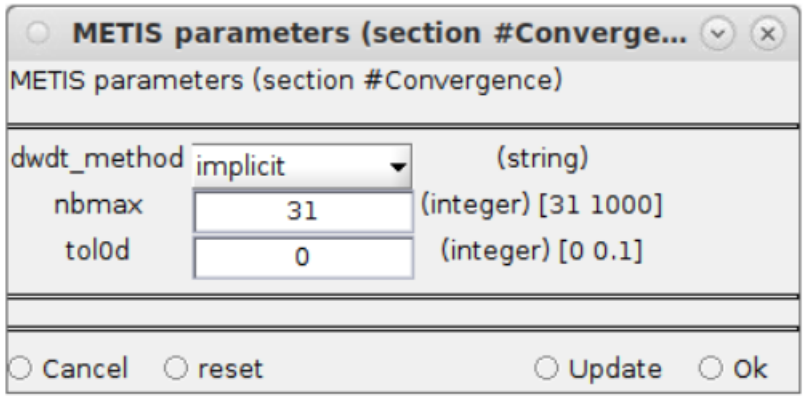
METIS parameters GUI (Miscellaneoous)



- Main Keys (for first use):
  - shot: shot number
  - machine: machine name
  - orientation: direction of magnetic toroidal field
  - reference_parameters :
    file name of METIS parameters as created with the button “Create reference”.
    Override METIS internal parameters if used!

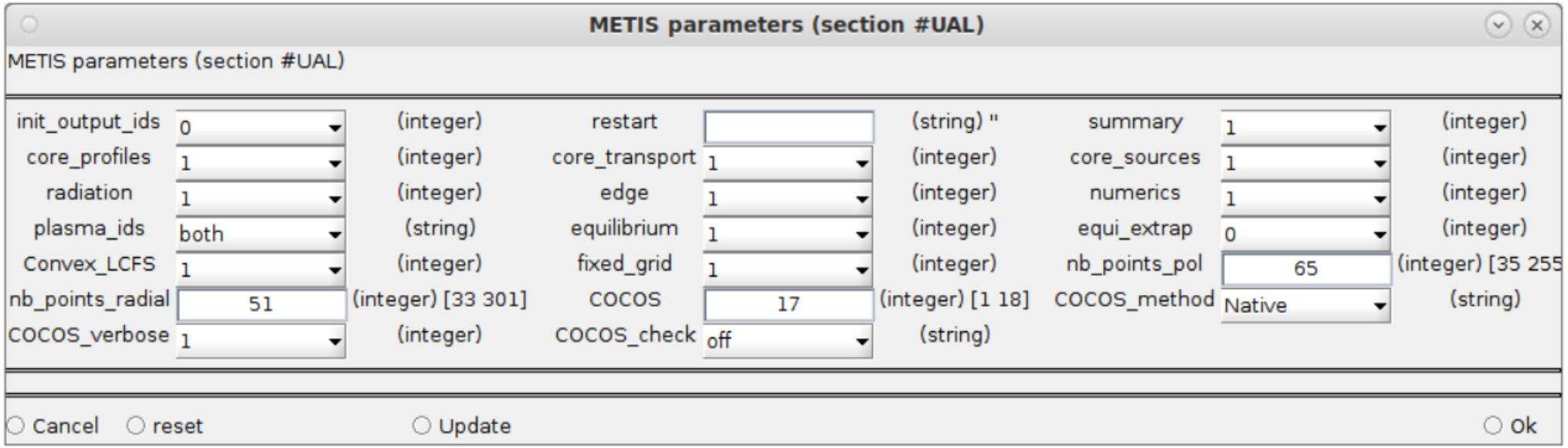
METIS parameters GUI (Axisymmetry - Convergence)

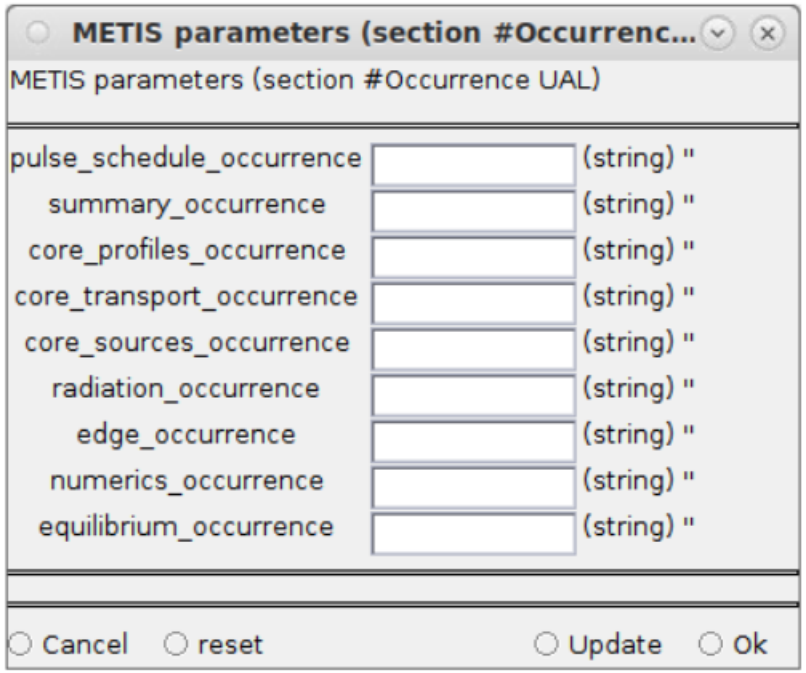




- This settings are reserved for advanced use only

METIS parameters GUI (UAL - Occurrences)





8. Advanced METIS Features

METIS inline MATLAB help

- In the GUI and in Matlab command line:
  - ✓ Parameters tooltip
  - ✓ Data description
- For evolution mode in Matlab command line:
  - ✓ Simulink help : simmetis, zerodevolution
  - ✓ IMAS help : metis4imas
- For external data used :
  - ✓ external_data_rule_for_METIS.m

METIS in MATLAB workspace

- **Standalone METIS data structure:**
  - **Input data: z0dinput**
  - **Output data: post**
    - **post.zerod: 0d data**
    - **post.profil0d: profiles**
    - **post.z0dinput: input data used during simulation**

- **Evolution mode data structure:**

  z0dstruct

METIS input data structure

- z0dinput structure:
  - Information:
    - z0dinput.info: parameter tooltips
    - z0dinput.zsinfo: 0D data descriptions
    - z0dinput.profinfo: 1D data descriptions
  - Data:
    - z0dinput.option: parameters
    - z0dinput.cons: references (Ip, powers, …)
    - z0dinput.geo: plasma geometry
    - z0dinput.exp0d: experimental data + LCFS pts

METIS in MATLAB CLI

- METIS commands:
  - metis_load: load and update a simulation
  - metis_save: save a simulation (from workspace)
  - metis_fast: fast simulation (selected time slices)
  - metis_run: full simulation (all time slices)
  - metis_fitlh: search for LHCD current drive efficiency
  - metis_upadate_tests: update non regression tests
  - z0working_point: compute steady state solution for a selected time slice

Remark: under MATLAB used “help <function name>” to get function documentation.

METIS call from a MATLAB script

- zerod_init: initialize a simulation
- zerod_fast: fast simulation (selected time slices)
- zerod: full simulation (all time slices)
- zerodevolution: one time step simulation
- metis4imas: run metis inside IMAS framework
- simmetis: Simulink interface for METIS

9. METIS & Simulink

METIS in SIMULINK

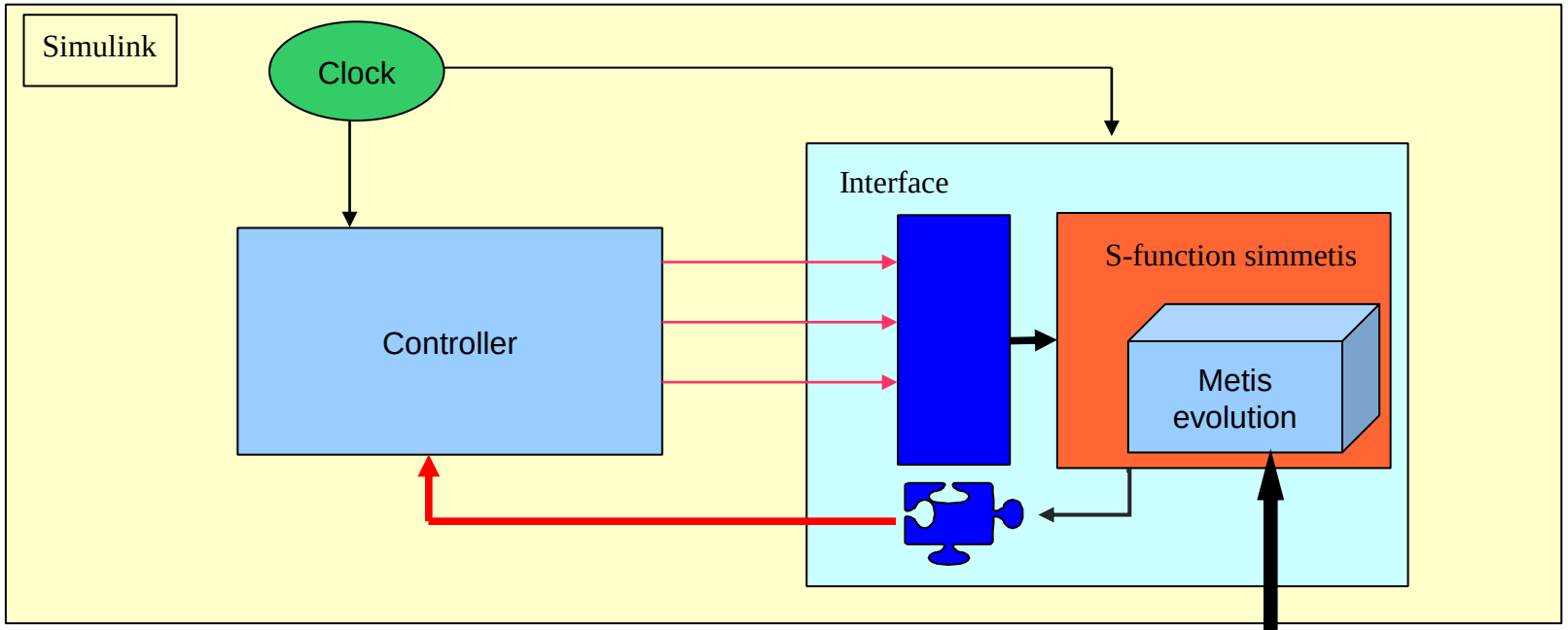
- ✓ METIS call from Simulink = S-function simmetis
- ✓ Standard S-function Input & output

- ○ simmetis use one time step METIS computation: zerodevolution

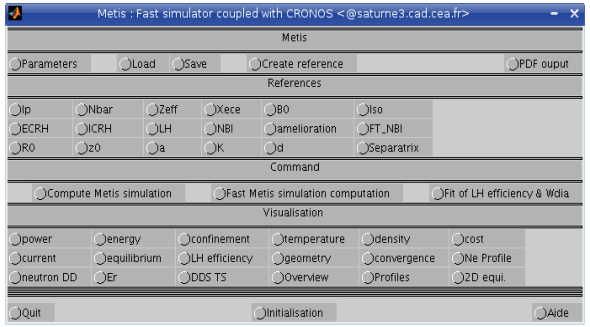
- ➢ Some documentation is written inside functions
  - ✓ help simmetis or help zerodevolution (in Matlab)
  - ✓ S-function documentation :

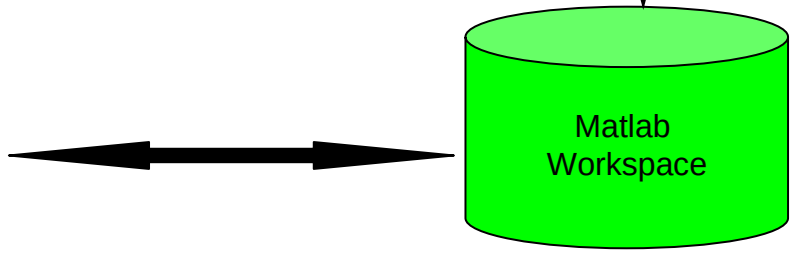
    http://www.mathworks.fr/fr/help/simulink/slref/sfunction.html

METIS evolution in Simulink



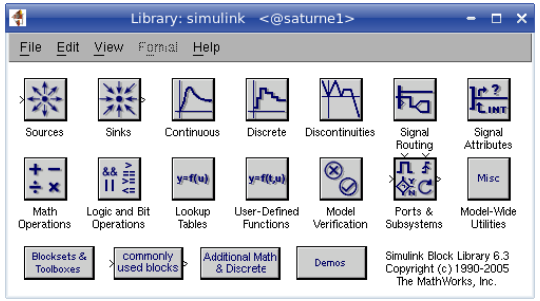
METIS : main interface

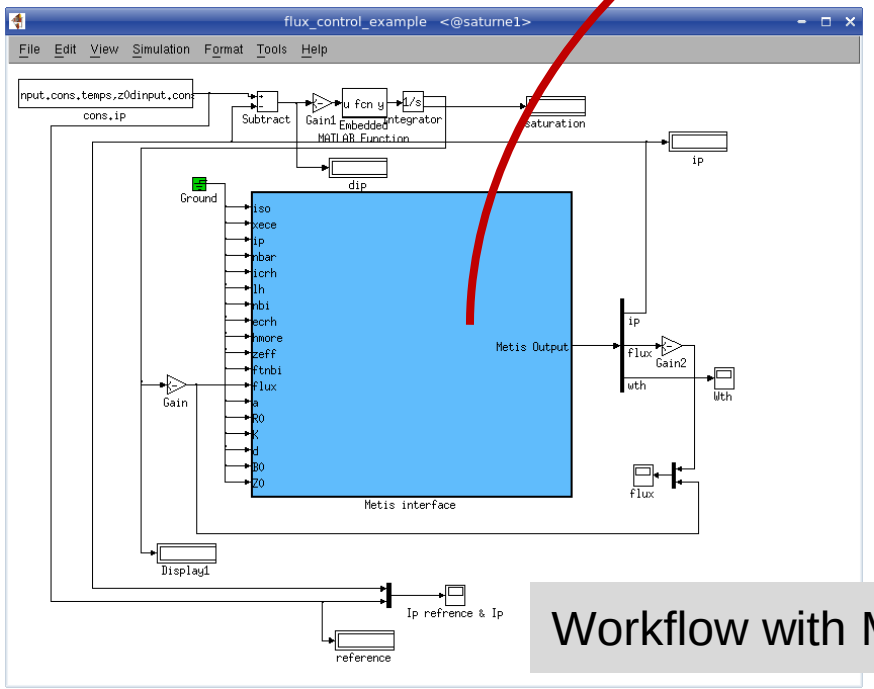


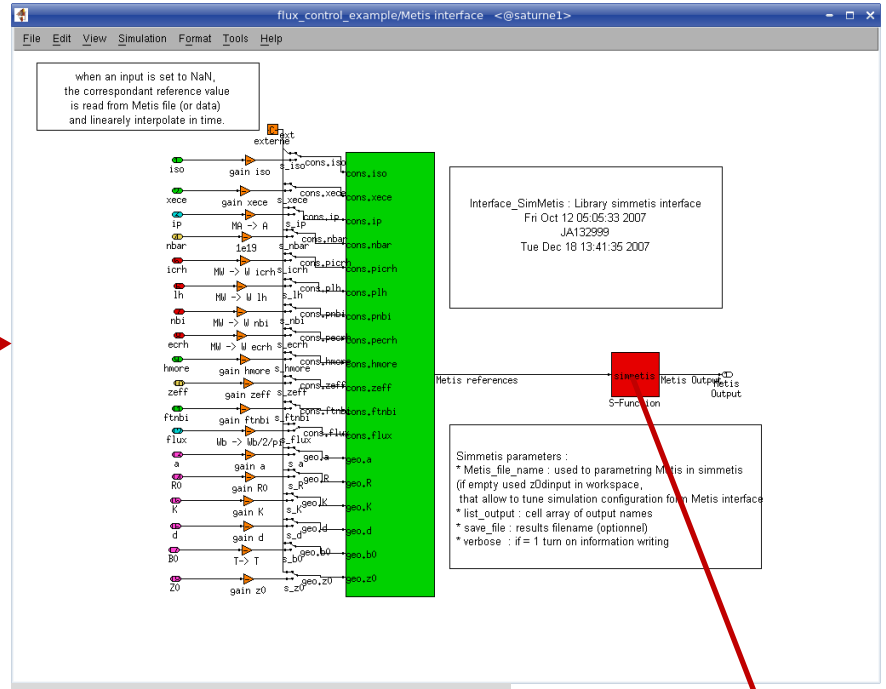


METIS interface to Simulink

Simulink







Interface to simmetis

S-function simmetis

Workflow with METIS

Sample Simulink workflows

- Sample workflow for simmetis testing (under …/zerod/):
  - test_simmetis.mdl : simple test
  - flux_control_example.mdl : poloidal flux feedback test

- Sample interface between Simulink and simmetis (under …/zerod/):
  - interface_simmetis.mdl : switches, multiplexor + S-function call

- For testing:
  - Launch METIS + load a simulation
  - Open Simulink workflow, tune simulation parameters, and run it

Matlab workspace (in evolution mode)

- METIS takes default parameters and references in Matlab workspace

➢ METIS GUI still active with Simulink

✓ Only active selected references and parameters are override by Simulink data

- All METIS internal data copy in Matlab workspace

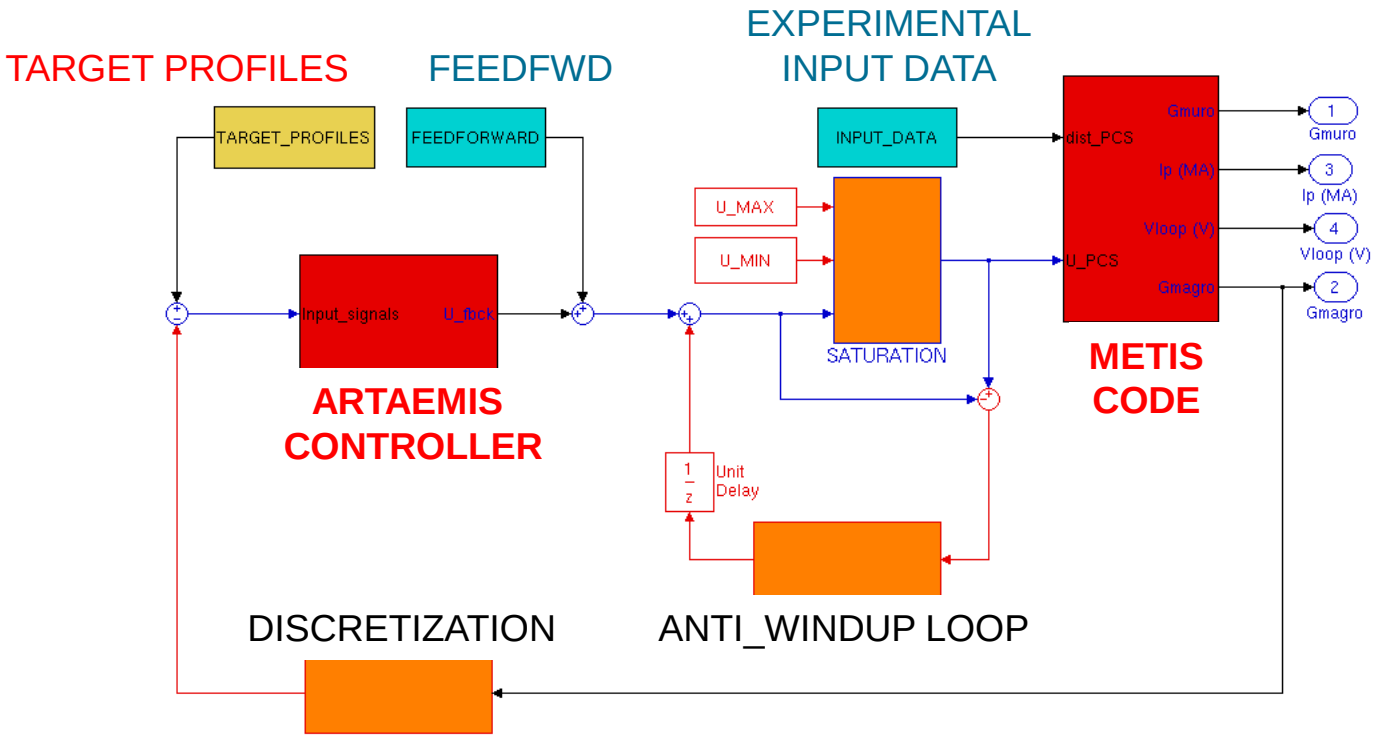
➢ Simulink can access to any METIS data

METIS data structures for Simulink

- Last (current time) METIS data are available in Matlab workspace from Simulink:
  - zerod_simmetis: 0D METIS data
  - profil0d_simmetis: 1D METIS data
  - cons_simmetis: references (use by METIS)
  - geo_simmetis: geometry (use by METIS)
  - sepa_simmetis: if LCFS given by points

- More data available in global structure:
  see simmetis online help

METIS & Simulink use case

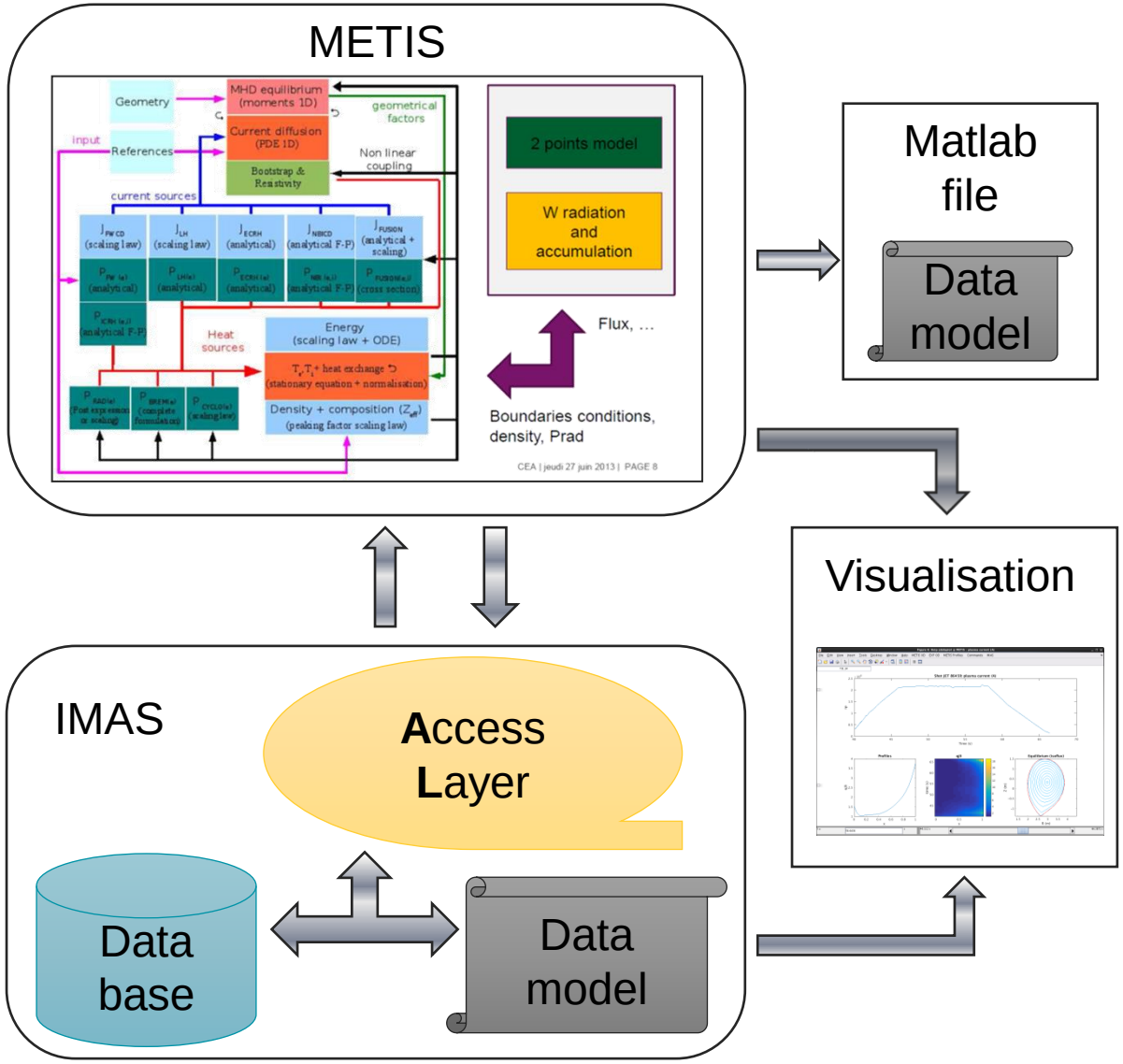


- ✓ Closed loop ARTAEMIS/ METIS simulations to extend control to steady
- ✓ Experimental input data is taken from DIII-D shot

[ITR/P1-20 Integrated Magnetic and Kinetic Control of Advanced Tokamak Scenarios Based on Data-Driven Models D. Moreau et al, 24th IAEA Fusion Energy Conference]

10. METIS IMAS Interface

METIS AL connection



- Allows to call IMAS tools using METIS data set.
- METIS can get its inputs from IMAS
- METIS can write IDSs in IMAS database.
- Can be used for analysis shot data, mixing both experimental data and METIS prediction

- IDSs:
  - Core_profiles / Plasma_profiles
  - Core_sources / Plasma_sources
  - Core_transport / Plasma_transport
  - Equilibrium
  - Pulse_schedule
  - Summary
  - Dataset_description
  - transport_solver_numerics
  - Edge IDSs for some METIS configurations

Export data in IMAS like format

- Button export:
  - Create IMAS data structure with all requested IDSs and store it in a “matfile”.

- “matfile” format depends on Matlab preferences:
  - See command “preferences” in Matlab (https://fr.mathworks.com/help/matlab/ref/save.html)

Value of `version`	Loads in MATLAB Versions	Supported Features	Compression	Maximum Size of Each Variable
`"-v7.3"`	7.3 (R2006b) or later	Saving parts of variables, and all Version 7 features.	Yes (default)	≥ 2 GB on 64-bit computers
`"-v7"`	7.0 (R14) or later	Unicode® character encoding, which enables file sharing between systems that use different default character encoding schemes, and all Version 6 features. Version 7 also supports saving variables without compression using the `"-nocompression"` option.	Yes (default)	2^{31} bytes per variable
`"-v6"`	5 (R8) or later	Saving *N*-D arrays, cell arrays, and structure arrays; variable names longer than 19 characters; and all Version 4 features.	No	2^{31} bytes per variable
`"-v4"`	All	Saving 2-D `double`, character, and `sparse double` arrays.	No	100,000,000 elements per array, and 2^{31} bytes per variable

METIS4MAS overview

- metis4imas.m = interface UAL ↔ METIS
  - Drives METIS (call methods)
  - Reads and converts input IDS data
  - Maps IDSs data and writes in database

- Remark:
  - use imasdb command under Matlab to set the default user, machine and version that will be used by metis4imas later.

METIS4IMAS inputs

- shot = shot number.
- run = run number for this.
- occurrence = input IDS scenario occurrence in Kepler (default = 0).

- Method = METIS command (see below).
- time = time scalar or vector for which METIS must be run. If empty, read all time slices of the IDS and use IDS vector time as input.

- codeparam_filename = codeparam (see below).
- interpolation_methode = interpolation method used for data extraction (see UAL user guide).

METIS4IMAS methods

- test = test the function (use test data and create IDSs).
- auto_test = complete test of metis4imas (whole functions)
- full = whole shot computation, full METIS mode
- fast = whole shot computation using fast METIS mode
- Init = initialization of the evolution mode of METIS.
- one_time = compute plasma evolution for on time step.
- read = read UAL data and return in Matlab structure

- <filename> = load a METIS simulation and create associated IDS Matlab data structures.

METIS4IMAS code parameters

- Code parameters contains:
  - all standard METIS parameters
  - UAL specific parameters that allow to control which IDS are written (+ occurrence choice)
- Sources of code parameters:
  - XML string
  - XML file
  - Input IDS pulse_schedule (if codeparam empty)
  - Default values if not specified (no input IDS pulse_schedule & codeparam empty)
- Can be stored in a Matlab structure (for testing only)

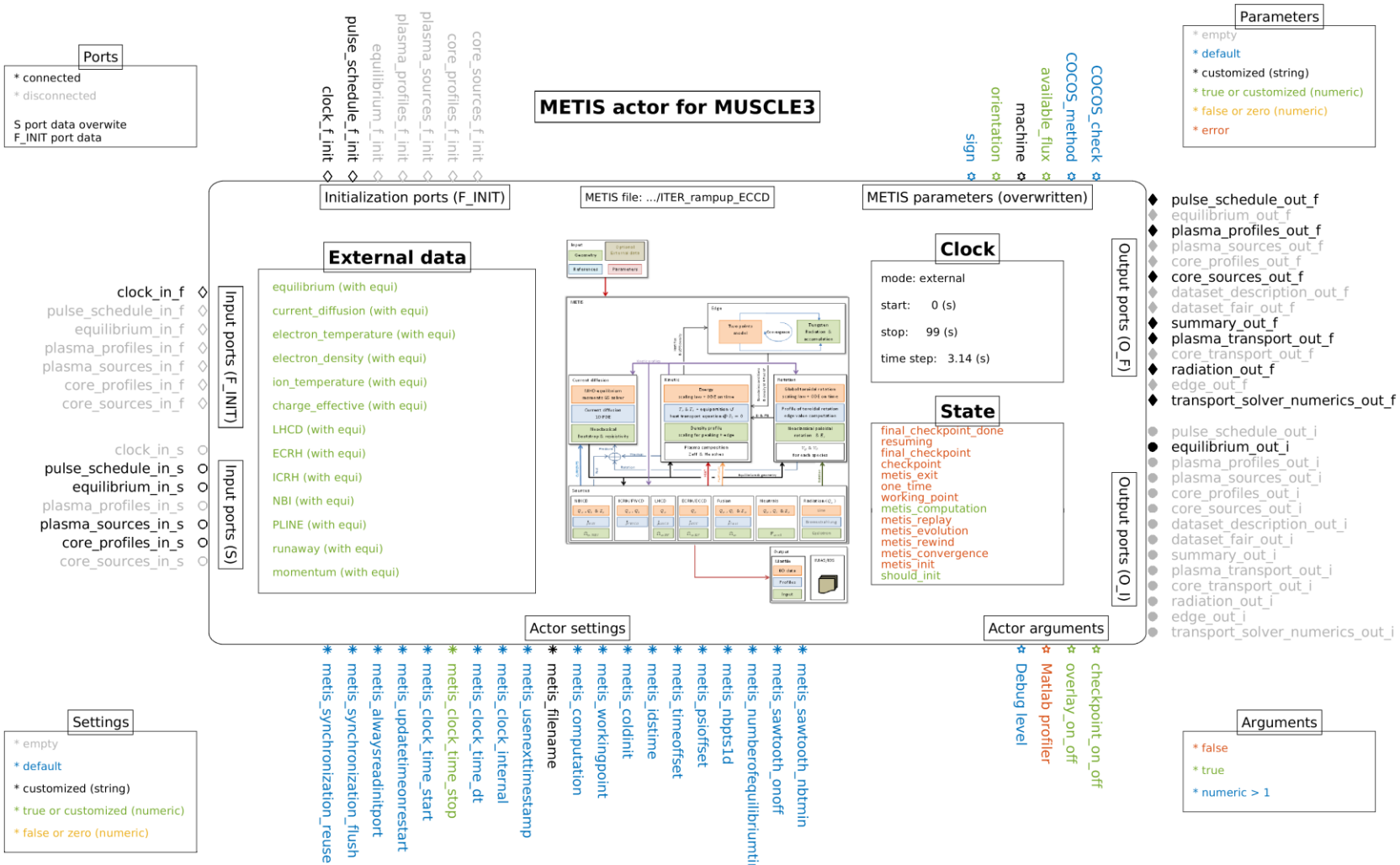
METIS4IMAS outputs

- Standard output:
  - err_flag = error code, must be = 0 if no error
  - output_data = Matlab structure containing the counterpart of IDSs structures in Matlab compatible format.

- METIS internal state:
  - stored in appdata IMAS_Z0DSTRUCT
  - allows to access to any METIS data outside of METIS.

11. METIS MUSCLE3 ACTOR

METIS MUSCLE3 ACTOR OVERVIEW



METIS MUSCLE3 ACTOR DESCRIPTION

- Code source is in <METIS_root>/workflow/
  - Matlab code is in in <METIS_root>/muscle3/workflow/mfile
  - Examples of MUSCLE3 workflow with METIS can be found in <METIS_root>/workflow/muscle3/YMMSL
  - Examples of launch scripts for workflows can be found in <METIS_root>/workflow/muscle3/run_script

- The Matlab gateway function between MUSCLE3 and METIS is ”metis4muscle3”

- The actor documetation is here: <METIS_root>/ doc/METIS actor for MUSCLE 3.pdf

- More workflow examples can be found in the PDS project:
  https://git.iter.org/projects/SCEN/repos/pds/browse

- MUSCLE3 documentation can be found here:
  https://muscle3.readthedocs.io/en/latest/

Outlook on METIS MUSCLE3 YMMSL file

1. Header

```
ymmsl_version: v0.1
```

2. Checkpointing control

```
checkpoints:
  at_end: true
  simulation_time:
  - at: [30,60,90]
```

3. Model declaration

```
model:
  name: METIS4MUSCLE3_TEST
  components:
  metis:
      implementation: metis
```

METIS MUSCLE3 YMMSL file (ports declaration)

4. Ports declaration (4 operators, all optional)

```
    ports:
# Init and input ports used:
# clock port is used to force the synchronization (use only time stamp information)
# F_INIT ports are duplicated on init and in.
# after initialization port with operator F_INIT or S are used to send data to METIS
# If both are connected, data from input S ports overwrite data from input F_INIT ports

        f_init: [clock_f_init,pulse_schedule_f_init,equilibrium_f_init,plasma_profiles_f_init,plasma_sources_f_init,core_profiles_f_init,
                 core_sources_f_init,clock_in_f,pulse_schedule_in_f,equilibrium_in_f,plasma_profiles_in_f,
                 plasma_sources_in_f,core_profiles_in_f,core_sources_in_f]

        s: [clock_in_s,pulse_schedule_in_s,equilibrium_in_s,plasma_profiles_in_s,plasma_sources_in_s,
            core_profiles_in_s,core_sources_in_s]

        o_f: [equilibrium_out_f,pulse_schedule_out_f,plasma_profiles_out_f,plasma_sources_out_f,
              core_profiles_out_f,core_sources_out_f,dataset_description_out_f,dataset_fair_out_f,
              summary_out_f,plasma_transport_out_f,radiation_out_f,edge_out_f,transport_solver_numerics_out_f]

        o_i:[equilibrium_out_i,pulse_schedule_out_i,plasma_profiles_out_i,plasma_sources_out_i,
             core_profiles_out_i,core_sources_out_i,dataset_description_out_i,dataset_fair_out_i,
             summary_out_i,plasma_transport_out_i,radiation_out_i,edge_out_i,transport_solver_numerics_out_i]
```

METIS MUSCLE3 YMMSL file (conduits, input)

5. Connections between actors (all optional)

conduits:

5.1 - on init with ports f_init (use at actor initialization)

source.clock: metis.clock_f_init
source.pulse_schedule: metis.pulse_schedule_f_init
source.equilibrium: metis.equilibrium_f_init
source.plasma_profiles: metis.plasma_profiles_f_init
source.plasma_profiles: metis.core_profiles_f_init
source.plasma_sources: metis.plasma_sources_f_init
source.plasma_sources: metis.core_sources_f_init

5.2 - on input with ports F_INIT (if an IDS message is empty, previous data are kept)

source.clock: metis.clock_in_f
source.pulse_schedule: metis.pulse_schedule_in_f
source.equilibrium: metis.equilibrium_in_f
source.plasma_profiles: metis.plasma_profiles_in_f
source.plasma_profiles: metis.core_profiles_in_f
source.plasma_sources: metis.plasma_sources_in_f
source.plasma_sources: metis.core_sources_in_f

5.3 Input with ports S (ports S data overwrite ports F_INIT data if both are connected)

source.clock: metis.clock_in_s
source.pulse_schedule: metis.pulse_schedule_in_s
source.equilibrium: metis.equilibrium_in_s
source.plasma_profiles: metis.plasma_profiles_in_s
source.plasma_profiles: metis.core_profiles_in_s
source.plasma_sources: metis.plasma_sources_in_s
source.plasma_sources: metis.core_sources_in_s

METIS MUSCLE3 YMMSL file (conduits, output)

5.4 Output on O_I ports (ports can be O_I or O_F or both)

```
metis.equilibrium_out_i: sink.equilibrium_output
metis.pulse_schedule_out_i: sink.pulse_schedule_output
metis.plasma_profiles_out_i: sink.plasma_profiles_output
metis.core_profiles_out_i: sink.plasma_profiles_output
metis.plasma_sources_out_i: sink.plasma_sources_output
metis.core_sources_out_i: sink.plasma_sources_output
metis.dataset_description_out_i: sink.dataset_description_output
metis.dataset_fair_out_i: sink.dataset_fair_output
metis.summary_out_i: sink.summary_output
metis.plasma_transport_out_i: sink.plasma_transport_output
metis.core_transport_out_i: sink.plasma_transport_output
metis.radiation_out_i: sink.radiation_output
metis.edge_out_i: sink.edge_output
metis.transport_solver_numerics_out_i:
                                sink.transport_solver_numerics_output
```

5.5 Output on O_F ports (ports can be O_I or O_F or both)

```
metis.equilibrium_out_f:              sink.equilibrium_output
metis.pulse_schedule_out_f:           sink.pulse_schedule_output
metis.plasma_profiles_out_f:          sink.plasma_profiles_output
metis.core_profiles_out_f:            sink.plasma_profiles_output
metis.plasma_sources_out_f:           sink.plasma_sources_output
metis.core_sources_out_f:             sink.plasma_sources_output
metis.dataset_description_out_f:      sink.dataset_description_output
metis.dataset_fair_out_f:             sink.dataset_fair_output
metis.summary_out_f:                  sink.summary_output
metis.plasma_transport_out_f:         sink.plasma_transport_output
metis.core_transport_out_f:           sink.plasma_transport_output
metis.radiation_out_f:                sink.radiation_output
metis.edge_out_f:                     sink.edge_output
metis.transport_solver_numerics_out_f:
                                sink.transport_solver_numerics_output
```

METIS MUSCLE3 YMMSL file (MATLAB cmd)

6. Command to launch the Matlab actor

```
implementations:
  metis:
      executable: matlab
```

6.1 without checkpointing and with debug level switch to 2 (opening a Matlab console)

```
args: -r "[s,t] = unix('which python');pyenv('Version',strtrim(t),'ExecutionMode','InProcess'); …
          addpath(getenv('DIR_METIS4MUSCLE3'));metis4muscle3(2,false);"
```

6.2 with checkpointing and with debug level switch to 2 (opening a Matlab console)

```
args: -r "[s,t] = unix('which python');pyenv('Version',strtrim(t),'ExecutionMode','InProcess'); …
          addpath(getenv('DIR_METIS4MUSCLE3'));metis4muscle3(2,true);“
```

6.3 with checkpointing and debug switch off (Matlab launch in batch mode)

```
args: -batch "[s,t] = unix('which python');pyenv('Version',strtrim(t),'ExecutionMode','InProcess'); …
          addpath(getenv('DIR_METIS4MUSCLE3')); metis4muscle3(0,true);"
```

The variable DIR_METIS4MUSCLE3 should be defined in the launch script and point to the directory containing the Matlab .m files for METIS MUSCLE3 actor (<METIS_root>/workflow/muscle3/mfile).

METIS MUSCLE3 YMMSL file (workflow)

6. METIS actor settings

6.1. workflow synchronization

```
# synchronisation on reuse_instance : METIS actor can be forced to be synchronized or not on reuse_instance state
   metis_synchronization_reuse: 0

# Force buffer to be flushed when O_I ports are used to prevent workflow dead lock or message synchronization problem
#  on some system with MUSCLE3
  metis_synchronization_flush: 0

# if = 1, read also init ports on every iteration in addition of input ports, not only at initialization
  metis_alwaysreadinitport: 0

# if =1, update time on restart with resuming message time stamp
  metis_updatetimeonrestart: 0
```

METIS MUSCLE3 YMMSL file (workflow)

6.2. workflow datation

```
# start of the simulation in relay mode  (if inside simulated data time, otherwise start from first time
# of the simulation provide by MUSCLE3 message)
  metis_clock_time_start: 0.0

# end time of the simulation (if in replay mode, the simulation stop at the minimum of this time and the end of simulated data.
# Can be set to large number to switch it off)
  metis_clock_time_stop:  99.0

# time step (s): used if no source of time is provided to the actor or if internal clock is used
  metis_clock_time_dt: 3.14

# switch on or off  internal actor clock use
  metis_clock_internal: off

# if = 1, METIS compute for next_timestamp instead of timestamp information provided in MUSCLE3 message
  metis_usenexttimestamp: 0
```

METIS MUSCLE3 YMMSL file (computation)

6.3. METIS initialization and behavior

```
# METIS file name (with full path) for replay mode
  metis_filename:  /Home/JA132999/Zone_Travail/METIS4ITM/eufus.Git/metis/certification/metis/ITER_rampup_ECCD.mat

# for testing METIS computation can be switch off and data are interpolated from METIS simulation file defined
# by setting metis_filename
  metis_computation: 1

# if = 1, METIS is run in working point mode for power plant design
  metis_workingpoint: 0

# cold initialisation: if =1, if = 1, METIS is initialized from METIS simulation file defined by setting metis_filename
# using interpolated data instead of making a first computation run
  metis_coldinit: 0
```

METIS MUSCLE3 YMMSL file (synchro)

6.4. External data management (see next section for selection of external data used in the workflow)

```
# metis_idstime: if = 1, external data will use time store in IDS instead of time stamp from MUSCLE3 message
  metis_idstime: 0

# metis_timeoffset: offset between METIS internal time (time stamp in METIS output message)
# and input message time stamp or IDS time (s).
  metis_timeoffset: 0.0

# metis_nbpts1d: number of radial points in 1D profile used for external except for the equilibrium (default = 101).
  metis_nbpts1d: 101

# polodial flux offset between METIS and external equilibrium psi_metis = cocos_sign * psi_equilibrium / (2*pi) - psioffset
  metis_psioffset: 0.0

# number of external equilibrium time slices stored in METIS actor. The minimum is 3.
# Should be set to Inf if the workflow is indented to rewind. It has a impact on memory used by the actor.
  metis_numberofequilibriumtimeslices: 3
```

METIS MUSCLE3 YMMSL file (model configuration)

6.5. Tuning of sawtooth mimicking mechanism (to be used with NICE code in evolutive mode)

```
# Turn on or off mechanism for mimicking sawtooth when METIS is used in link with NICE code solving Grad-Shafranov
#  and diamagnetic function evolution (equivalent to current diffusion equation).
  metis_sawtooth_onoff: off

# minimum number of time steps between two sawtooth triggers
  metis_sawtooth_nbtmin: 3
```

6.6 METIS internal parameters (can be expanded with any METIS internal parameter)

```
# METIS optional parameter that will overload field of structure z0dinput.option. Any field of z0dinput.option can be used.
# The prefix “metis_option” is mandatory.
  metis_option_machine: test_muscle3
  metis_option_available_flux: 130.0

# orientation should be enforced as COCOS is handle by IMAS IDS (sign=1 & orientation = -1 for ITER)
  metis_option_sign: 1
  metis_option_orientation: -1

# if other IDS than equilibrium and plasma_profiles are exported from METIS,
# the intrinsic METIS COCOS function should be used as the tool from O.Sauter take in charge only these two IDSs.
  metis_option_COCOS_method: Native
  metis_option_COCOS_check: off
```

METIS MUSCLE3 YMMSL file (external data on/off)

6.7. external data use in METIS

```
# METIS switch for the use of external data read in IDSs

# From equilibrium IDS
  metis_external_data_equilibrium: 1
  metis_external_data_current_diffusion: 1

# From IDS plasma_profiles (need equilibrium IDS  for precise coordinates transformation)
  metis_external_data_electron_temperature: 1
  metis_external_data_electron_density: 1
  metis_external_data_ion_temperature: 1
  metis_external_data_charge_effective: 1
  metis_external_data_momentum: 1

# From IDS plasma_sources (need equilibrium IDS  for precise coordinates transformation)
 metis_external_data_LHCD: 1
  metis_external_data_ECRH: 1
  metis_external_data_ICRH: 1
  metis_external_data_NBI:  1
  metis_external_data_PLINE: 1
  metis_external_data_runaway: 1
```

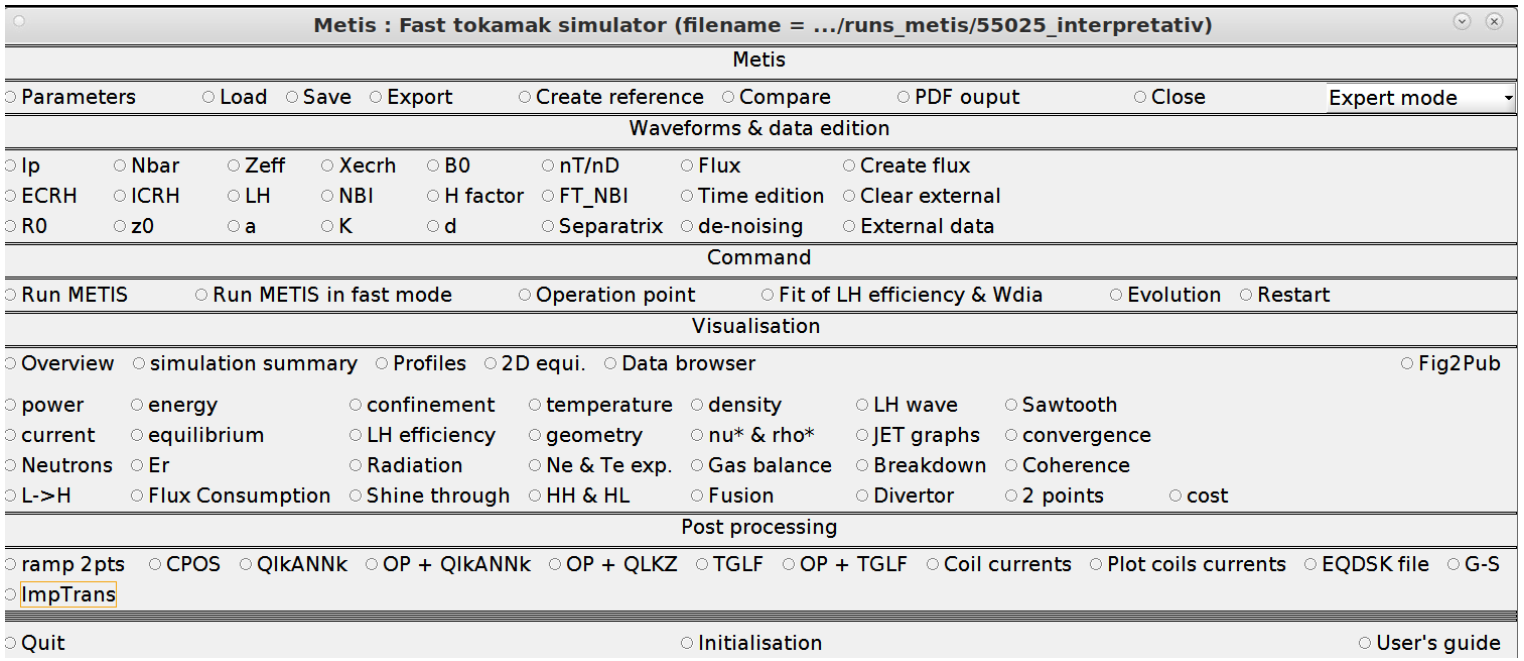
12. Post processing connected to METIS

Metis post processing connected to the GUI

METIS graphical interface allows to call various post processing functions:

- Need external codes to be installed separately to METIS
- Outputted data can be used as external data in METIS when it is relevant

Post processing



List of post processing connected to METIS

- Native Matlab post processing
  - ”ramp_2pts”: compute the result of two-point model versus the core plasma density (need two-point model be switch on in METIS).
  - CPOS: export METIS data in a matfile at the format of former data standart CPO defined during ITM EUROfusion activities (obsolete format, the IMAS standart should be use now).

- Requiring FEEQS.M code installation
  - “Coil currents”: Compute estimation of current in poloidal coils using fast inverse module of FEEQS.M (not suitable for tokamak with iron core)
  - “Plot coils currents”: Plot results of current in poloidal coils computed with fast inverse module of FEEQS.M
  - “EQDSK file”: Computation of EQDSK files using fast fixed boundary equilibrium solver of FEEQS.M. The extrapolation outside the LCFS will not included real x-point and
    is valid only far from external current sources.
  - “G-S”: Creation of an external equilibrium structure for METIS by recomputing Grad-Shafranov equation based on fast fixed boundary mode of FEEQS.M code and using METIS data as input (LCFS points, Ip, P’ and FF’).

- Requiring QuaLiKiz neutral network installation
  - ”OP + QlkANNk”: Compute operating point for one time slice using kinetic profiles predicted with Qualikiz neural network (steady state, all d/dt = 0, only dPsi/dt # 0). Creates external data for Te, Ti and Ne (external data must be cleared before next call of this tool)
  - ”QlkANNk”: Compute and plot flux from Qualikiz neural network.

- Requiring QuaLiKiz code installation
  - ”OP + QLKZ””: Compute operating point for one time slice using kinetic profiles predicted with Qualikiz (steady state, all d/dt = 0, only dPsi/dt #0). Creates external data for Te, Ti and Ne (external data must be cleared before next call of this tool

- Requiring TGLF code installation
  - ”TGLF”: Compute and plot results from TGLF simulations including e.g. fluxes, diffusivities, growth rates/frequencies spectra (Currently only available on IRFM/intra)
  - ”OP + TGLF”: Compute operating point for one time slice using kinetic profiles predicted with TGLF (steady state, all d/dt = 0, only dPsi/dt #0; currently only available on IRFM/intra.). Creates external data for Te, Ti and Ne (external data must be cleared before next call of this tool)

- Requiring NEO, FACIT & QuaLiKiz code installations
  - FACIT: Compute and plot results from FACIT simulations
  - NEO: Compute and plot results from NEO simulations
  - ”ImpTrans”: Compute and plot results associated to impurity transport based on QuaLiKiz, NEO and FACIT codes

External codes for METIS post processing

- FEEQS.M
  - Infomation about FEEQS.M are available here: https://www-sop.inria.fr/members/Holger.Heumann/Software.html
  - FEEQS.M code is available here: https://git.iter.org/projects/EQ/repos/feeqs/browse/FEEQS.M or https://gitlab.inria.fr/hheumann/feeqs
  - For the version on INRIA git server, the branch “iter_to_inria” should be used.

- QuaLiKiz Neural Network
  - QLK-NN can be installed whith the help of the script install_qlkANN10D.m in ”coef” directory.
    The script should be updated for the local environment.
  - Information available here: https://gitlab.com/qualikiz-group/QuaLiKiz/-/wikis/QLKNN/QLKNN-overview

- QuaLiKiz
  - QLK can be installed whith the help of the script install_QLKZ.m in ”coef” directory.
    The script should be updated for the local environment.
  - The code is available here: https://gitlab.com/qualikiz-group/QuaLiKiz

- TGLF
  - Information about TGLF code are available here: https://gacode.io/tglf.html
  - The code is available here: https://github.com/gafusion/gacode/tree/master/tglf

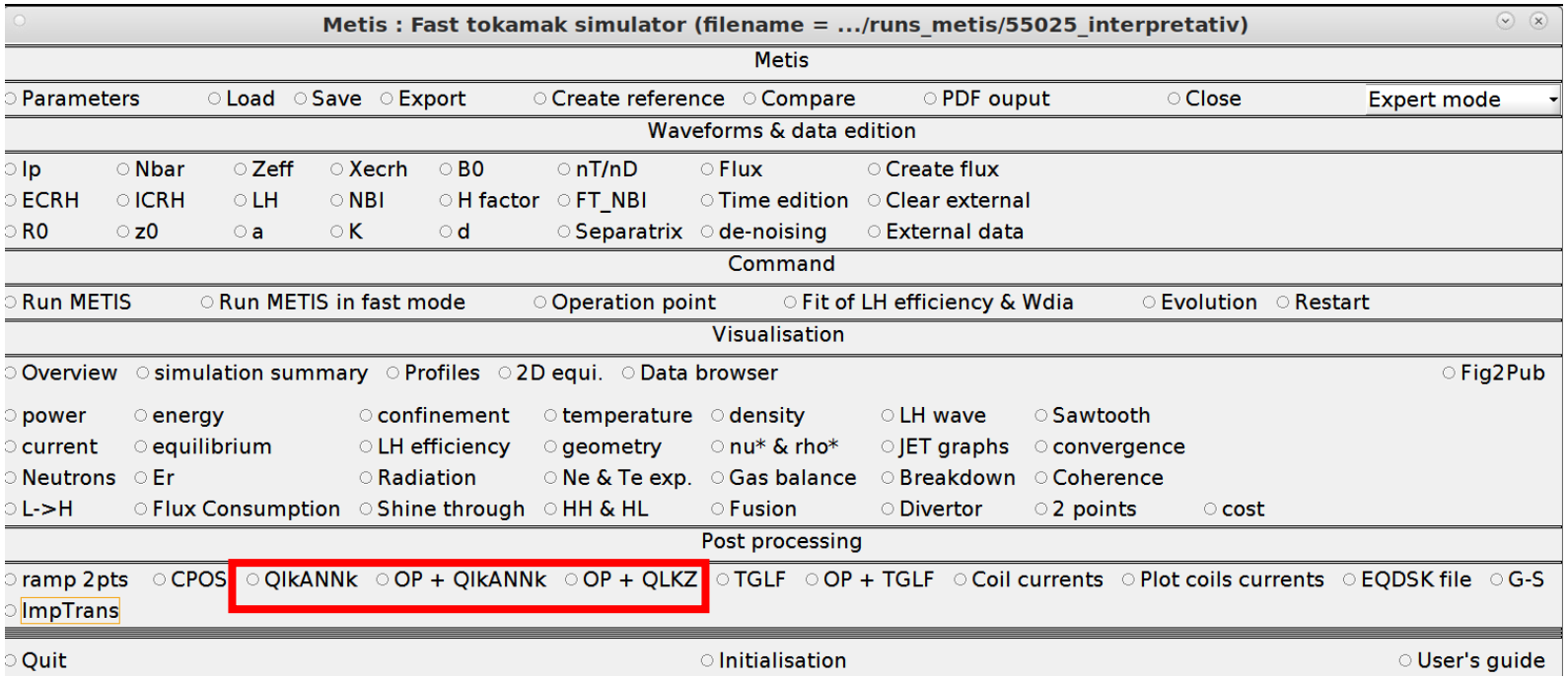
- NEO
  - Information about NEO code are available here: https://gacode.io/neo.html
  - The code is available here: https://github.com/gafusion/gacode/tree/master/neo

- FACIT
  - Information about FACIT code are available here: https://iopscience.iop.org/article/10.1088/1361-6587/ac5b4d
  - The code is available here: https://drf-gitlab.cea.fr/pmaget/facit

13. METIS & QuaLiKiz Neural Network

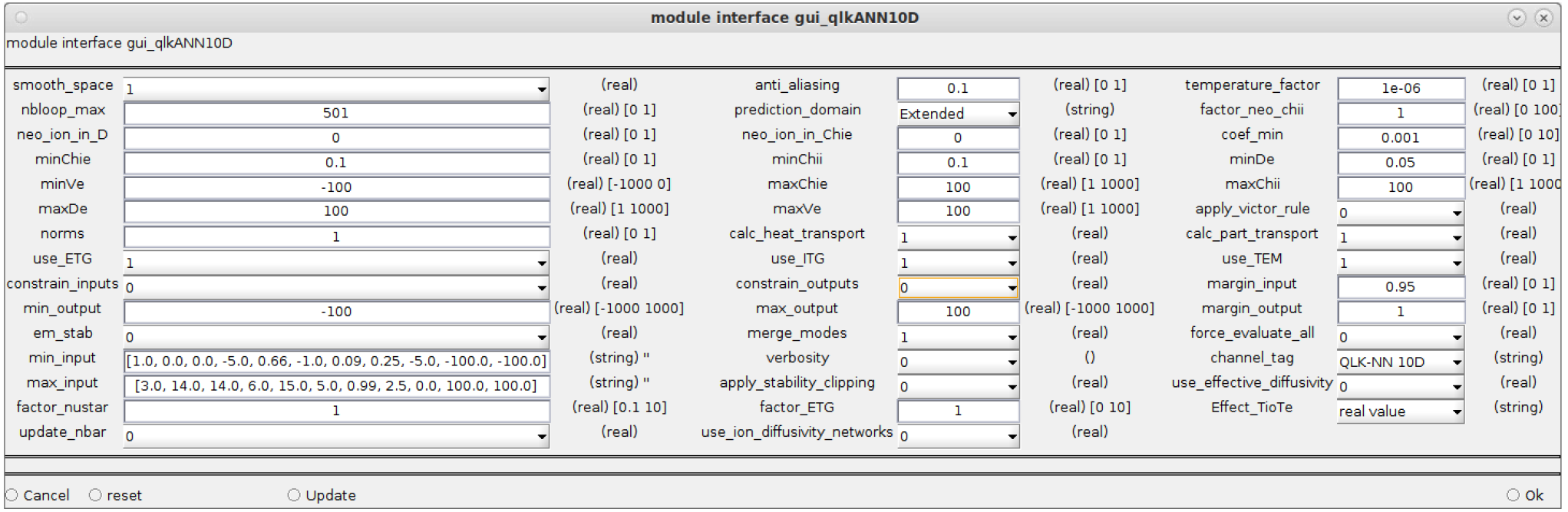
How to use QLK and QLK-NN in METIS

- **Current implementation of QLK-NN (QuaLiKiz Neural Network):**
  - Can be used for METIS simulation to compute transport coefficients or to solve the steady state transport equation at a chosen time slice, updating sources during the convergence
  - Uses the main ions and electrons to compute turbulent fluxes
- **Current implementation of QLK (QuaLiKiz):**
  - Can be used to solve the steady state transport equation at a chosen time slices
  - Uses the detailed plasma composition to compute turbulent fluxes: main ions, electrons, imp1, imp2, W and minority species



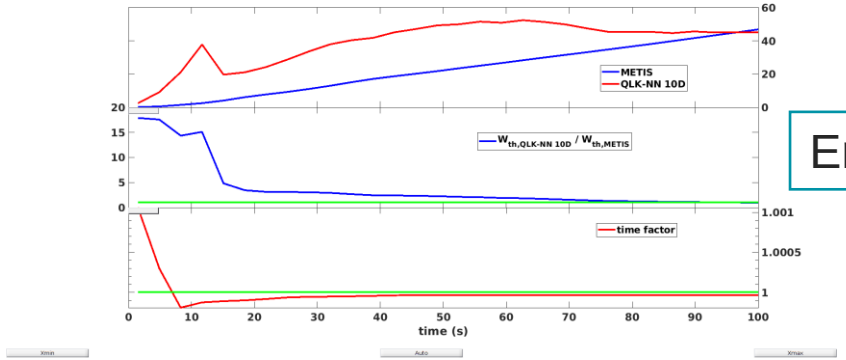
Using QLK-NN for the whole simulation : inputs

- Button “QLK-NN”
- Ask for QLK-NN specific input parameters (a set of default parameters are shown below)

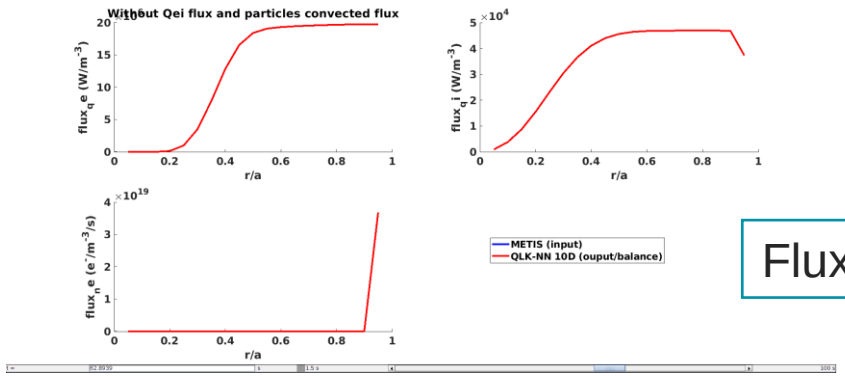


Using QLK-NN for the whole simulation: outputs

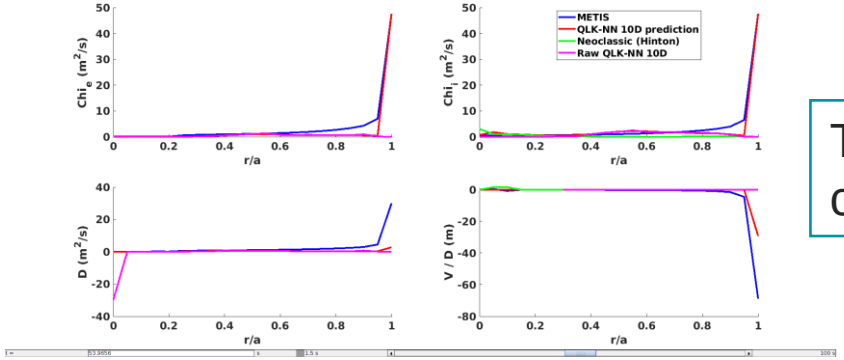
- Outputs the radial profiles for flux, transport coefficients for electrons and ions and particle diffusivities together with separate D and V for electrons
- Provides temperature profiles (electron and ion) and electron density profile for the steady state transport equation at fixed sources (corrected of dW/dt).
- Also ouputs energy content evolution compared to the METIS one



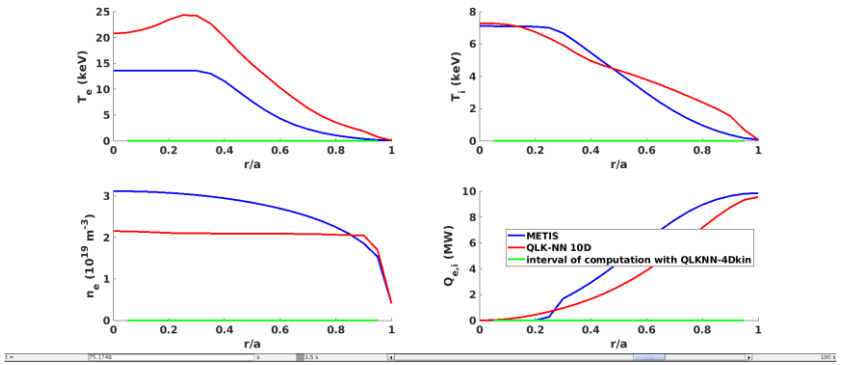
Energy



Flux



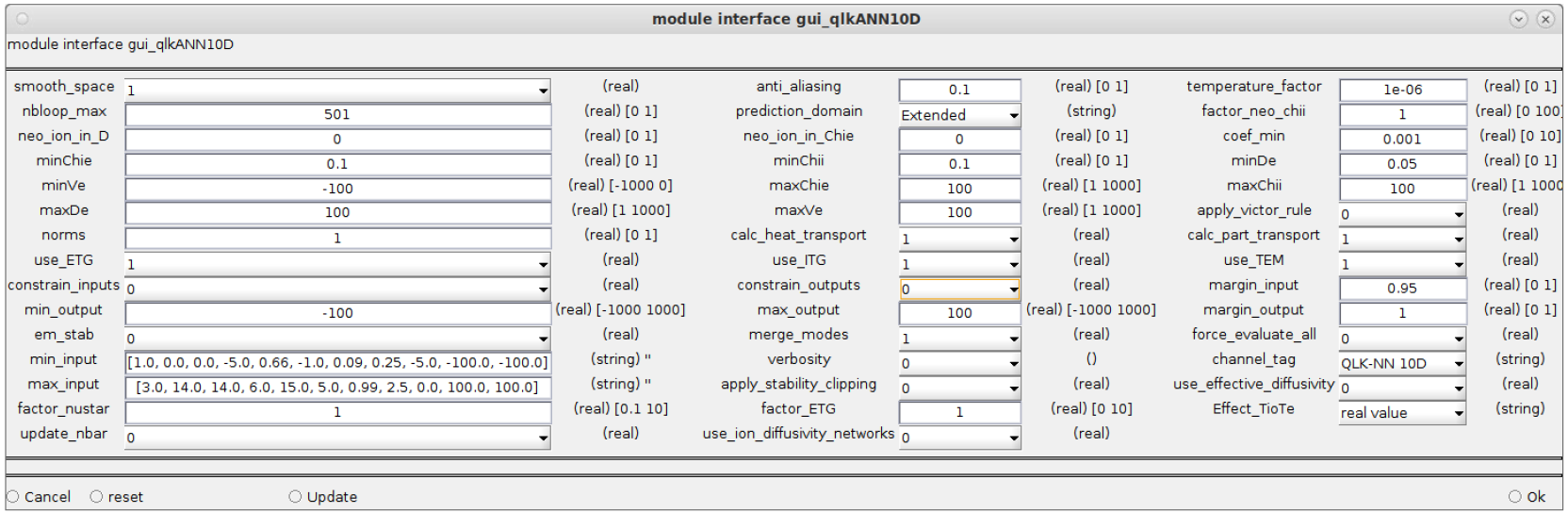
Transport coefficients

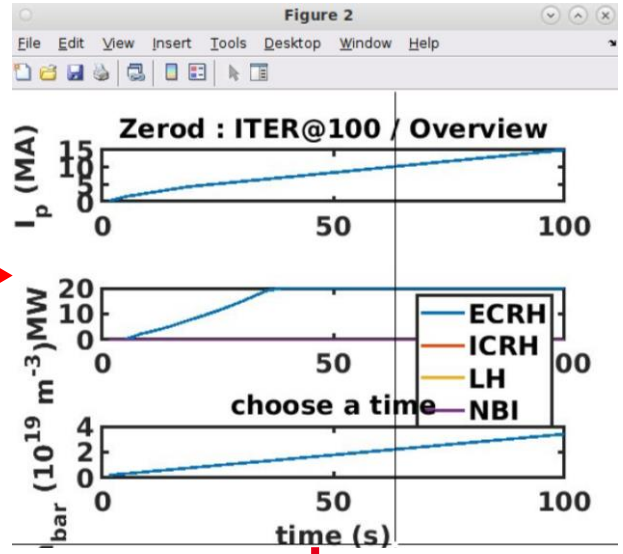


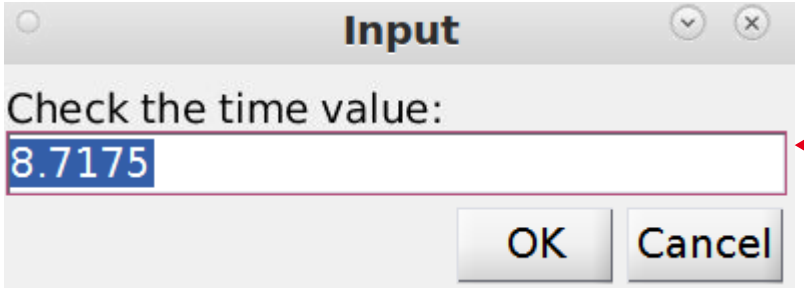
Kinetic profiles

Using QLK-NN for one time slice steady state solution : inputs

- Button “OP + QLK-NN”
- Ask for QLK-NN specific input parameters (a set of default parameters are shown below)
- Select a time slice and confirme the exact time if needed

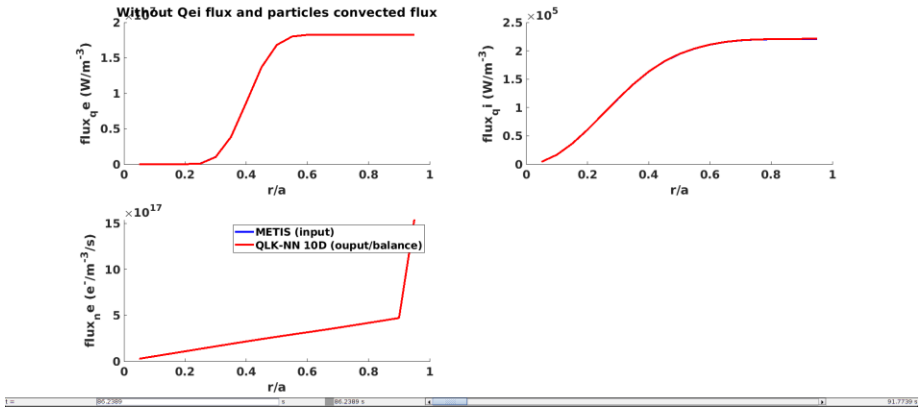




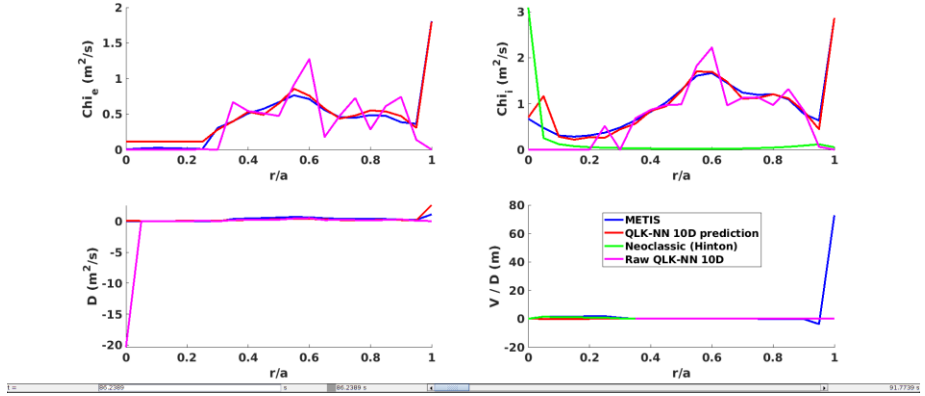


Using QLK-NN for one time slice: outputs

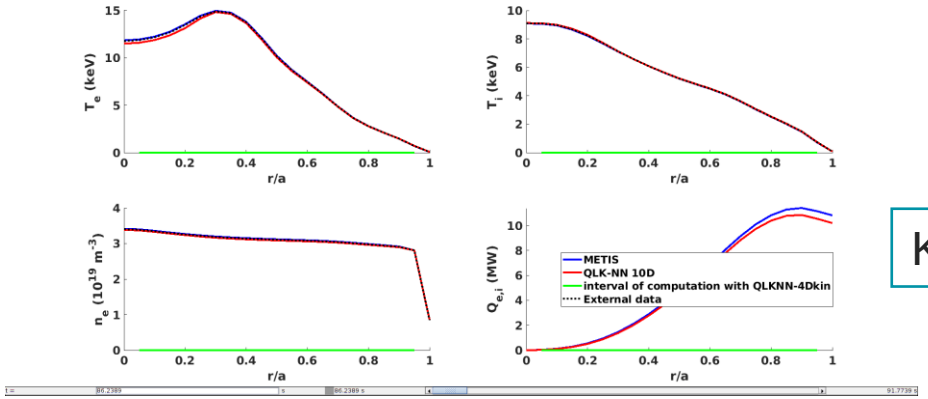
- Outputs the radial profiles of transport coefficients for electron and ion heat and particle diffusivities and convection coefficients for electrons
- Outputs flux
- Outputs recomputed electron temperature and density profile and ion temperature profile solution of steady state transport equation with self consistent sources.



Flux



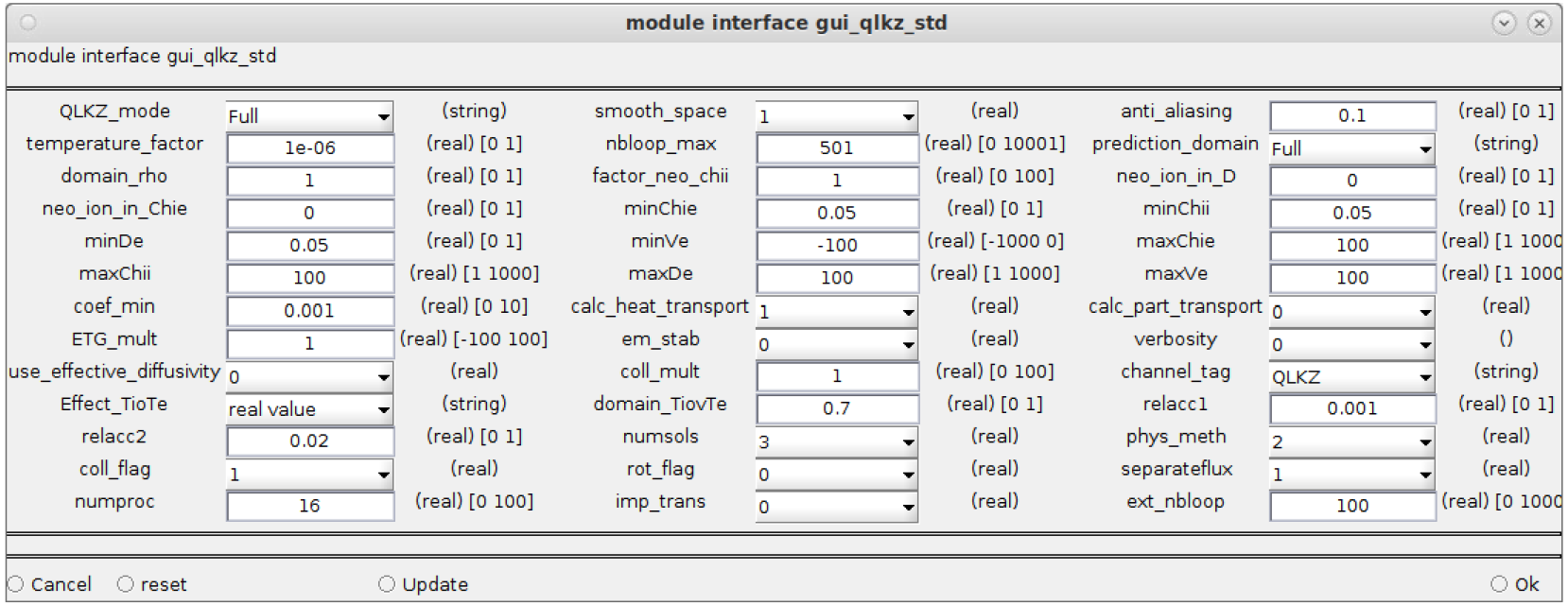
Transport coefficients

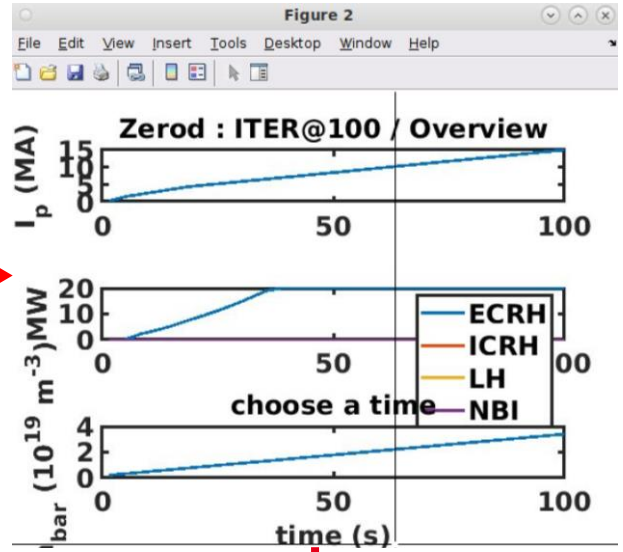


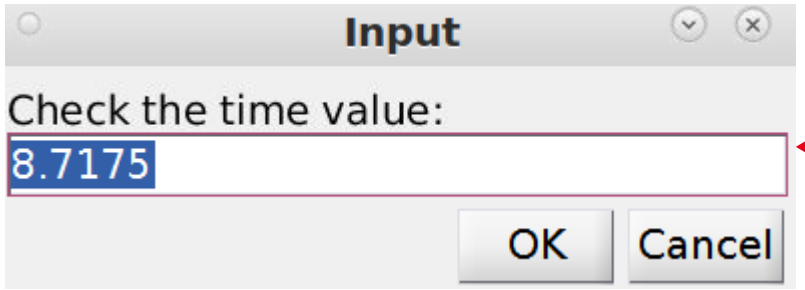
Kinetic profiles

Using QLK for one time slice steady state solution : inputs

- Button “OP + QLK”
- Ask for QLK specific input parameters (a set of default parameters are shown below)
- Select a time slice and confirm the exact time if needed

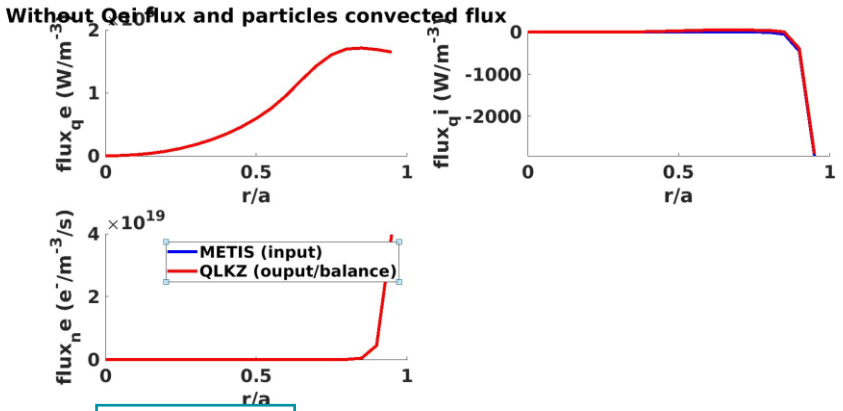




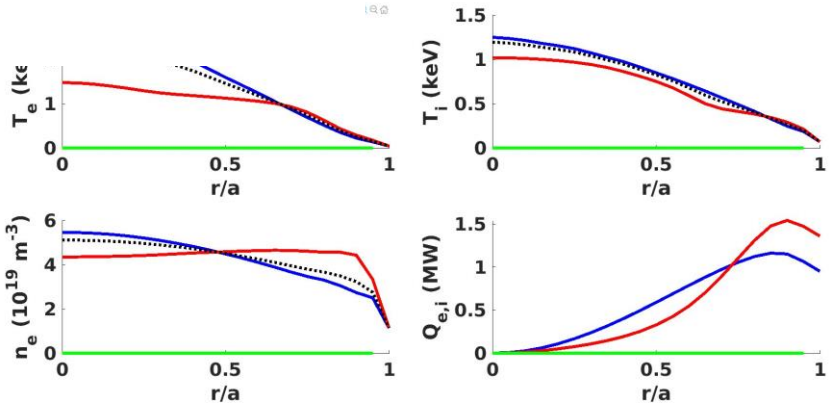


Using QLK-NN for one time slice: outputs

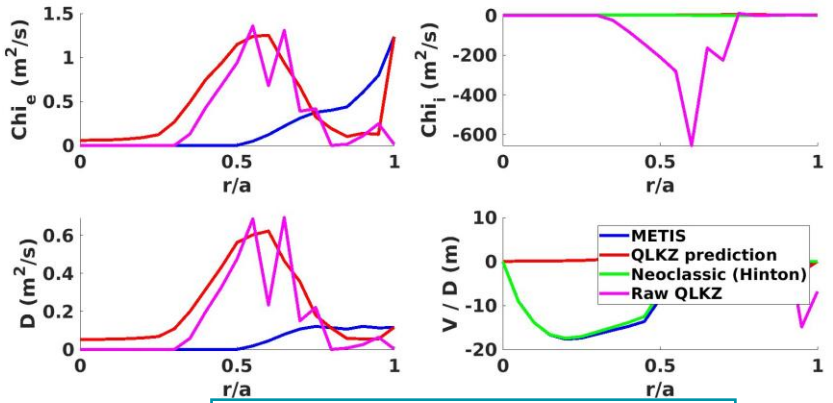
- Outputs the radial profiles of transport coefficients for electron and ion heat and particle diffusivities and convection coefficients for electrons
- Outputs flux
- Outputs recomputed electron temperature and density profile and ion temperature profile solution of steady state transport equation with self consistent sources.



Flux



Kinetic profiles



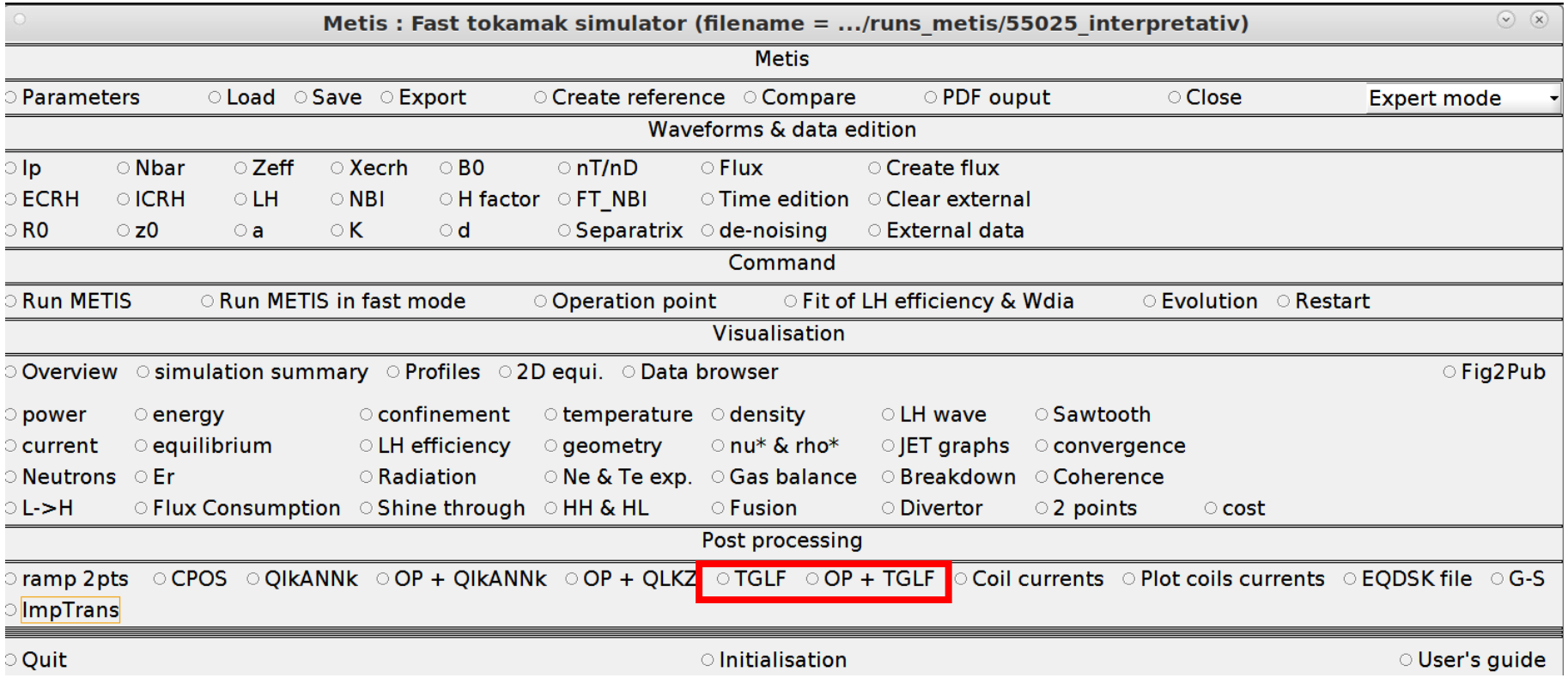
Transport coefficients

14. METIS & TGLF

How to use TGLF in METIS

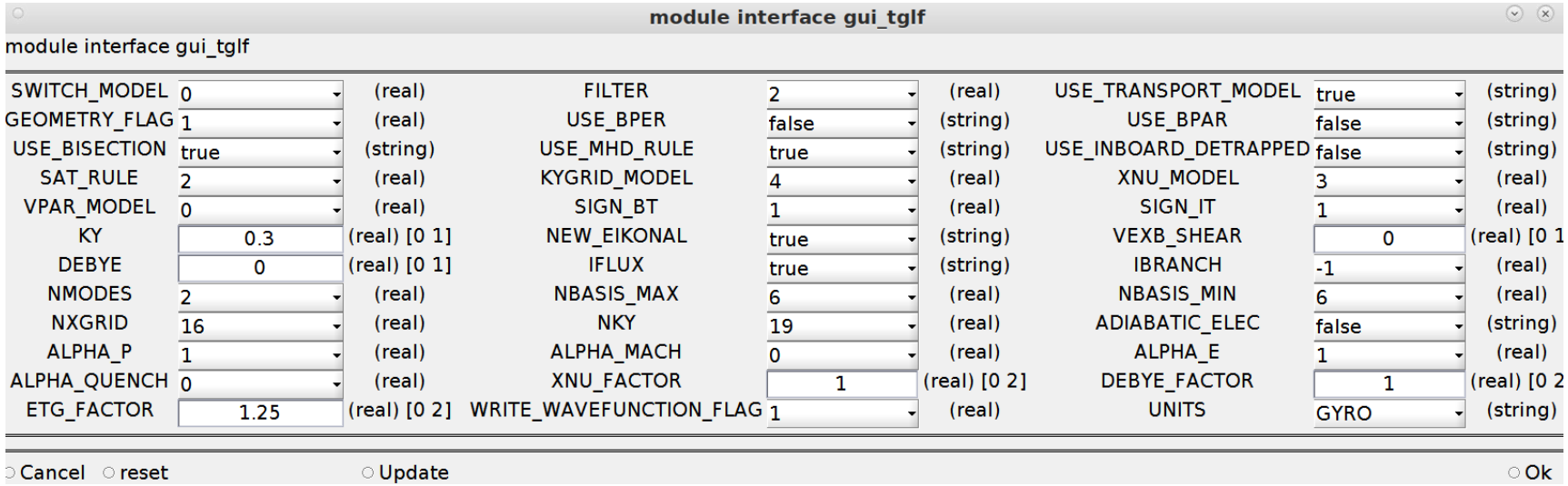
Current implementation:

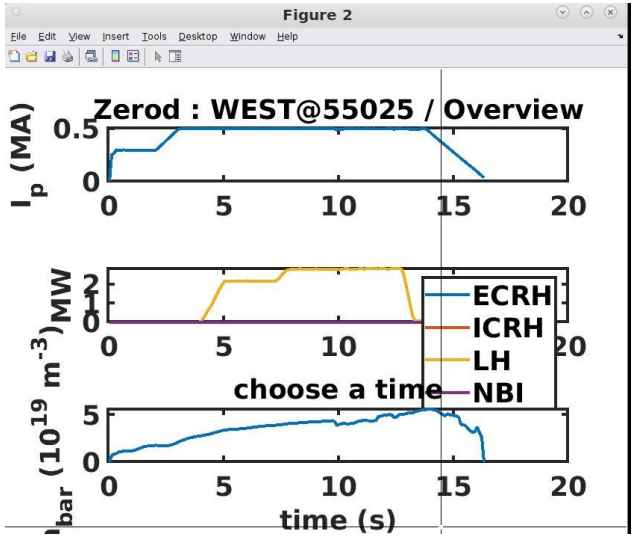
- Can be used for a given time slice of a METIS simulation or to solve the steady state transport equation
- Uses the main ions, electrons and the dominant light impurity to compute turbulent fluxes

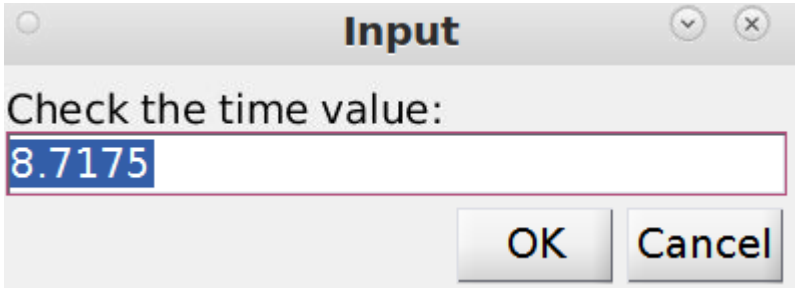


Using TGLF for a time slice: inputs

- Button “TGLF”
- Ask for TGLF specific input parameters (a set of default parameters are shown below)
- Select a time slice and confirme the exact time if needed

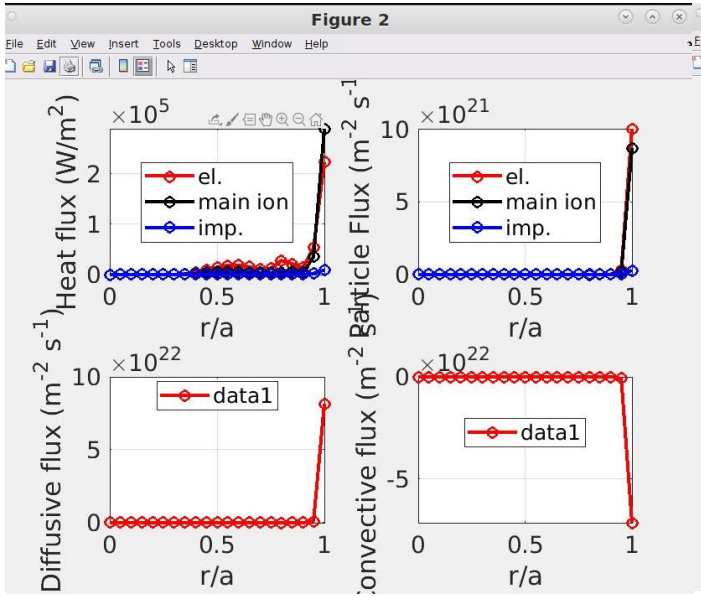


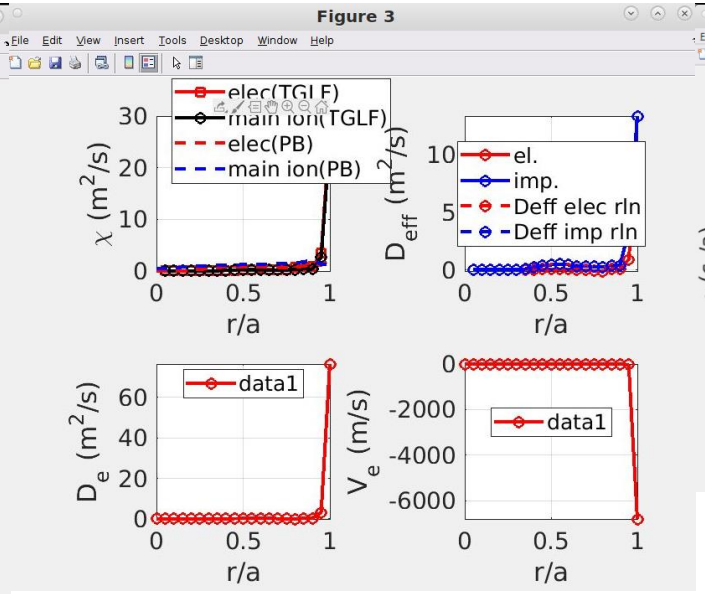


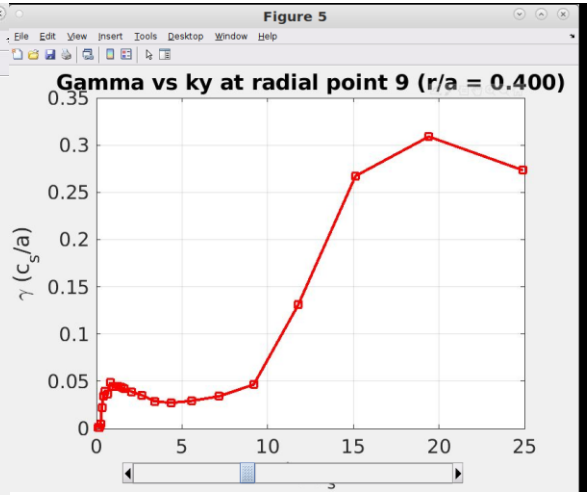


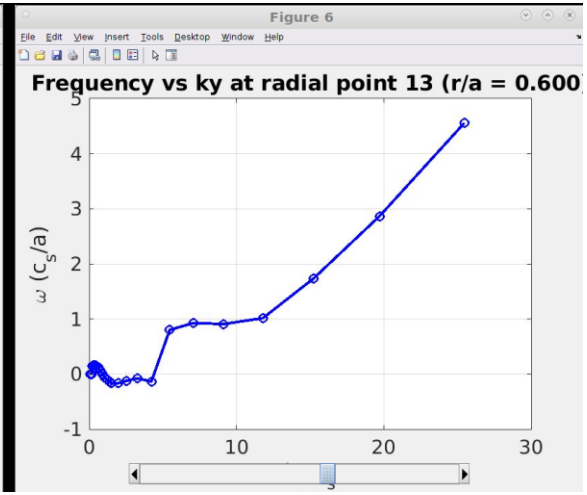
Using TGLF for a time slice: outputs

- Output the radial profiles of effective heat and particle diffusivities together with separate D and V for electrons
- Also ouputs growth rate and frequencies for each radial location



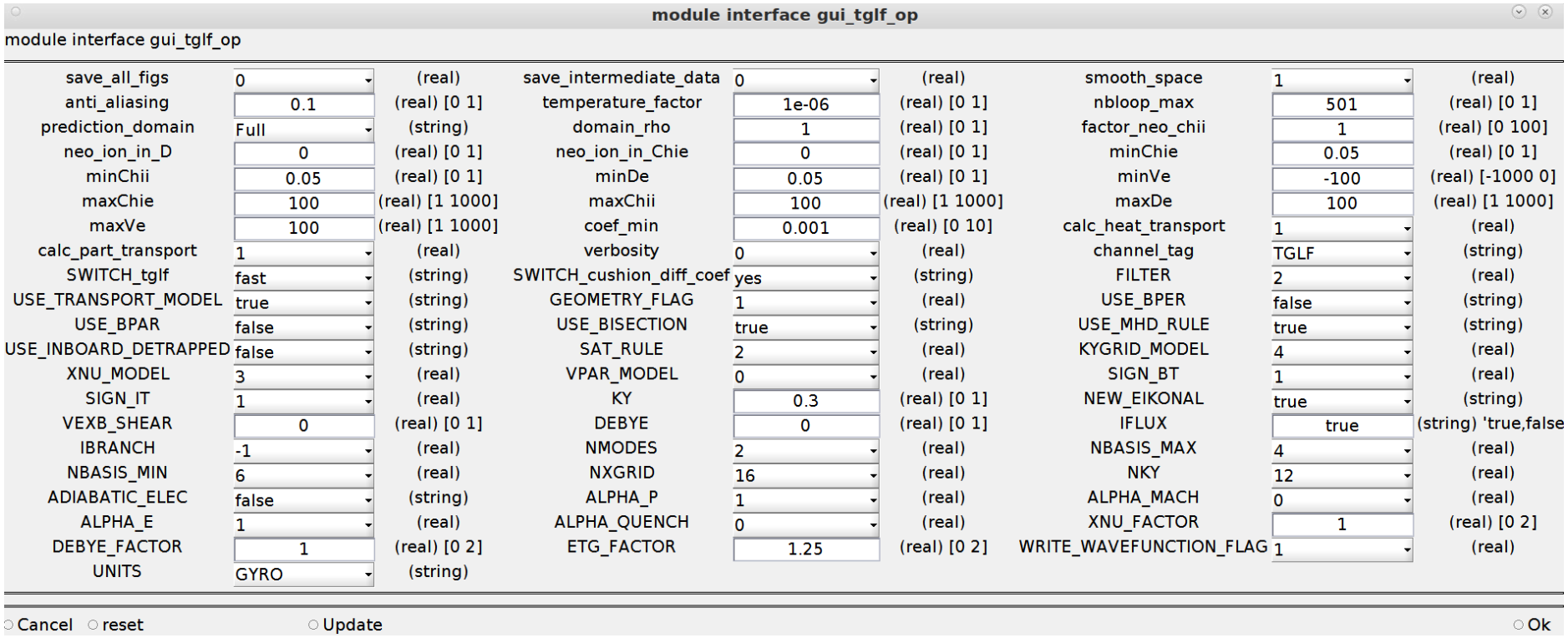






Using TGLF for steady state

- Additional inputs for the transport equations
- Currently is set up for full radius simulations with sat2
- Outputs are the converged temperature and density profiles for the main ions and the electrons
- Intermediate outputs allow to follow the convergence loop for the predicted profiles and the error between iterations

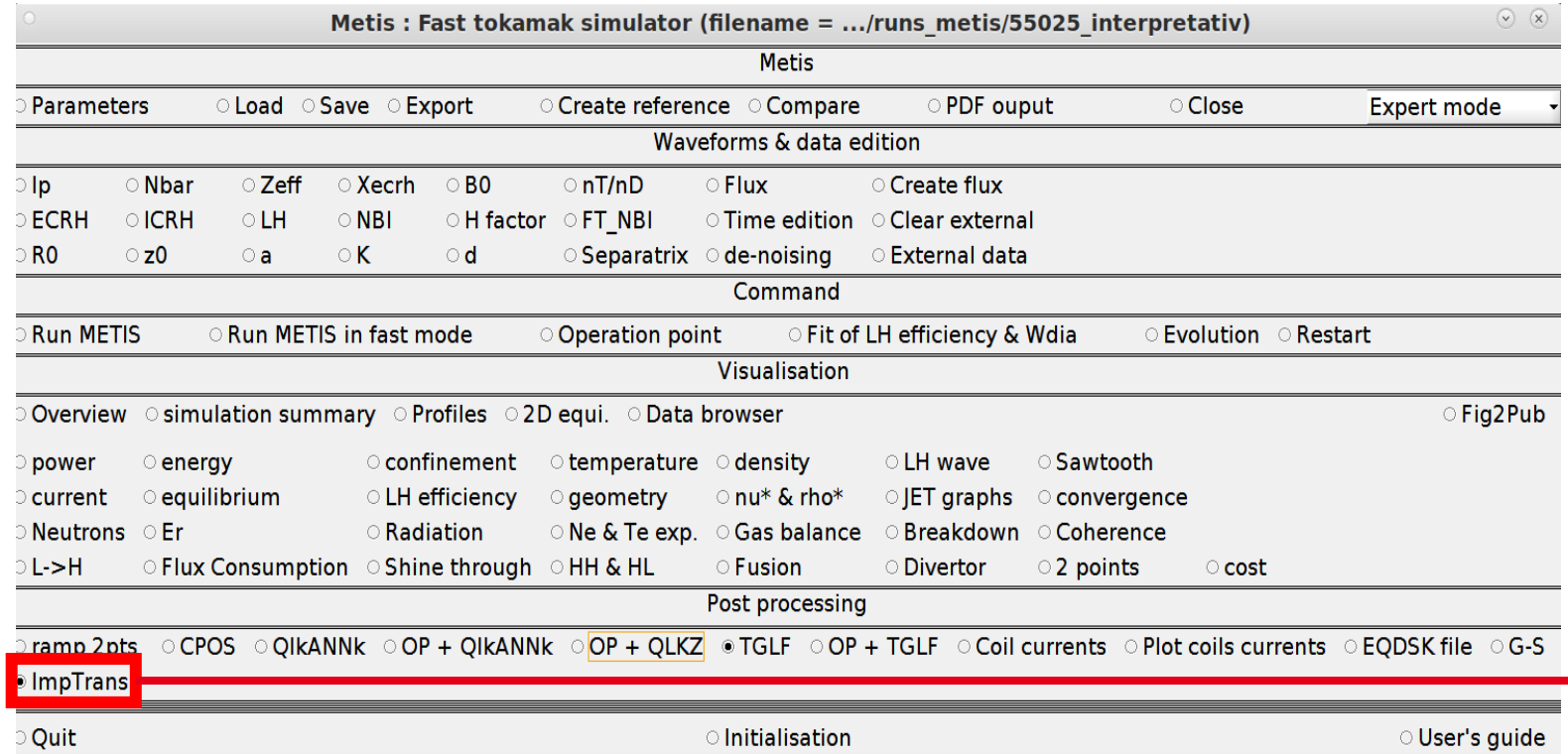


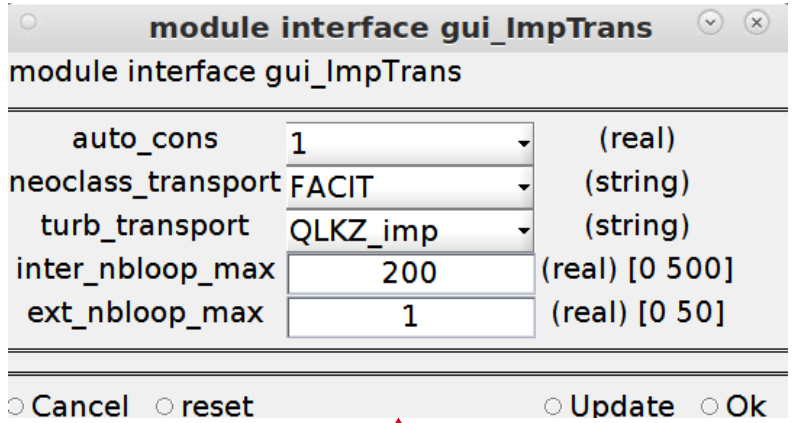
15. Impurity transport in METIS

Impurity transport in METIS

Current implementation:

- ❑ Standalone (auto_cons=0) computations of D and V with FACIT/NEO/QLKZ (any combination)
- ❑ Steady state (auto_cons=1) impurity transport equations with transport model of choice

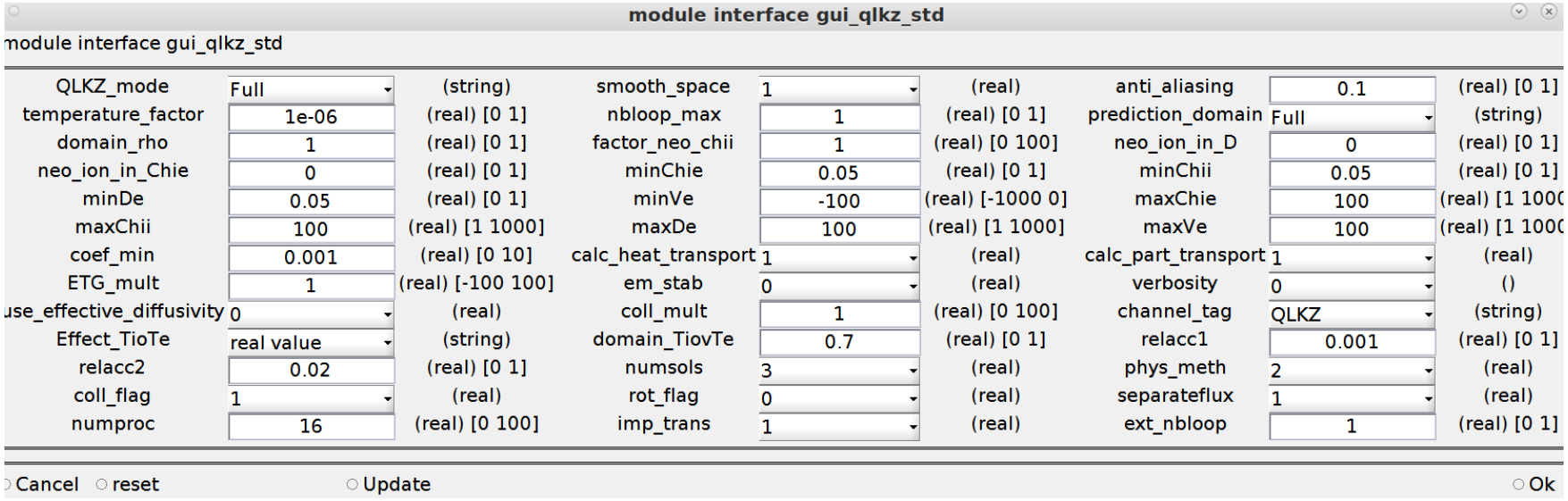


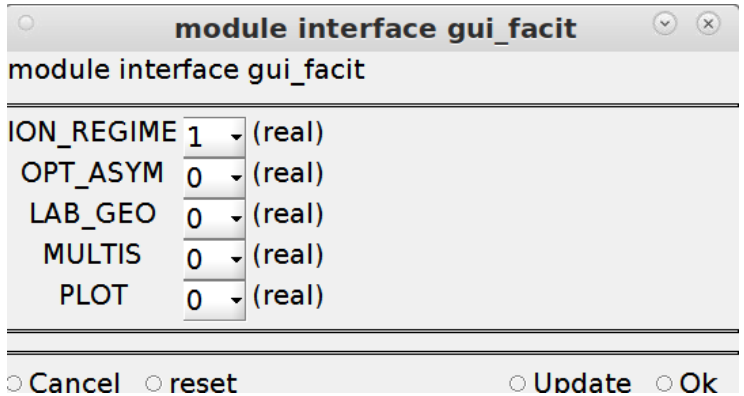


Example standalone FACIT+QLKZ: inputs

Input options for QLKZ and FACIT.

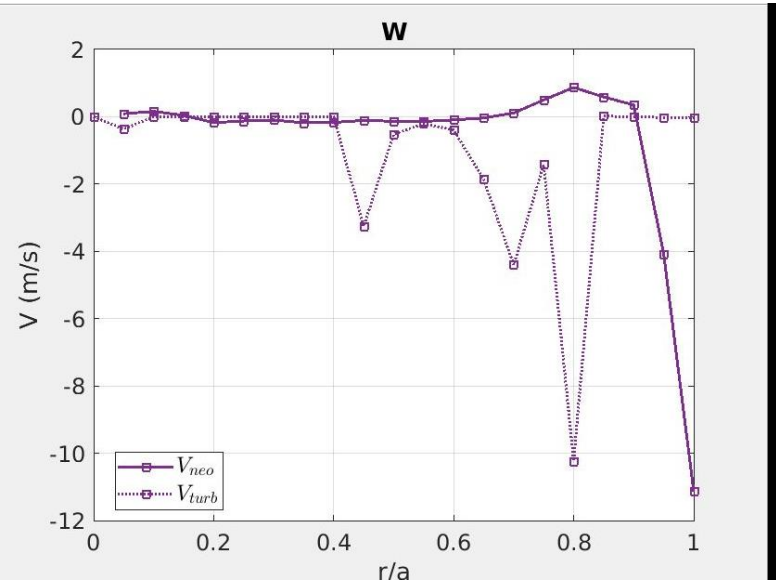
- ❑ Multiple species collisions taken into account in FACIT based on Zeff contributions (can be forced with MULTIS=1)
- ❑ imp_trans option defined consistently but can be still modified for other transport model options

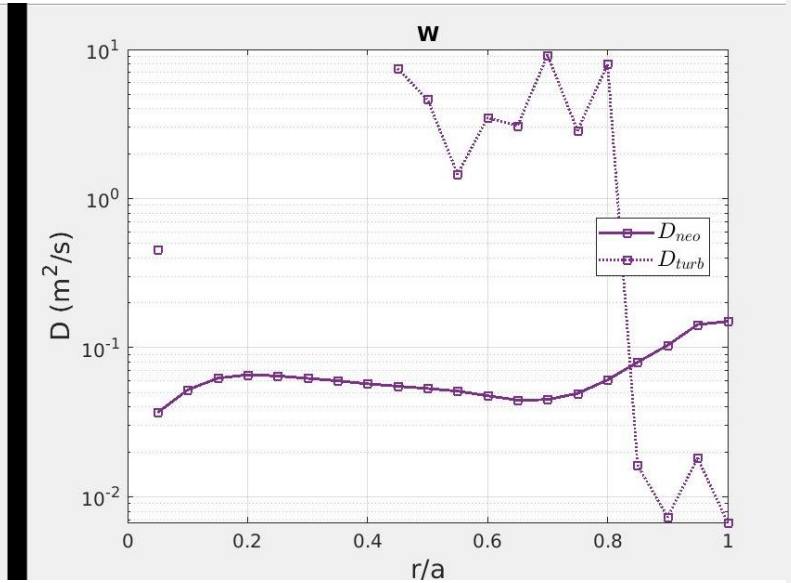


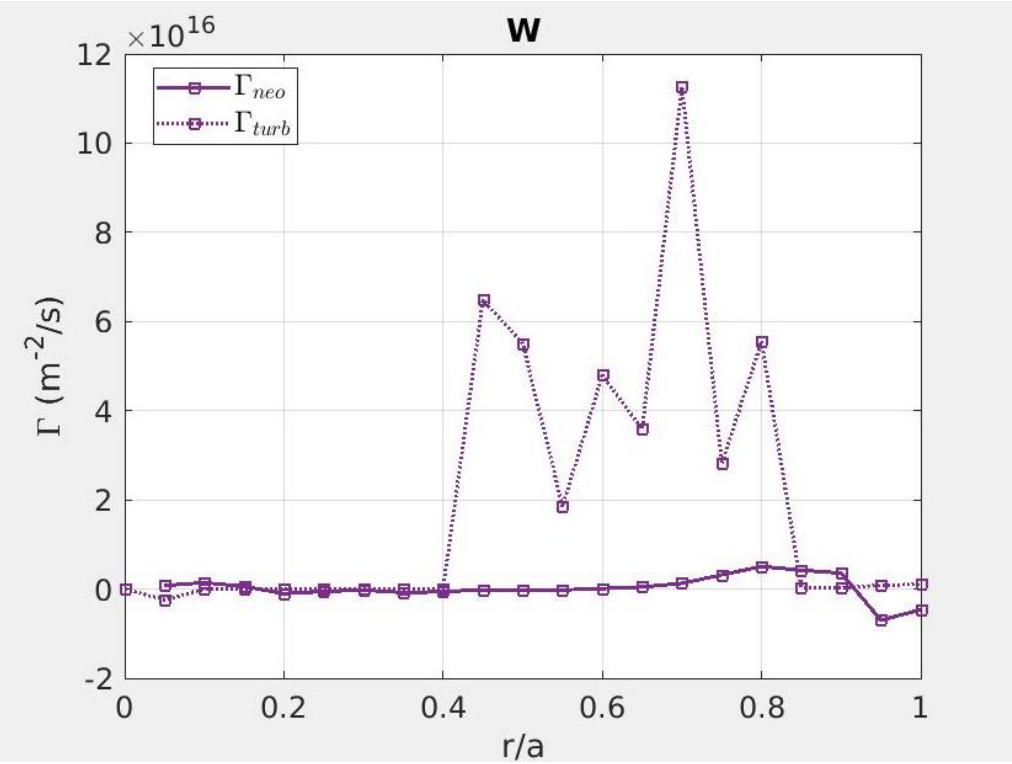


Example standalone FACIT+QLKZ: outputs

Data shown for W but data for all species stored in /coef/data/QLKZimp_neocl_data.mat

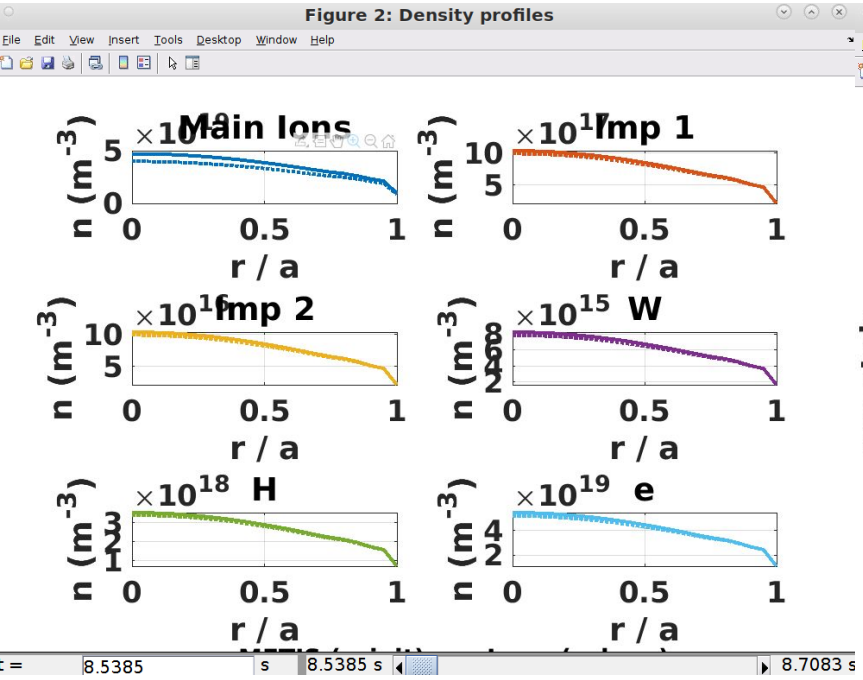


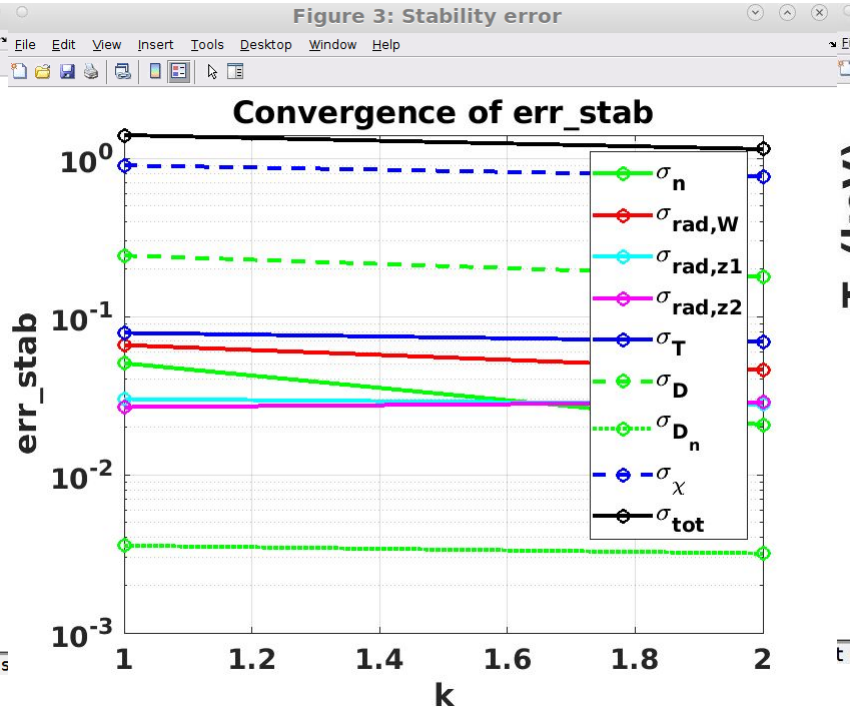


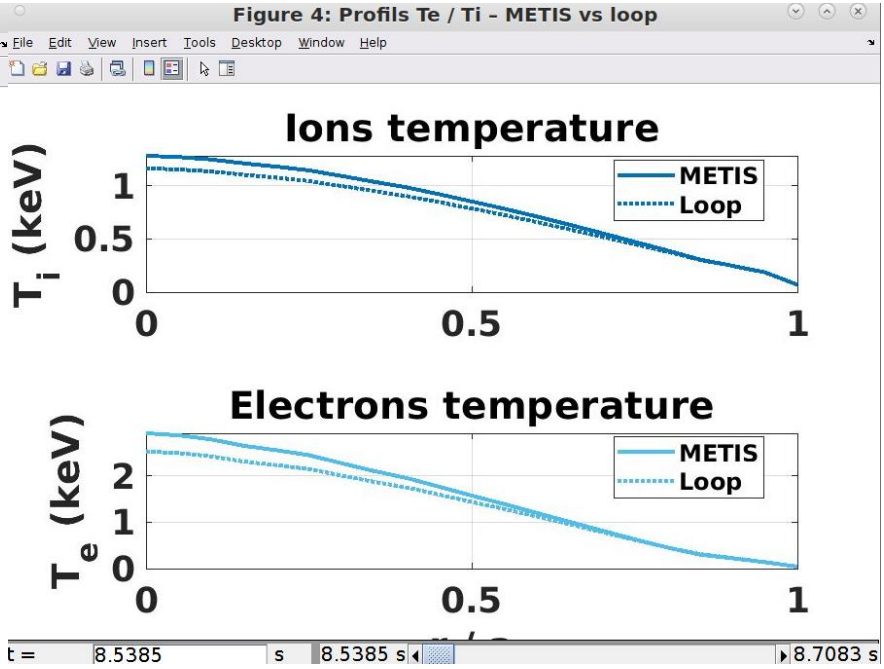


Example steady state FACIT+QLKZ

- Input options same as OP+QLKZ
- Evolves impurity z1 and W, main ions from quasineutrality, minority ions defined from fraction, imp z2 as well
- Radiated power evolved in internal loop. Error evolution decomposed for different quantities.







16. Tuning METIS moments equilibrium

Specific settings for equilibrium

- Limitation of internal moment equilibrium in METIS: could be imprecise with default settings if
  - High beta
  - Compact or spherical tokamak
  - Very high shapping
  - Anisotropic pressure
  - Some reversed shear profile

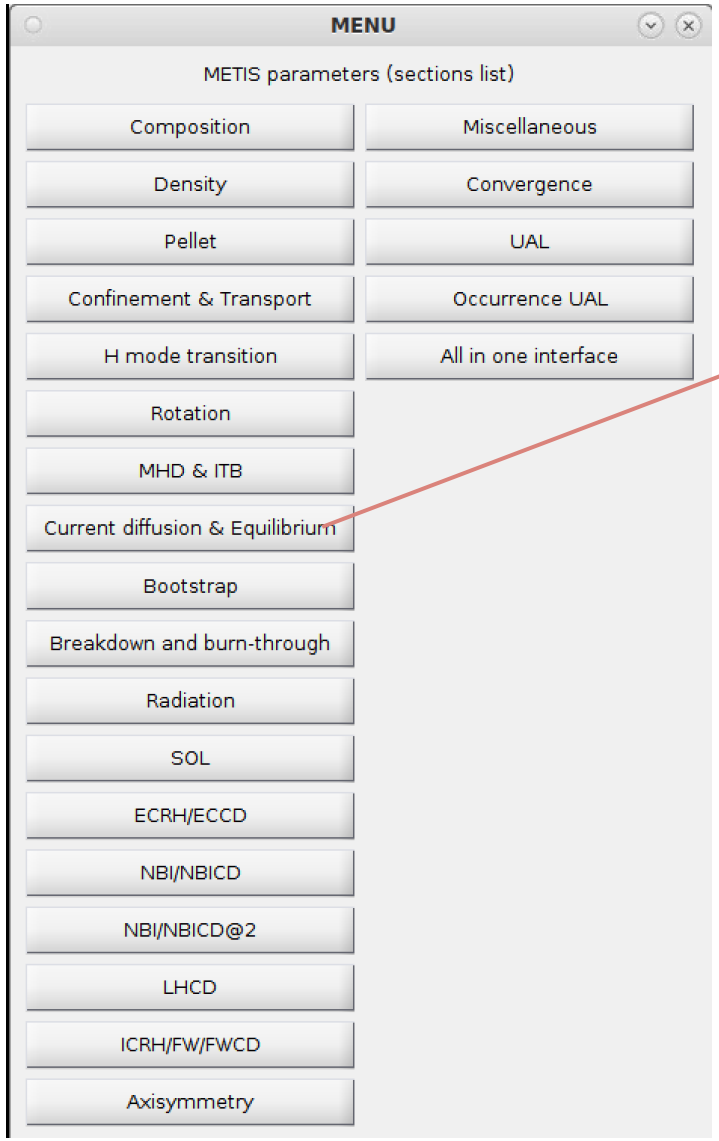
- Specific settings in METIS to tune the equilibrium
  - For radial variation of moments: ”moment_mode” → can be set to 0 to get flat K(rho), mainly for spherical tokamak
  - Radial evolution of shaping: ”morphing” → can be reduce for high beta plasma
  - Value of safety factor on magnetic axis: ”q(0) regularisation” → change the formula for extrapolation, can be set to 5 to prevent unwanted reversed shear profile
  - If pressure is anisitropic: ”equi_ppar” → change the pressure expression used in G-S equation.

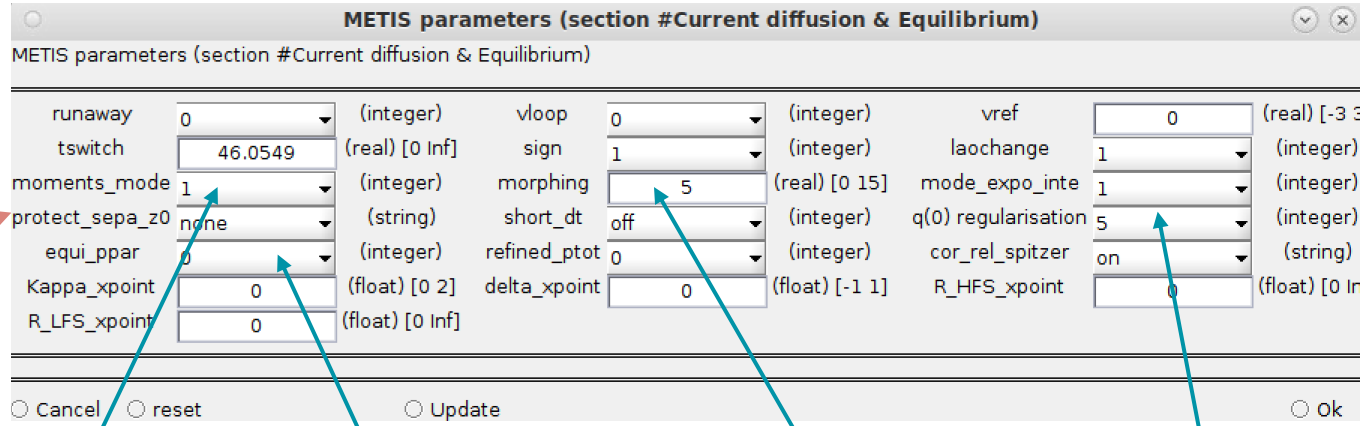
- Used of exernal equilibrium code to tune parameters
  - FEEQS.M is directely link to METIS and can be for this purpose, but other code can be used.

- FEEQS code can be used to provide equilibrium to METIS (external equilibrium)

FEEQS.M repository: https://git.iter.org/projects/EQ/repos/feeqs/browse
https://gitlab.inria.fr/hheumann/feeqs

Where are these parameters in METIS GUI





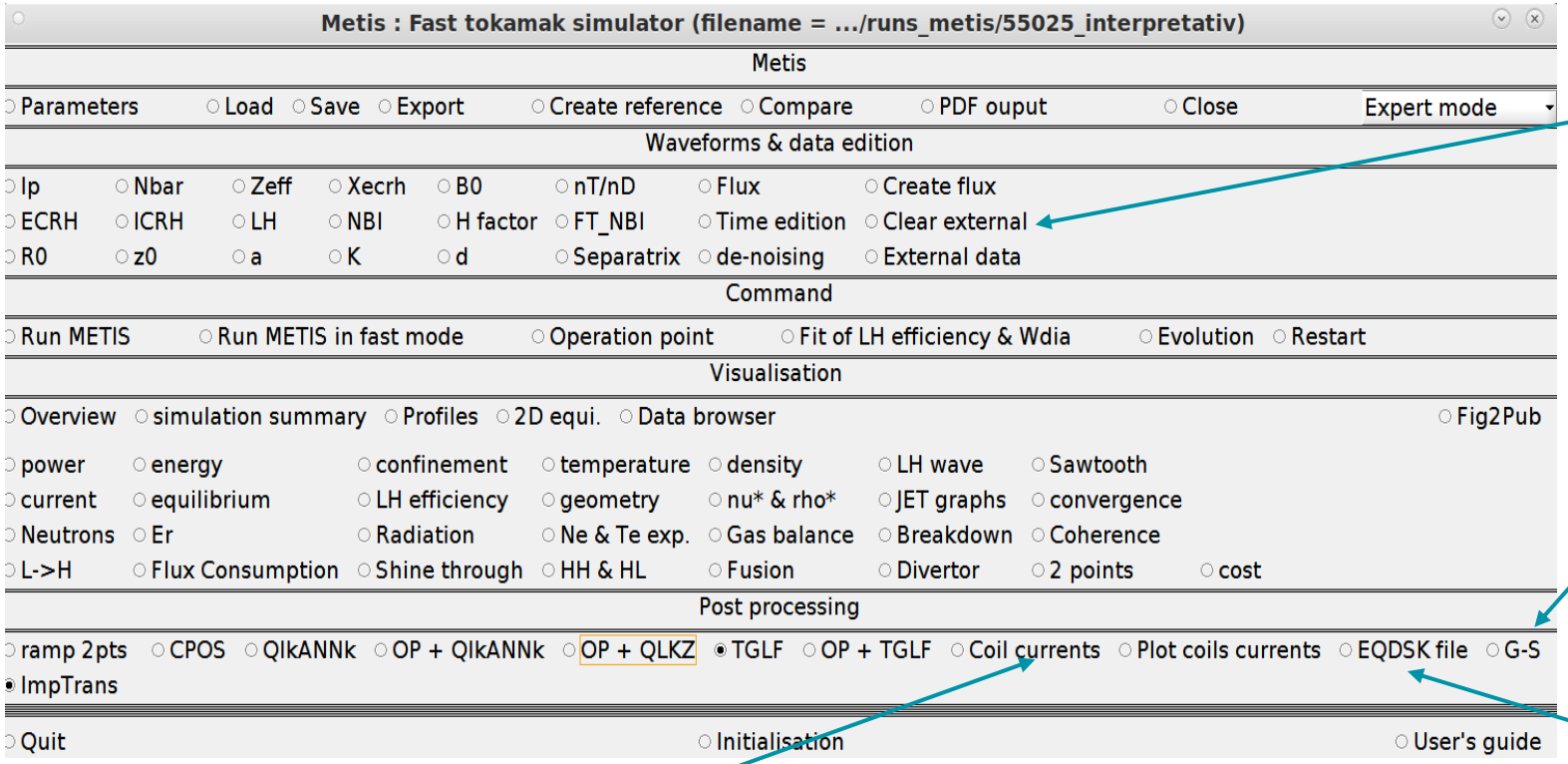
If pressure is anisitropic

Radial evolution of shaping

Radial variation of moments

Value of safety factor on magnetic axis

METIS post processing with FEEQS.M



To manage the use of external equilibrium created with the help of button ”G-S”

Compute Grad-Shafranov equation solution with FEEQS.M fixed boundary solver for each METIS time slice and set it as an external equilibrium for METIS (will be used in the next run of METIS instead of the METIS internal equilibrium).

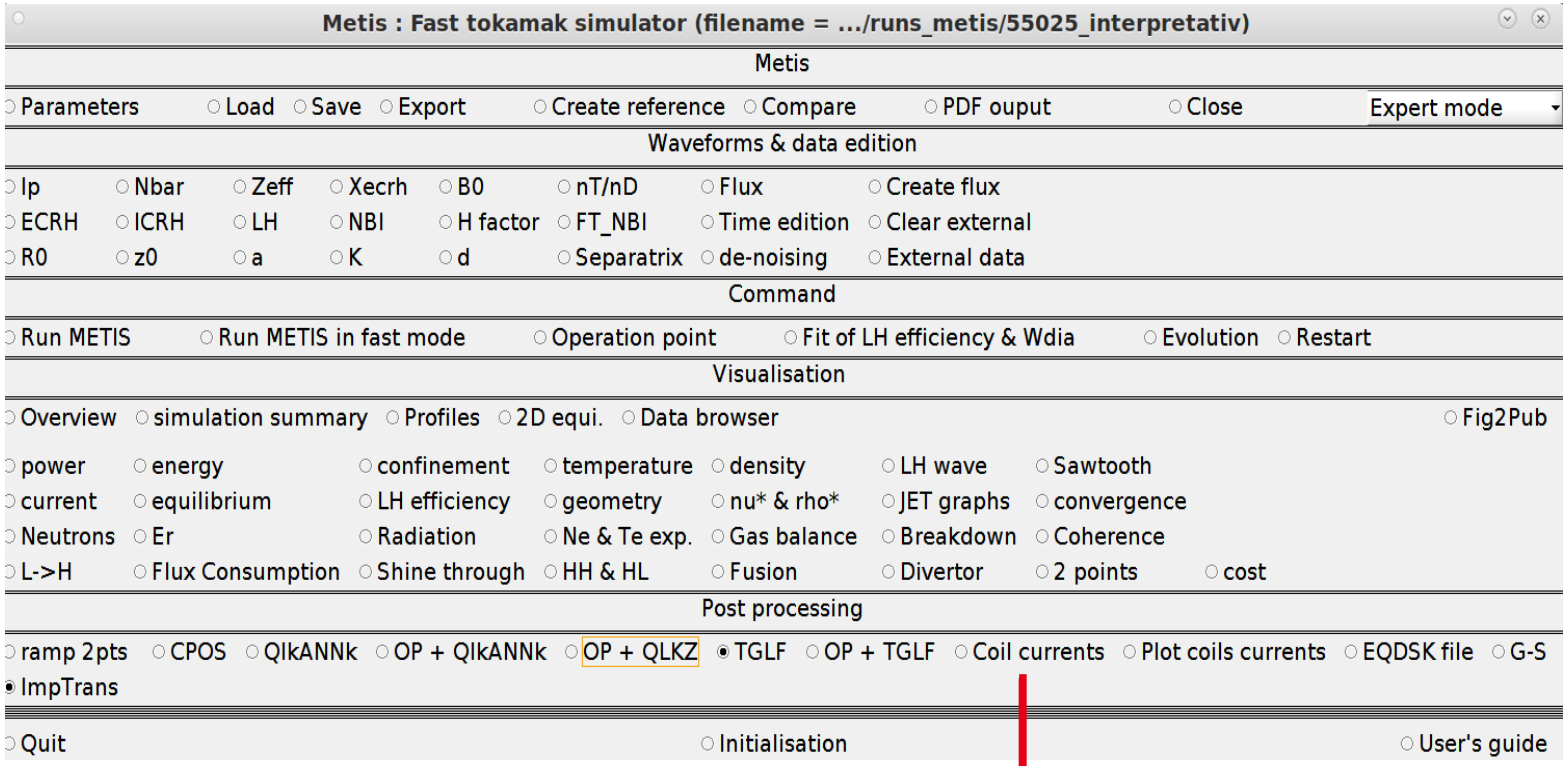
This tool has no parameters

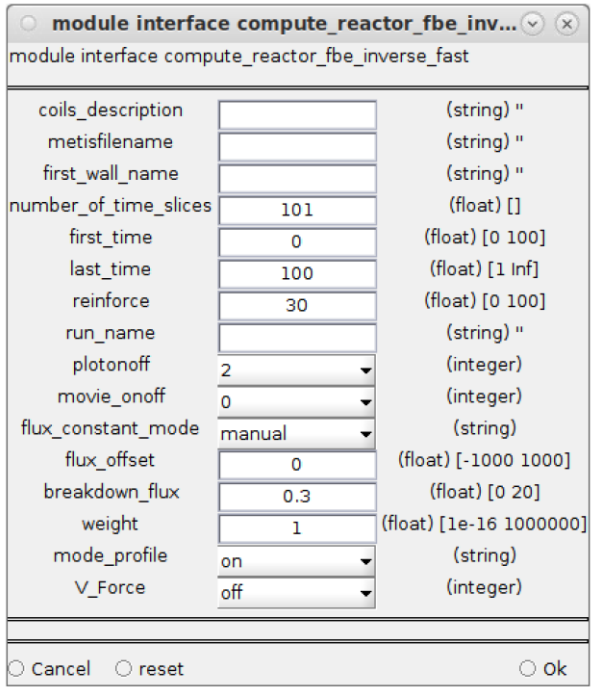
Compute coil currents wih fixed boundary inverse

Compute EQDSK files with fixed boundary mode

Warning: not suitable for JT-60SA: use dedicated tools instead (see JT-60SA section)

Fast inverse with FEEQS.M





Parameters for fast inverse (1/2)

- **coils_description**: path and name of coils description file used for the computation.
  If empty, a file selector will be open at the next step
  (see example in directory of project Reactor in FEEQS sources for the format of this file)

- **metisfilename:** path and name of METIS simulation file used for the computation. If empty, a file selector will be open at the next step

- **first_wall_name:** path and name of first wall description file used for the computation. Can be left empty, optional parameter)
  see example in directory of project Reactor in FEEQS sources for the format of this file

- **number_of_time_slices**: Number of time slices extracted from METIS simulation at which inverse equilibrium would be computed

- **first_time**: First time of the METIS simulation at which inverse equilibrium would be computed

- **last_time:** Last time of the METIS simulation for which inverse equilibrium would be computed

- **reinforce:** weights of extrema in LCFS (to have a better match on X-points and RIG and ROG)

- **run_name**: run_name is added to name of save files (can be leave empty)

- **plotonoff**: Set level of displayed graphs: 0 = no graphs; 1 = results only; 2 = all graphs (debug)

- **movie_onoff**: If = 1, record movie made of 2D equilibrium time slice graphs
  if = 2, record movie made of 2D equilibrium time slice graphs and save a .fig file for each time slice

Parameters for fast inverse (2/2)

- **flux_constant_mode**: This parameter allows to choose the way poloidal flux offset used to match METIS current diffusion LCFS poloidal flux and FEEQS.M LCFS poloidal flux is computed:
  if = **manual**, offset is prescribed by the user.
  In other modes, offset is automatically computed taking into account flux consumption estimation for breakdown computed in METIS and flux leakage computed with the help of current diffusion equation between end of breakdown and first_time compare to flux computed by premagnetisation tool:
  - **start_of_flux**: offset is computed to have maximum available flux for the first plasma, taking into account breakdown flux
  - **balanced**: offset is computed to have flux margin for ramp-up and ramp down if possible
  - **end_of_flux**: offset is computed to have minimum flux value (negative) at the end of ramp-down

- **flux_offset**: Poloidal flux offset used to match METIS current diffusion LCFS poloidal flux and FEEQS.M LCFS poloidal flux (Wb)

- **breakdown_flux**: Poloidal flux consumed for plasma break-down and burn-through (Wb).
  If METIS breakdown model is turn on, take the maximum of this value and METIS value

- **weight**: regularization term weight in cost function (sum(I^2) and dipole)

- **mode_profile**: If = on, use P'' and FF'', otherwise use beta-p and li_3’

- **V_Force**: Switch on or off the computation of voltages and forces (based on boboz code).

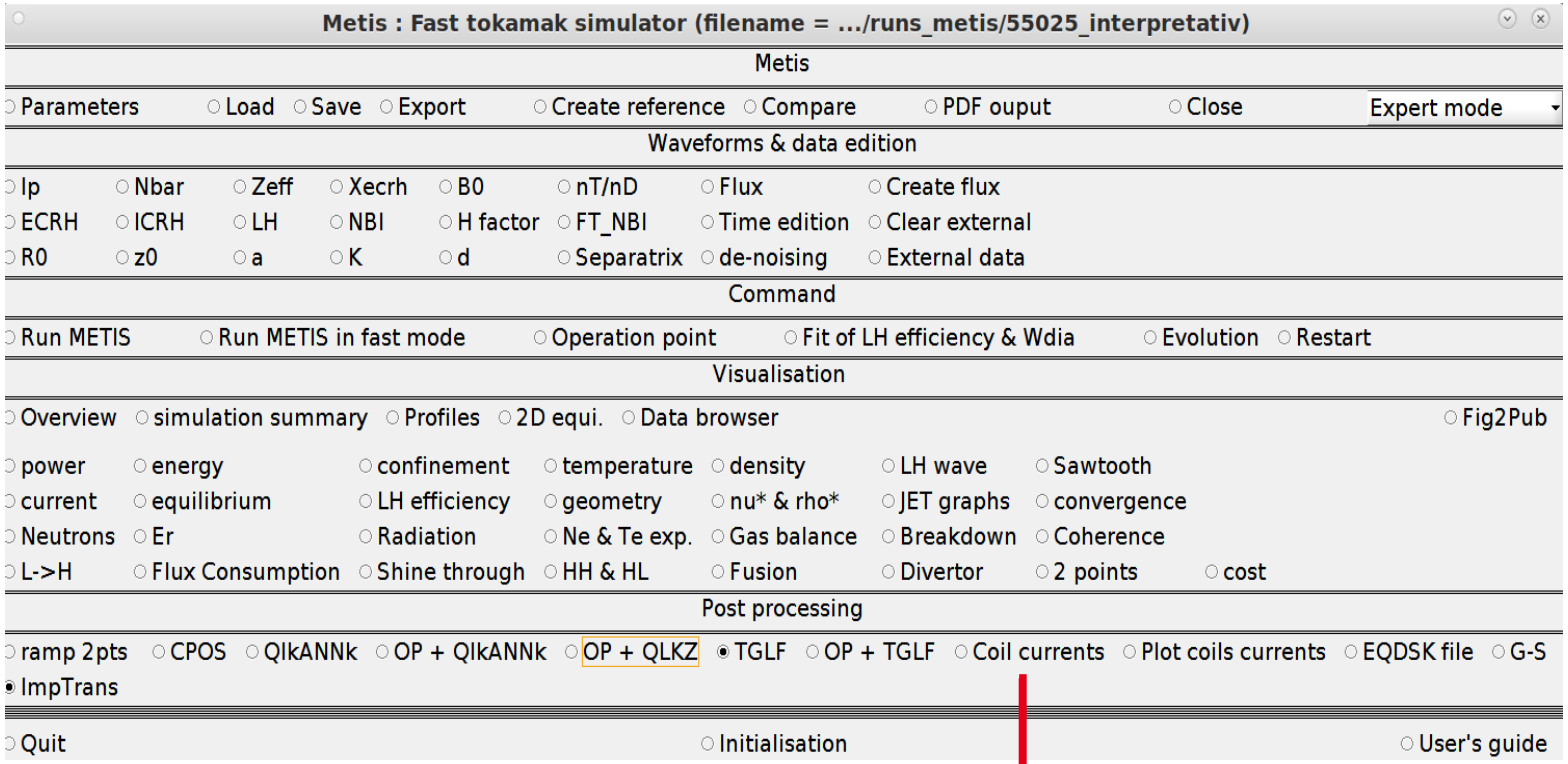
Example of file contents for coils description

ITER	Rcentre	Zcentre	DR	DZ	#turns	#PS
Name	m	m	m	m	none	number
CS3U	1.722	5.313	0.719	2.075	554	1
CS2U	1.722	3.188	0.719	2.075	554	2
CS1U	1.722	1.063	0.719	2.075	554	3
CS1L	1.722	-1.063	0.719	2.075	554	3
CS2L	1.722	-3.188	0.719	2.075	554	4
CS3L	1.722	-5.313	0.719	2.075	554	5
PF1	3.9431	7.5641	0.9590	0.9841	248.6	6
PF2	8.2851	6.5298	0.5801	0.7146	115.2	7
PF3	11.9919	3.2652	0.6963	0.9538	185.9	8
PF4	11.9630	-2.2436	0.6382	0.9538	169.9	9
PF5	8.3908	-6.7369	0.8125	0.9538	216.8	10
PF6	4.3340	-7.4765	1.5590	1.1075	459.4	11

Should be a in a simple ASCII text file

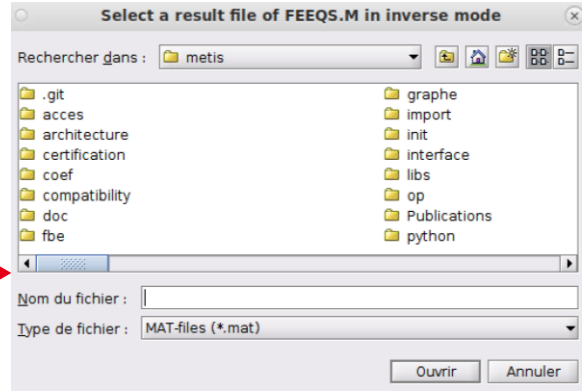
- **Name** → Machine name and then coil name
- **Rcentre** → radius of geometrical center of the coil (m)
- **Zcentre** → altitude of geometrical center of the coil (m)
- **DR** → width of the coil (m)
- **DZ** → high of the coil (m)
- **#turns** → number of turns in the coil
- **#PS** → index of the power supply powering the coil

Plot graphs results for fast inverse

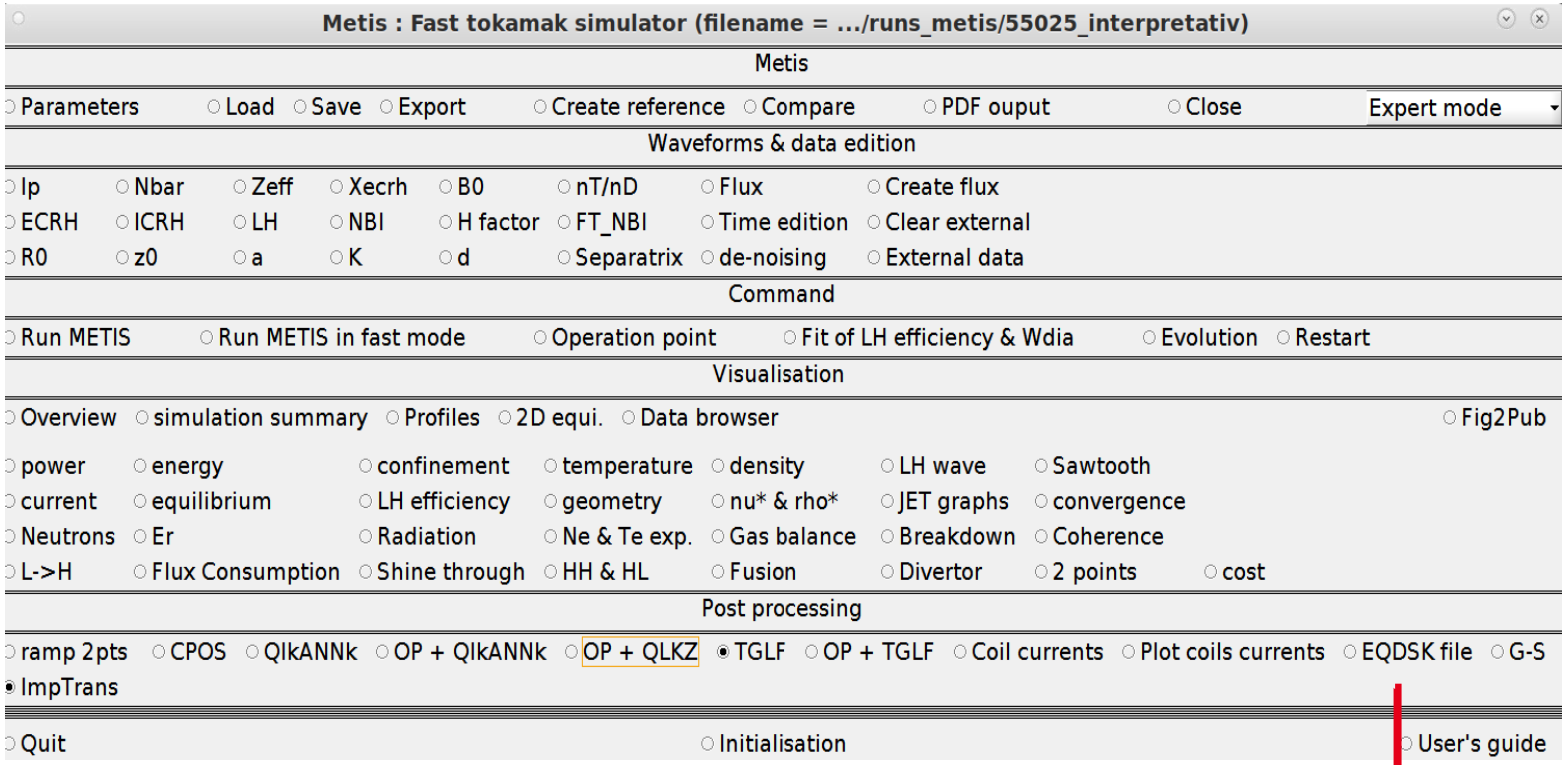


- Just select a result file named "*_inverse4metis_results_fast"

✓ Plot graphs for coil currents, convergence, and optionally for voltages and forces



Compute EQDSK file with FEEQS.M fixed boundary solver

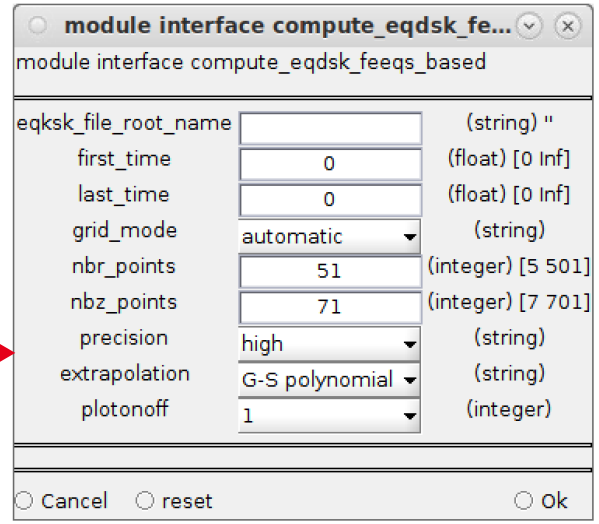


- Compute EQDSK files for METIS time slice in the selected time interval

✓ Generate file .eqdsk with time slice index and time stamp

✓ Extrapolate outside the LCFS with polynomial vacuum G-S solution

❖ Do not describe correctly flux map outside the LCFS near X-point



Parameters for EQDSK file generation

- **eqksk_file_root_name:** Root name for EQDSK files; leave it empty for automatic name generation based on METIS information (machine name and shot name)

- **first_time:** First time of the METIS simulation at which EQDSK file would be computed.
- If both first_time and last_time =0, graphical time selection will be proposed at the next step.

- **last_time:** Last time of the METIS simulation for which EQDSK file would be computed.
  If equal to first_time, only one EQDSK file will be created

- **grid_mode:** If = automatic, grid definition will be automatically generated, otherwise manual radial and vertical number of point will be used

- **nbr_points:** If grid_mode = manual, number of radial points in the grid

- **nbz_points:** If grid_mode = manual, number of vertical points in the grid

- **precision:** Precision of equilibrium computation and interpolation (computational time increases with precision)

- **extrapolation:** Method used to extrapolate outside the LCFS:
  - **Interpolation** = use simple interpolation method
  - **G-S polynomial** = fit a polynomial solution of G-S constrained with flux and magnetic field on each point of LCFS

- **plotonoff:** Set level of displayed graphs (for multi time slices computation, it is recommended to switch off graphics display):
  - 0 = no graphs
  -1 = plot results

17. ICRH model in METIS

METIS ICRH models and use

- Three models are available in METIS
  - Simplified model for Fast Wave Electron heating and Fast Wave Current Drive
    - All heat goes to electrons
    - Source shape is scale on absorption parametric dependances (see F. Meo et al, 27th EPS Conference on Controlled Fusion and Plasma Physics (ECA, Budapest, 2000), Vol. 24, pp. 1613–1616)
    - Current drive efficiency is fitted of data from ITER basis physics 1999 (https://iopscience.iop.org/issue/0029-5515/39/12) , Chapter 6, page 2512
  - Simplified mode for minority heating scheme (named: “PION_fit-Stix”)
    - Wave propagation is reduced to a simple fit of some data from PION code (L-G Eriksson and T Hellsten, 1995, Phys. Scr. 52 70, DOI 10.1088/0031-8949/52/1/014).
    - The distribution function is computed using Stix formulation (T.H. Stix, 1975, Nucl. Fusion 15 737, DOI 10.1088/0029-5515/15/5/003) at the maximum of the absorption.
  - Simplified mode for minority heating scheme based of R. Dumont formulation (see https://github.com/IRFM/METIS/blob/main/doc/description_model_ICRH_Dumont-Vu.pdf)
    - Use simplified model for wave absorption
    - Solve simplified semi analytical Fokker-Planck equation at each radial position.
- There no model for three ions scheme or other schemes than minority heating and fast wave.
- The fusion power induced by fast ions generated by ICRH for D/T reactions in minority heating scheme can be also computed. It is based on integration using the distribution function and the reaction rate for beam-thermal reaction. The power obtains should be compared to more precise code. The parameter f_icrh_salpha can be used to calibrate METIS prediction.

METIS ICRH models and use

- Parameters for ICRH models
  - **fwcd:** selection of heating scheme
  - **icrh_model:** see previous slide
  - **mino & cmin:** type and concentration of minority. If the minority specie is declared in “gas” composition (i.e. DT, pB11 or DHe3), cmin should be set to 1
  - **nphi:** toroidal wave number (depending on antenna geometry and spectrum)
  - **freq:** frequency of the injected wave (set the position of deposition, MHz)
  - **icrh_width:** focalization of wave in plasma to mimic wave propagation and absorption in plasma
  - **fact_mino:** fraction of the minority ions interacting effectively with the wave (set to 1 by default)
  - **orbit_width:** if switched on, broaden heat deposition accordingly to orbit effect using simplified formulation (M. Schneider et al 2015 Nucl. Fusion 55 013003, DOI 10.1088/0029-5515/55/1/013003)
  - **ifast_icrh:** possibility to generate current source if the wave has a non null parallel index. This parameter set the level of this effect.
  - **fabs_fw:** set the fraction of inputted power that are directly absorbed by electons (FWEH).
  - **MC_onoff:** switch on or off the mechanism that mimic mode conversion if only a fraction of the wave power is absrobed
  - **fMC_loss:** fraction of the power that come from mode conversion that is lost and not heat the plasma. The complement is assumed to be transfered to the electron.
  - **Icrh_mode_fast:** convergence mode of Dumont-Vu model
  - **f_icrh_salpha:** factor applied to the fusion source induced by ICRH fast ions.

18. METIS internal runaway electrons model

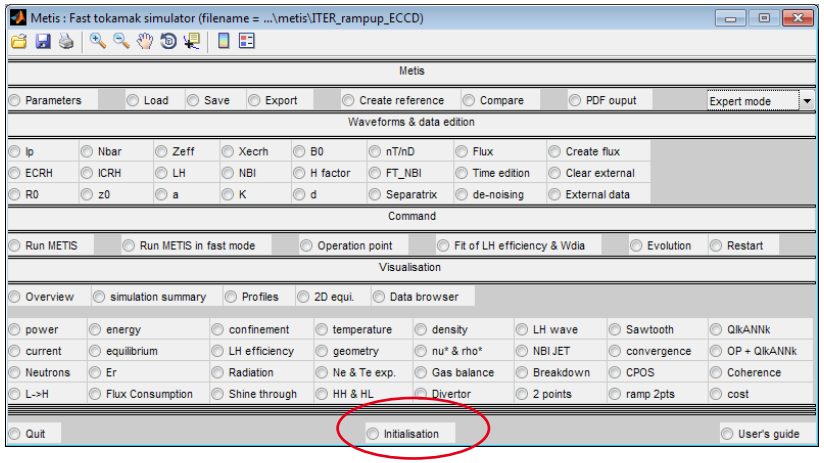
METIS model for runaway electrons

- METIS have already a simplified model for runaway
  - Total current is predicted:
  - 0D model for total runaway density/current
    [H. Smith et al, Physics of Plasma 13 (2006) 102502]
  - Current density shape given by seed shape
    (with exponential dependence):
    [F. Sourd, Phd, Université de Provence, 2004]
  - Taking into account as a source in current diffusion solver of METIS

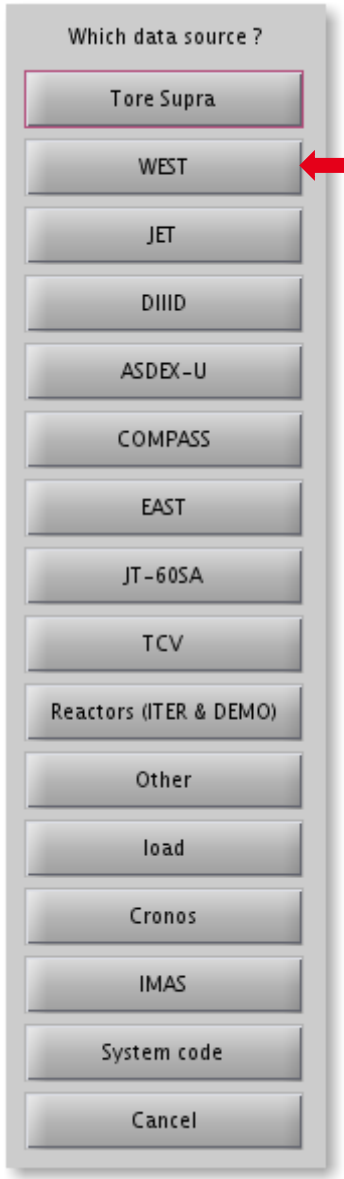
- Mainly use for ramp-up:
  - Change of effective resistivity during ramp-up

19. WEST scenario preparation

ACCESS to WEST scenario preparation tools

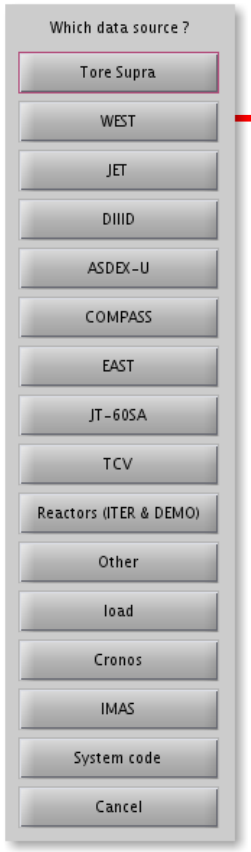


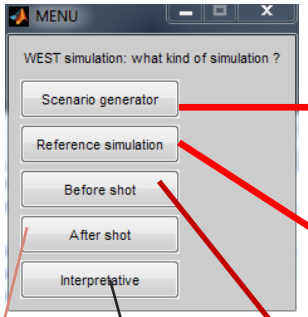


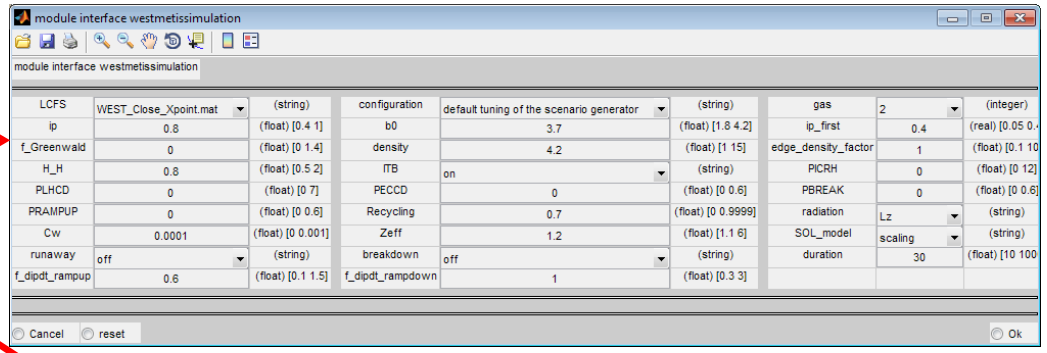


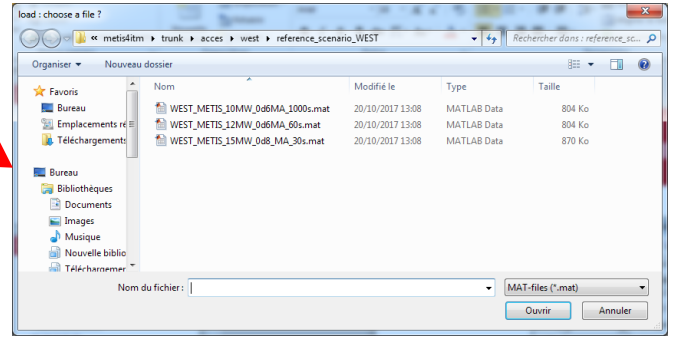
Menu for WEST

METIS initialization menu for WEST









Prepare an interpretative simulation = After shot + measured profiles & parameters

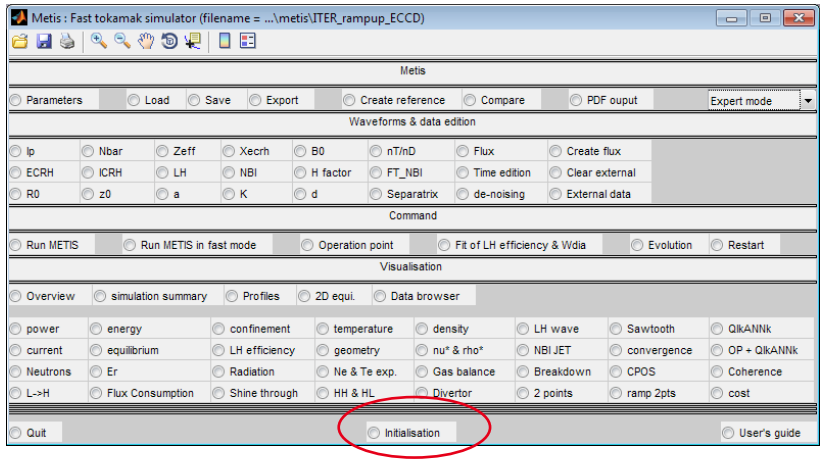
Read data in ARCAD & IMAS database to prepare a METIS simulation

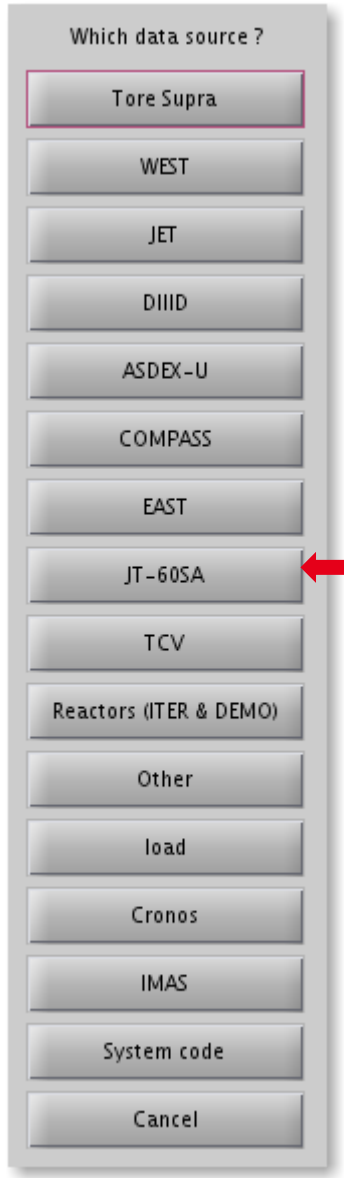
Read PCS data before shot or to replay a shot as tune in PCS and prepare a METIS simulation

- Menu to choose between:
  - Scenario generator
  - PCS data (before shot)
  - ARCAD data (after shot)
  - Interpretative (use ne,Te,…)

20. JT-60SA scenario preparation and postprocessing

ACCESS to JT-60SA scenario preparation tools

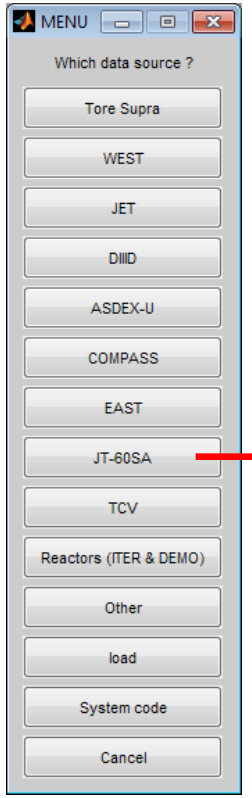


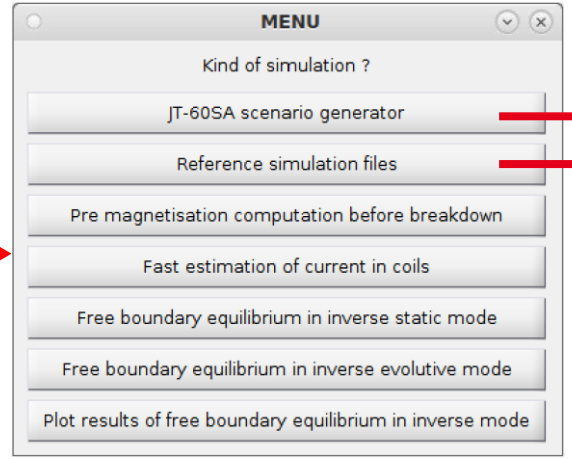


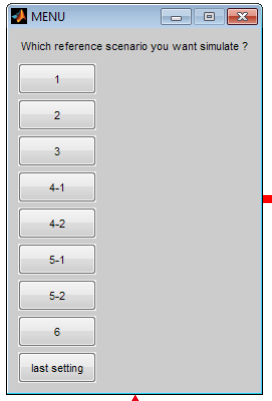
Menu for JT-60SA

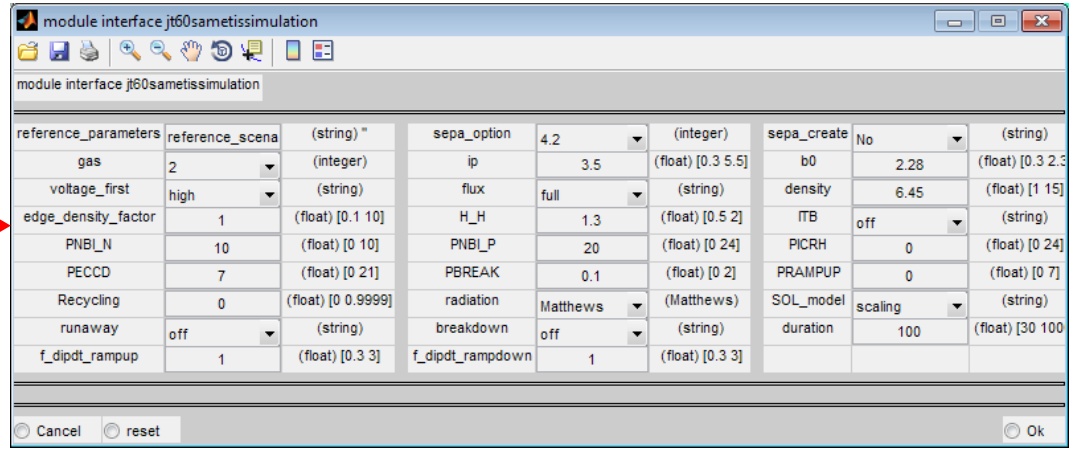
METIS initialization menu for JT-60SA

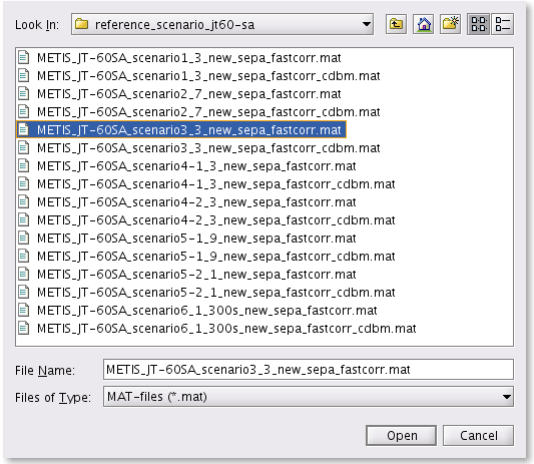
- Added menu to choose between:
  - reference scenario files
  - scenario generator











Ramp-up and Ramp-down constraints

- Documents :
  - JT-60SA Research
  - Development of operation scenarios for plasma breakdown and current ramp-up phases in JT-60SA tokamak , H. Urano et al, Fusion Engineering and Design 100 (2015) 345–356
  - Simulation of plasma current ramp-up with reduced magnetic flux consumption in JT-60SA, T Wakatsuki et al, Plasma Phys. Control. Fusion 57 (2015) 065005 (12pp)

- Data
  - Snap shots of LCFS during scenario 2 from CREATE team.

METIS scenario generator for JT60-SA

- We have developed a scenario template generator:
  - Create a full discharge scenario including:
    - break-down
    - burn through
    - ramp-up
    - flat-top
    - ramp-down.
  - Pre parametrization with a dedicated GUI.
  - Fine-tuning from METIS GUI.
  - Take into account constraints described in documentation.
  - Actuators are pre-tuned (i.e. NBI geometry, …).

Scenario generator parameters (1/4)

- sepa_option: reference scenario for LCFS parametrisation
  (JT-60SA Research Plan, Version 3.3, 2016, March)
- sepa_create: if = Yes, use LCFS provided by CREATE team (available for scenario 2);
  if = No, use LCFS generated by the program
- gas: Plasma composition: main gas species: 1 -> H, 2 -> D, 4 -> He
- ip: plasma current (MA)
- b0: vacuum magnetic field @ R_0 (T)
- voltage_first: Plasma initiation: burn through Vloop: high or low
  (H. Urano et al, Fusion Engineering and Design 100 (2015) 345-356)
- flux: available flux from pre-magnetisation: half CS current or full CS current
  (H. Urano et al, Fusion Engineering and Design 100 (2015) 345-356)
- density: line averaged electron density during flattop (10^19 m^(−3))

Scenario generator parameters (2/4)

- edge_density_factor: multiplication factor applied to edge density scaling law:
  if > 0, ne_edge = nea_factor * LCFS_denstity_scaling_law;
  if < 0, ne_edge = abs(nea_factor) * n_bar

- H_H: time confinement multiplication factor during H-mode phase

- ITB: allows or not ITB formation (on/off)

- PNBI_N: maximum power for NNBI during flattop (MW)

- PNBI_P: maximum power for PNBI during flattop (MW)

- PICRH: Optional ICRH source; maximum power for PICRH during flattop (MW);
  must be 0 by default (this source can be used for scope studies)

- PECCD: maximum power for ECRH/ECCD during flattop (MW);
  baseline maximum power is 7 MW

Scenario generator parameters (3/4)

- PBREAK: injected ECRH / ECCD power for assisted breakdown (MW);
  note that only a small fraction of this power is absorbed

- PRAMPUP: ECRH / ECCD assisted ramp-up: maximum power at the end of ramp-up (MW):
  power increases in time depending on density and plasma current

- Recycling: Recycling coefficient
- radiation: model for line radiative power in core plasma
  (Lz = cooling rate or Matthews = Matthews scaling law)

- SOL_model: scaling law or 2 points model

- runaway: allows or not runaway electrons in the discharge (on/off)

- breakdown: turn on or off the breakdown model for plasma initiation

- duration: shot duration: time of end of flat-top (s)

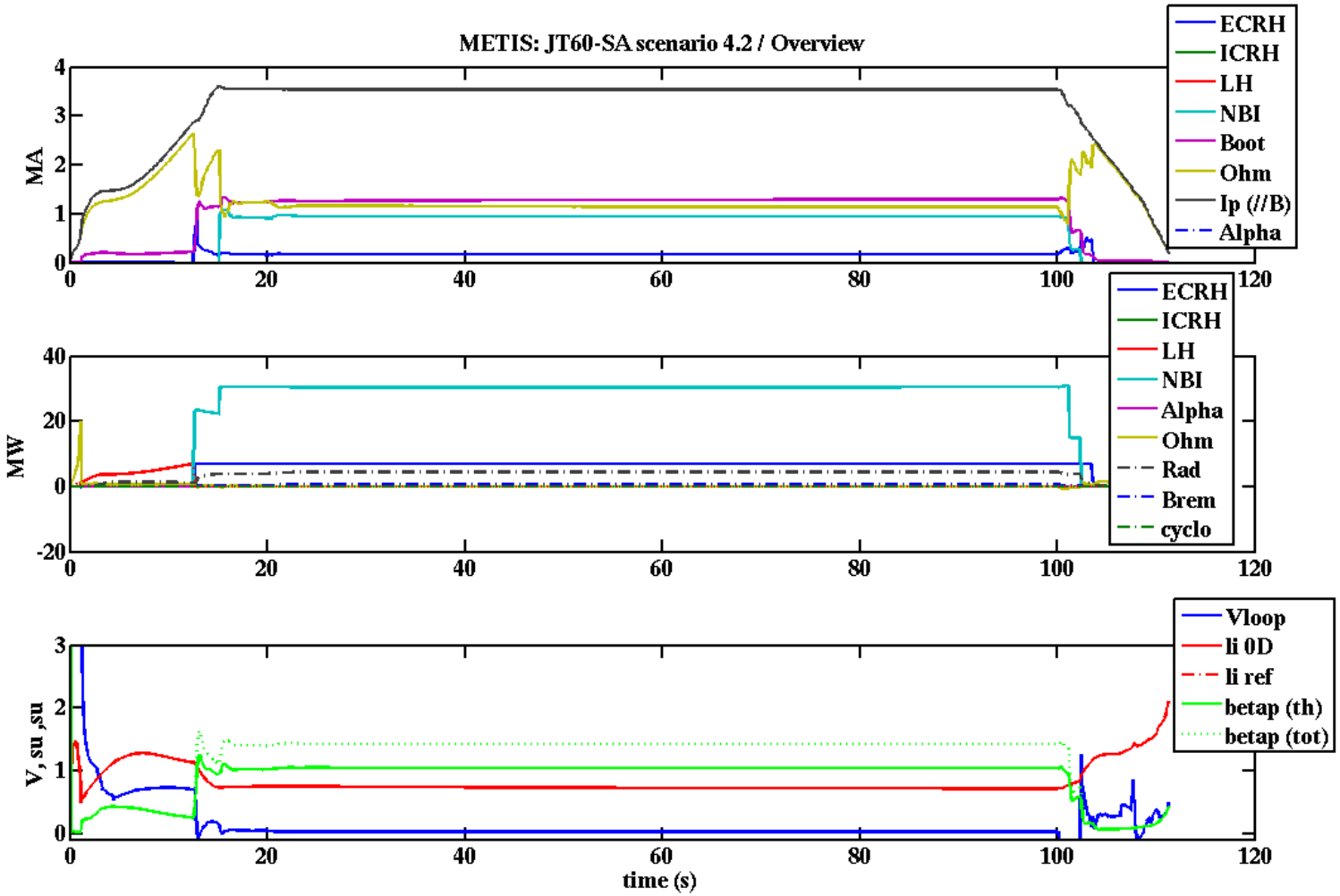
Scenario generator parameters (4/4)

- f_dipdt_rampup: Ramp-up rate; multiplication factor applied to dIp/dt for plasma current ramp-up with respect to reference case
  (H. Urano et al, Fusion Engineering and Design 100 (2015) 345-356)

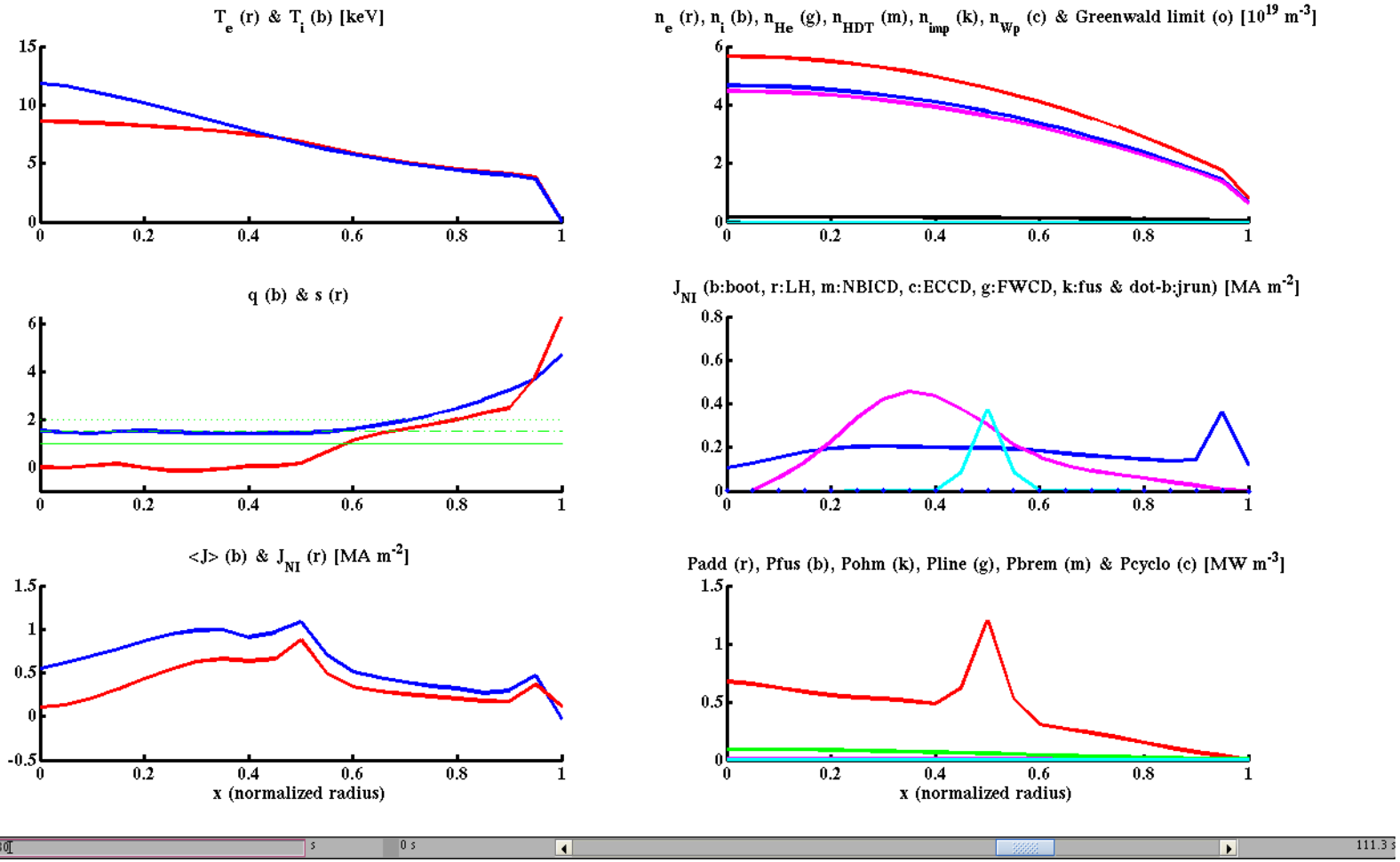
- f_dipdt_rampdown: Ramp-down rate; multiplication factor applied to dIp/dt for plasma current ramp-down with respect to reference case
  (H. Urano et al, Fusion Engineering and Design 100 (2015) 345-356)

- reference_parameters: file name that contains parameters that will overwrite user defined parameters (if empty, it is not used)

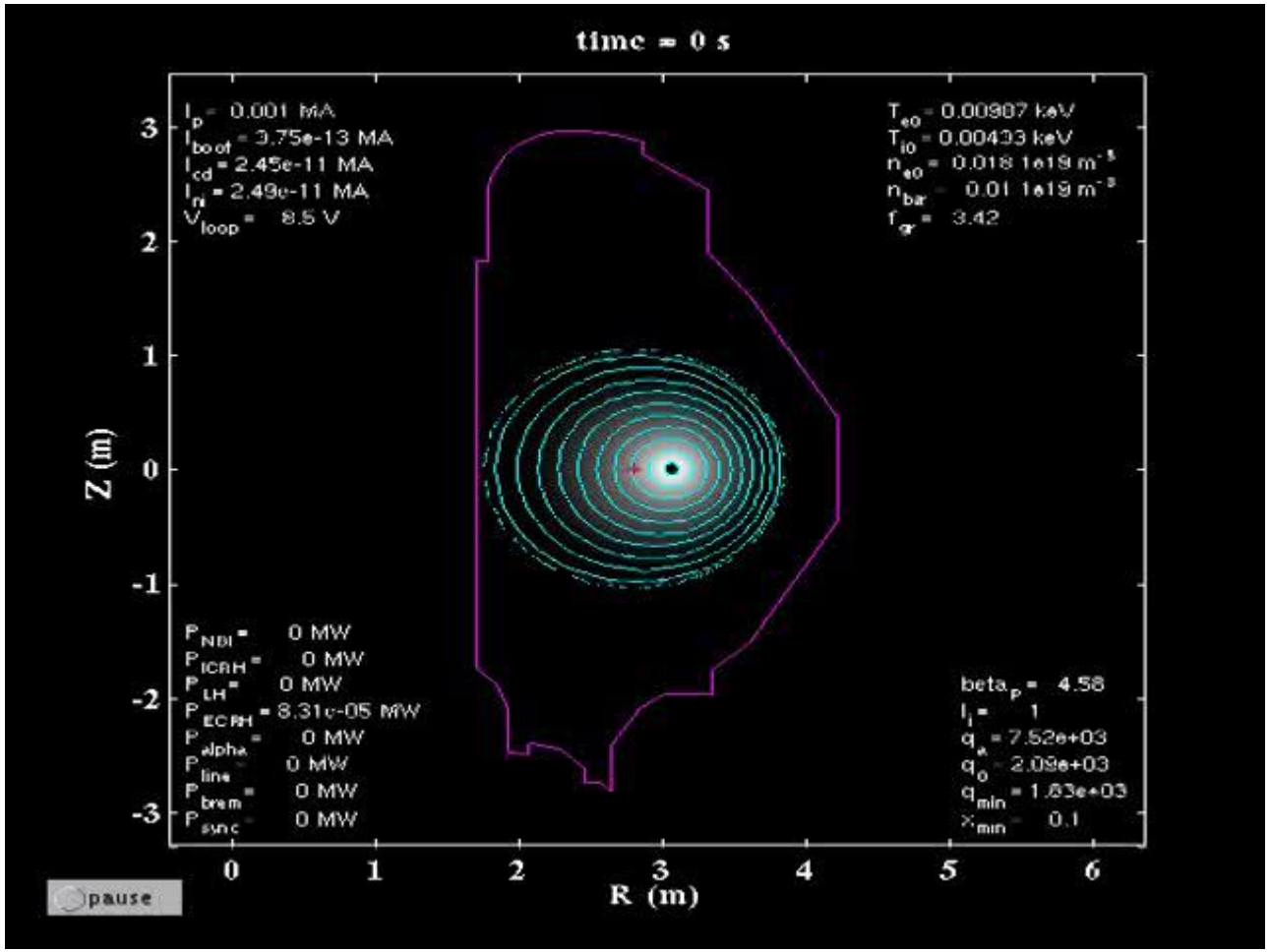
Example: scenario 4.2 (1/3)



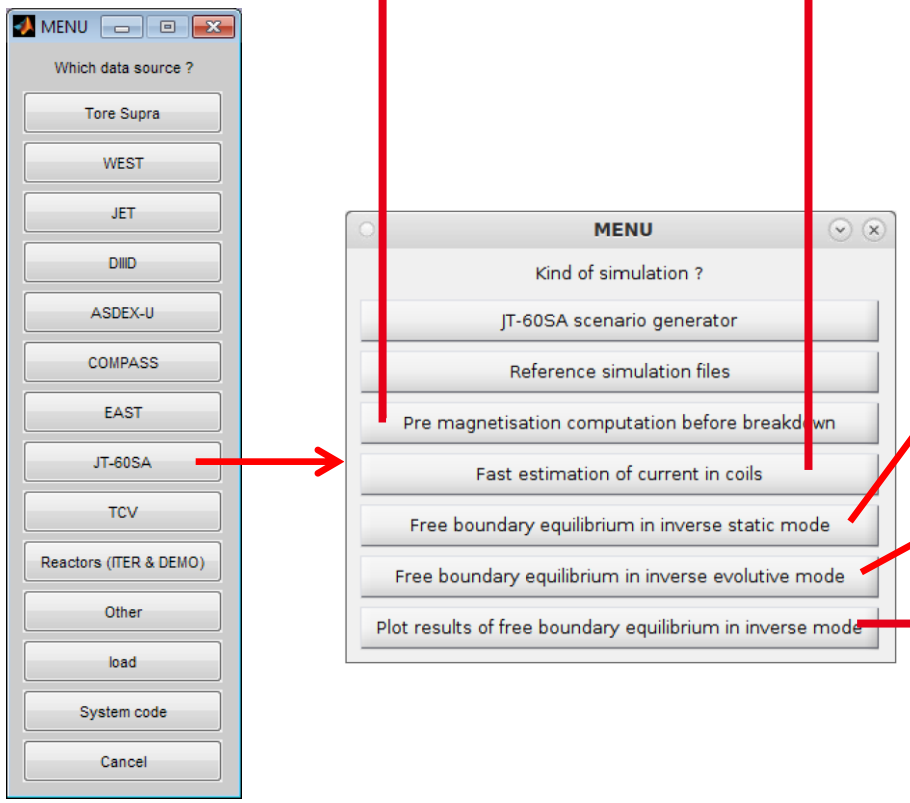
Example: scenario 4.2 (2/3)



Example: scenario 4.2 (3/3)

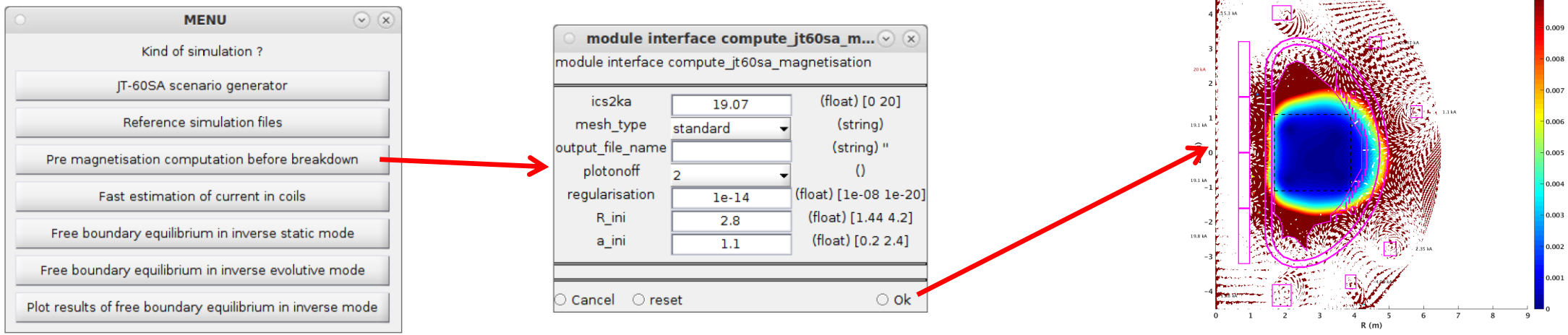


Postprocessing JT-60SA results with FEEQS.M



- FEEQS.M code should had been installed (independently from METIS)
- Menus allow to choose:
  - Computation of coil currents for plasma breakdown (pre-magnetization)
  - Computation of coil currents evolution in fast mode
    - Based on fixed boundary equilibrium code
    - No passive structure described
    - No update of the LCFS
  - Computation of coil currents evolution in standard mode
    - Based on inverse mode of free boundary equilibrium
    - No current computed in passive structures
    - LCFS updated
  - Computation with the inverse evolutive
    - Based on inverse evolutive mode of free boundary equilibrium
    - Current in passive structures are computed
    - LCFS updated
    - Difficult to tune
  - Visualization of results from previous postprocessing

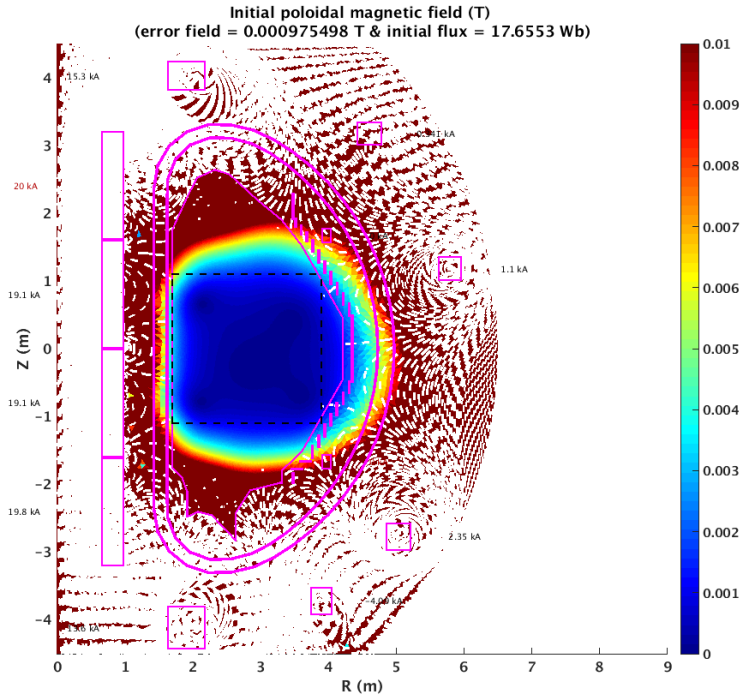
Breakdown coil currents (GUI)



FEEQS.m modes: inverse magnetostatic

- Use to compute initial current in coils (pre-magnetisation):
  - Computes static solution in vacuum suitable for breakdown
  - Constraints:
    - Minimize poloidal field in user defined zone
    - Maximize poloidal flux at planned magnetic axis position
    - Respect current limits for coils

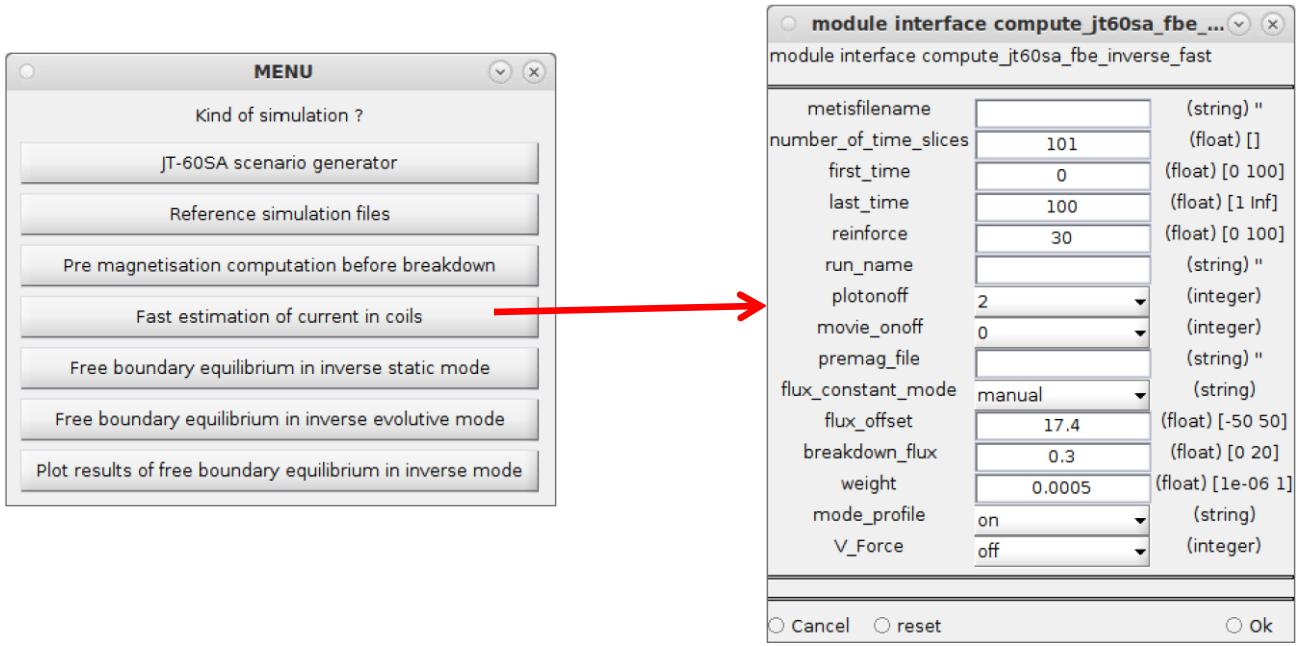
Initial current in coils before breakdown :
CS1 = 20 kA (limits = [-20,20] kA)
CS2 = 19.07 kA (limits = [-20,20] kA)
CS3 = 19.0576 kA (limits = [-20,20] kA)
CS4 = 19.8032 kA (limits = [-20,20] kA)
EF1 = 1.09903 kA (limits = [-20,10] kA)
EF2 = 0.341162 kA (limits = [-20,10] kA)
EF3 = 15.2554 kA (limits = [-20,20] kA)
EF4 = 13.5797 kA (limits = [-20,20] kA)
EF5 = -4.085 kA (limits = [-20,10] kA)
EF6 = 2.3451 kA (limits = [-20,10] kA)
VS1 = 0 kA (limits = [-5,5] kA)
VS2 = 0 kA (limits = [1e+35,-1e+35] kA)
Initial available flux = 17.6553 Wb
Error field in null zone = 0.000975498 T
Convergence: regularisation = 7.25591e-07
cost = 9.12912e-06



Pre-magnetization parameters

- **ics2ka**: initial current in CS2 coil (kA)
- **mesh_type**: select the number of elements in the 2D mesh (test, standard or fine; test is reserved for testing and must be not used for production)
- **output_file_name**: matfile name where results are stored (if empty, no file will be created)
- **plotonoff**: set level of displayed graphs: 0 = no graphs; 1 = results only; 2 = all graphs (debug)
- **regularisation**: weight of regularisation term (sum(I^2))
- **R_ini**: central postition of null zone at Z = 0 (m)
- **a_ini**: radius of null zone (m)

Feedforward coil currents fast inverse

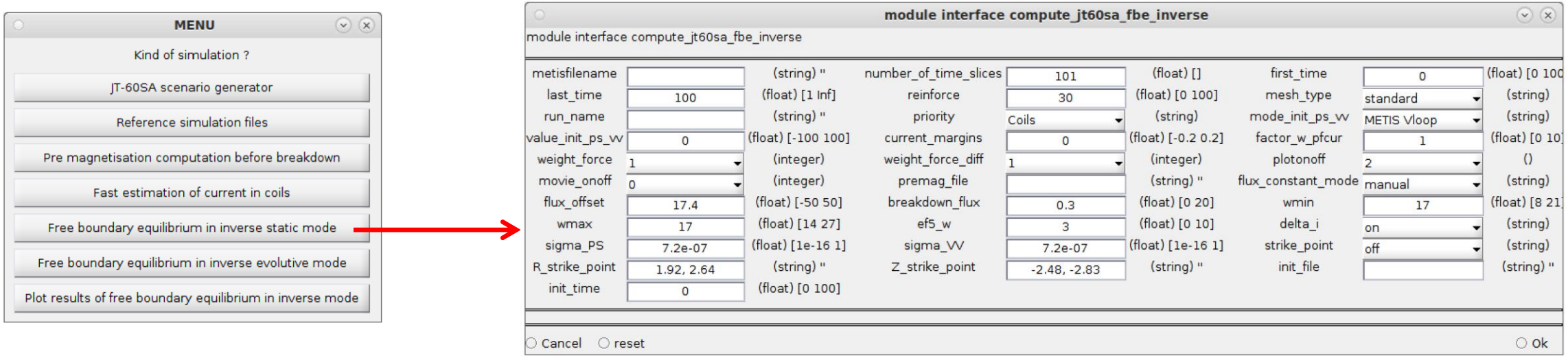


- Compute coil currents feedforward based on fast inverse fixed boundary G-S solver:
  - Search for best match of poloidal flux and its gradient on LCFS points.
  - Do not allow to compute updated LCFS
  - Do not include passive structure currents
  - In post processing, voltages, passive structure currents and coil forces are estimated.

Fast inverse parameters

- **metisfilename**: path and name of METIS simulation file used for the computation (if empty, a file selector will be open at the next step)
- **number_of_time_slices**: number of time slices extracted from METIS simulation at which inverse equilibrium would be computed
- **first_time**: first time of the METIS simulation at which inverse equilibrium would be computed
- **last_time**: last time of the METIS simulation for which inverse equilibrium would be computed
- **reinforce**: weigths of extrema in LCFS (to have a better match on X-points, strike points, RIG and ROG)
- **run_name**: run_name is added to name of save files (can be leave empty)
- **plotonoff**: set level of displayed graphs: 0 = no graphs; 1 = results only; 2 = all graphs (debug)
- **movie_onoff**: if = 1, record movie made of 2D equilibrium time slice graphs; if = 2, record movie made of 2D equilibrium time slice graphs and save a .fig file for each time slice
- **premag_file**: if available, name of file containing premagnetisation data (can be leave empty; if = workspace, reads data structure premagnetisation_jt60sa in Matlab workspace)
- **flux_constant_mode**: this parameter allows to choose the way poloidal flux offset used to match METIS current diffusion LCFS poloidal flux and FEEQS.M LCFS poloidal flux is computed:\
    - if = auto, offset is automatically computed taking into account flux consumption estimation for breakdown computed in METIS and flux leakage computed with the help of current diffusion equation between end of breakdown and first_time compare to flux computed by pre-magnetization tool
    - if = manual, offset is set by the user.
- **flux_offset**: poloidal flux offset used to match METIS current diffusion LCFS poloidal flux and FEEQS.M LCFS poloidal flux (Wb)
- **breakdown_flux**: estimation of the flux consumption during the breakdown phase (W).
- **mode_profile**: Select if P' and FF' are used in equilibrium solver or if the equilibrium is constraint by β_p and l_i
- **V_force**: Switch on or off the computation of voltages and forces in coils

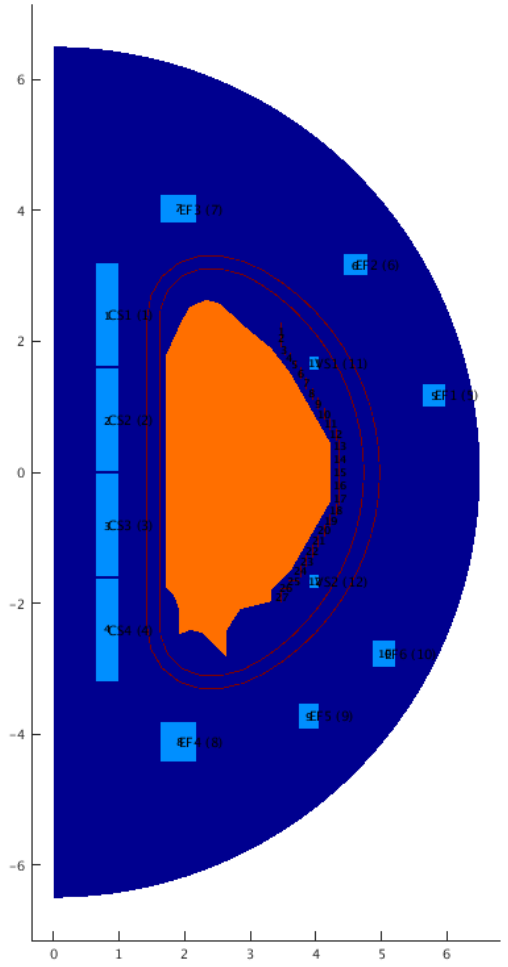
FEEQS.M mode: inverse static (GUI)



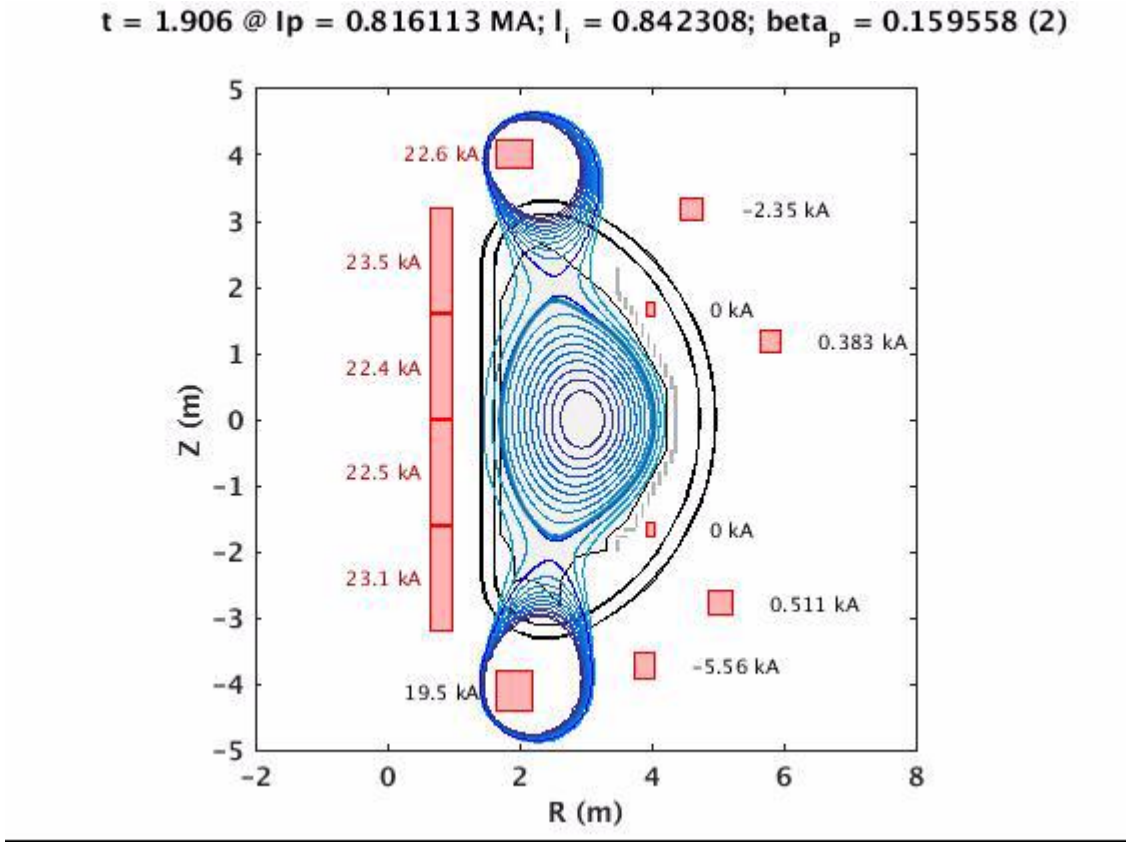
FEEQS.M mode: inverse Static (description)

- Use to compute time dependent currents in coils without passive structure currents
  - Computes FBE solutions under constraints
  - Don’t’ solve circuit equations for passive structures
  - Solves for a set of time slices → can be used for the whole shot
  - Uses METIS data as input
  - Constraints:
    - P’ and FF’ profiles from METIS
    - Plasma shape from METIS
    - Plasma currents
    - LCFS poloidal flux (Poynting constraint for flux consumption)
    - Coils limits (currents, forces and optionally maximum magnetic field on coils)
- Results
  - Real FBE solution → updated LCFS shapes
  - Coil currents → feedforward waveforms for full simulation
  - Forces and maximum magnetic field on coils
  - Estimation of voltage on coils and estimation of current in passives structures

Example of results in inverse static



Mesh



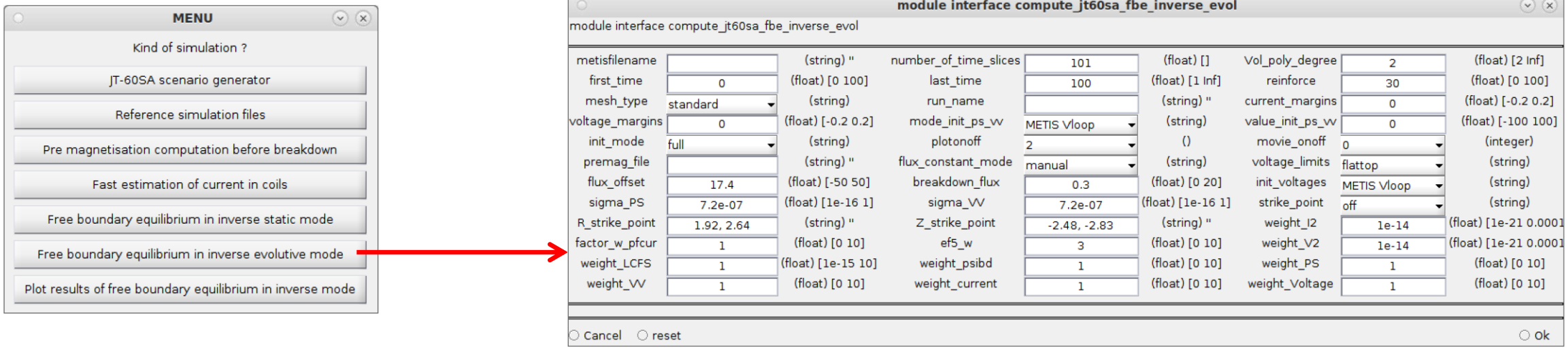
Inverse static parameters (1/2)

- **metisfilename**: path and name of METIS simulation file used for the computation (if empty, a file selector will be open at the next step)
- **number_of_time_slices**: number of time slices extracted from METIS simulation at which inverse equilibrium would be computed
- **first_time**: first time of the METIS simulation at which inverse equilibrium would be computed
- **last_time**: last time of the METIS simulation for which inverse equilibrium would be computed
- **reinforce**: weigths of extrema in LCFS (to have a better match on X-points, strike points, RIG and ROG)
- **mesh_type**: number of elements in the 2D mesh (test, standard or fine; test is reserved for testing and must be not used for production)
- **run_name**: run_name is added to name of save files (can be leave empty)
- **weight_force**: if = 1, constraint forces on coil to be in engineering limits
- **weight_force_diff**: if = 1, constraint difference of force between coils to be in engineering limits
- **plotonoff**: set level of displayed graphs: 0 = no graphs; 1 = results only; 2 = all graphs (debug)
- **movie_onoff**: if = 1, record movie made of 2D equilibrium time slice graphs; if = 2, record movie made of 2D equilibrium time slice graphs and save a .fig file for each time slice

Inverse static parameters (2/2)

- **premag_file**: if available, name of file containing premagnetisation data (can be leave empty; if = workspace, reads data structure premagnetisation_jt60sa in Matlab workspace)
- **flux_constant_mode**: this parameter allows to choose the way poloidal flux offset used to match METIS current diffusion LCFS poloidal flux and FEEQS.M LCFS poloidal flux is computed:\
  - if = auto, offset is automatically computed taking into account flux consumption estimation for breakdown computed in METIS and flux leakage computed with the help of current diffusion equation between end of breakdown and first_time compare to flux computed by pre-magnetization tool
  - if = manual, offset is set by the user.
- **flux_offset**: poloidal flux offset used to match METIS current diffusion LCFS poloidal flux and FEEQS.M LCFS poloidal flux (Wb)
- **wmin**: minimal weight of regularisation term in initialisation loop
- **wmax**: minimal weight of regularisation term in initialisation loop
  The best value for the weight will be chosen (to have similar contribution of each terms of cost function and keep a good convergence)
- **strike_point** (for diverted plasma):
  - if = off, no constraint on strike point position
  - if = on, add constraints on strike point positions
- **R_strike_point**: list of strike point radial positions (m, quote separated)
- **Z_strike_point**: list of strike point vertical positions (m, quote separated)

Inverse evolutive FBE



- Solve the optimization problem with the inverse evolutive free boundary equilibrium:
  - Search for optimal coil voltages solving G-S and circuit equations on an interval of time (starting from an inverse static).
  - Provide also coil currents, passive structure currents and forces applied to coils.
  - Quite difficult to tune to get good convergence
  - Interesting to use for transient phases.

Inverse evolutive parameters (1/3)

- **metisfilename:** Path and name of METIS simulation file used for the computation (if empty, a file selector will be open at the next step)
- **number_of_time_slices:** Number of time slices extracted from METIS simulation at which inverse equilibrium would be computed
- **Vol_poly_degree:** Degree of polynomial representation of coil voltage waveforms. Used to smooth the feedforward and speed up the computation.
  if equal to number of time step, switch to local times value (no more smoothing)
- **first_time:** First time of the METIS simulation at which inverse equilibrium would be computed
- **last_time:** Last time of the METIS simulation for which inverse equilibrium would be computed
- **reinforce:** Weights of extrema in LCFS (to have a better match on X-points and RIG and ROG)'
- **mesh_type:** Number of elements in the 2D mesh (test is reserved for testing and must be not used for production)
- **run_name:** run_name is added to name of save files (can be leave empty)
- **current_margins:** Margin to be keep to coil current limits (on absolute value, if < 0, allows to go above the limit).
  Pay attention to be compatible with plasma initialization value. Applied only to EF coils
- **voltage_margins:** Margin to be keep to coil voltage limits (on absolute value, if < 0, allows to go above the limit).
  Pay attention to be compatible with plasma initialization value. Applied on all coils
- **mode_init_ps_vv:** Method used to initialize currents in passive structures at first time of computation (at first_time, see tooltip in GUI).
- **value_init_ps_vv:** Value of loop voltage in passive structure elements at first time of computation (V)
- **init_mode:** Method used to initialize first equilibrium coil current and poloidal flux map (see tooltip in GUI for details)

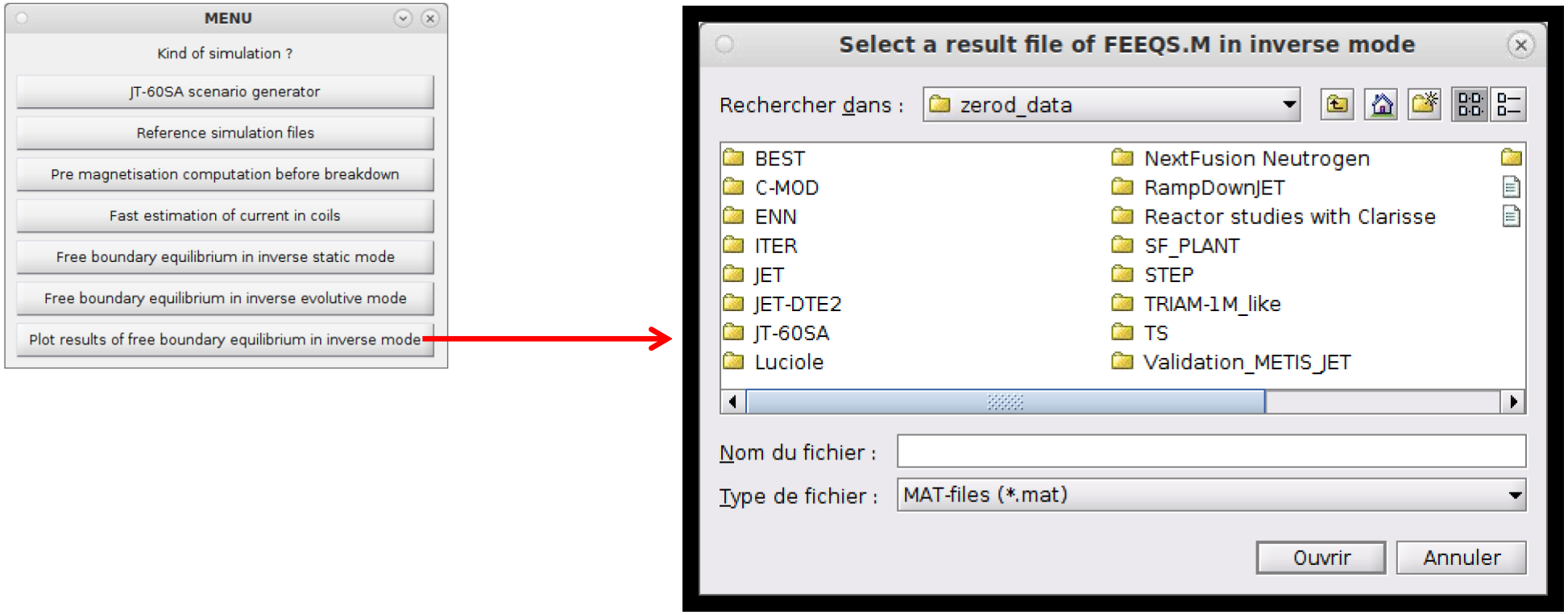
Inverse evolutive parameters (2/3)

- **plotonoff:** Set level of displayed graphs: 0 = no graphs; 1 = results only; 2 = all graphs (debug)
- **movie_onoff:** if = 1, record movie made of 2D equilibrium time slice graphs;
  if = 2, record movie made of 2D equilibrium time slice graphs and save a .fig file for each time slice'
- **premag_file:** if available, name of file containing pre-magnetization data (can be leave empty; if = workspace, reads data structure premagnetisation_jt60sa in Matlab workspace)
- **flux_constant_mode:** This parameter allows to choose the way poloidal flux offset used to match METIS current diffusion LCFS poloidal flux and FEEQS (see tooltip in GUI for details)
- **voltage_limits:** Voltage limits range: if = ramp-up, use extended value allowed by booster and discharge resistor;
  if = flattop, used limited range allowed by steady state power supply
- **flux_offset:** Poloidal flux offset used to match METIS current diffusion LCFS poloidal flux and FEEQS.M LCFS poloidal flux (Wb)
- **breakdown_flux:** Poloidal flux consumed for plasma break-down and burn-through (Wb).
  if METIS breakdown model is turn on, take the maximum of this value and METIS value.
- **init_voltages:** Method used to compute initial guess of voltages provided by power supplies for optimizer:
  if = zero, start with null values;\nif = resistive, start with voltages equal to resistive losses
  if = METIS Vloop, use METIS loop voltage to estimate first guess for voltages provided by power supplies
- **sigma_PS:** Resistivity of passive structure material (Ohm.m)
- **sigma_VV:** Resistivity of vacuum vessel material (Ohm.m)

Inverse evolutive parameters (3/3)

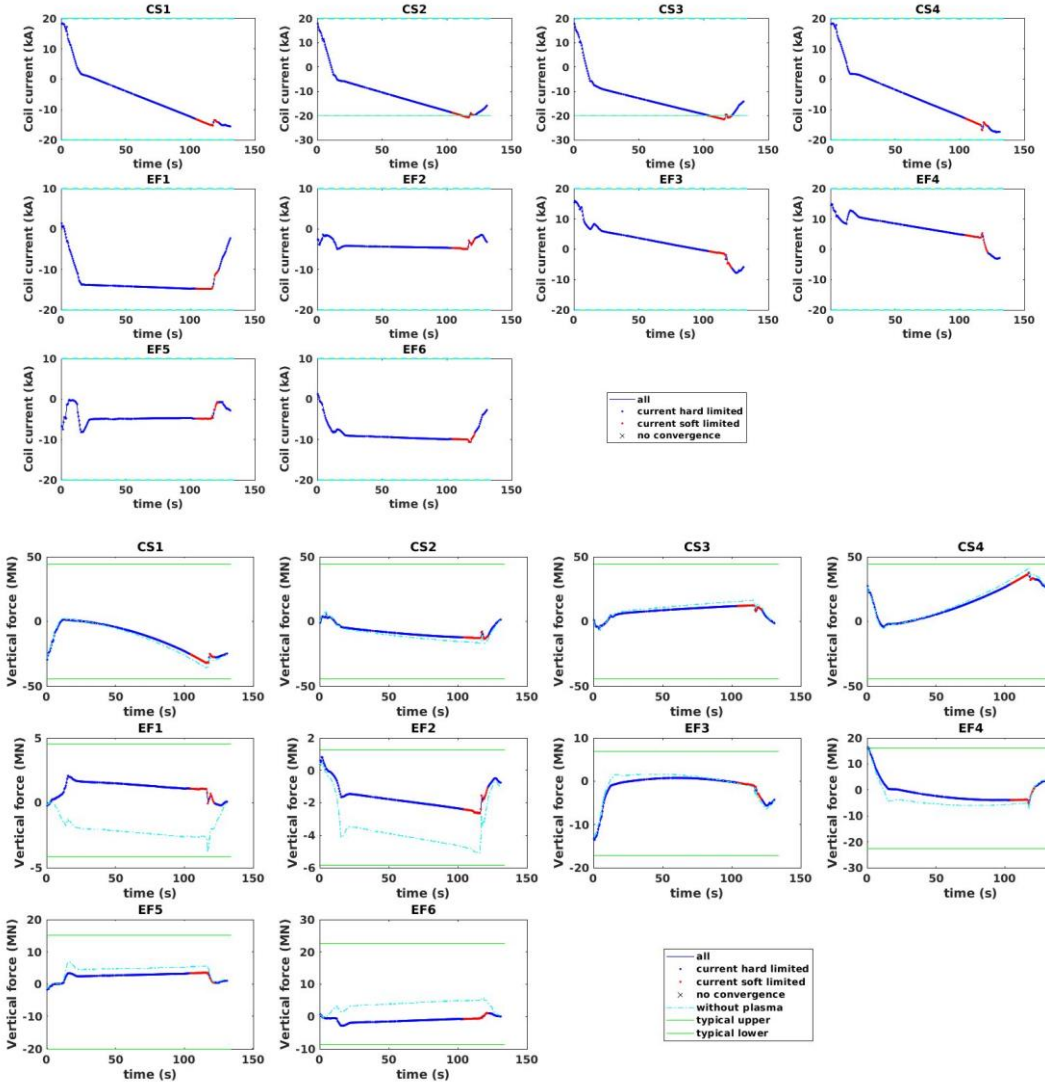
- **strike_point:** For diverted plasma: if = off, no constraint on strike point position
  if = on, add constraints on strike point positions
- **R_strike_point:** List of strike point radial positions (m, quote separated)
- **Z_strike_point:** List of strike point vertical positions (m, quote separated)
- **weight_I2:** Weight or regularization term son sum(Icoils^2) in first inverse mode
- **factor_w_pfcur:** Relative weight on penalization term for current limits in coils. Used only for first inverse equilibrium computation
- **ef5_w:** Penalization of current in EF5 coil to prevent it be used as main coil to make X-point (power of 10).
  Used only for first inverse equilibrium computation
- **weight_V2:** Weight or regularization term son sum(Voltages^2) in inverse evolutive mode
- **weight_LCFS:** Relative weight on penalization term on LCFS desired points in inverse evolutive mode
  (should be set to zero to switch off this term in cost function)
- **weight_psibd:** Relative weight on penalization term on prescribed poloidal flux at LCFS in inverse evolutive mode
  (should be set to zero to switch off this term in cost function)
- **weight_PS:** Relative weight on penalization term on integral(J_PS^2*dS) on passive structure elements in inverse evolutive mode
  (should be set to zero to switch off this term in cost function)
- **weight_VV:** Relative weight on penalization term on integral(J_VV^2*dS) on vacuum vessel in inverse evolutive mode
  (should be set to zero to switch off this term in cost function)'
- **weight_current:** Relative weight on penalization term for coil voltage limits compliance in inverse evolutive mode
  (must be set to zero to switch off this term in cost function)'

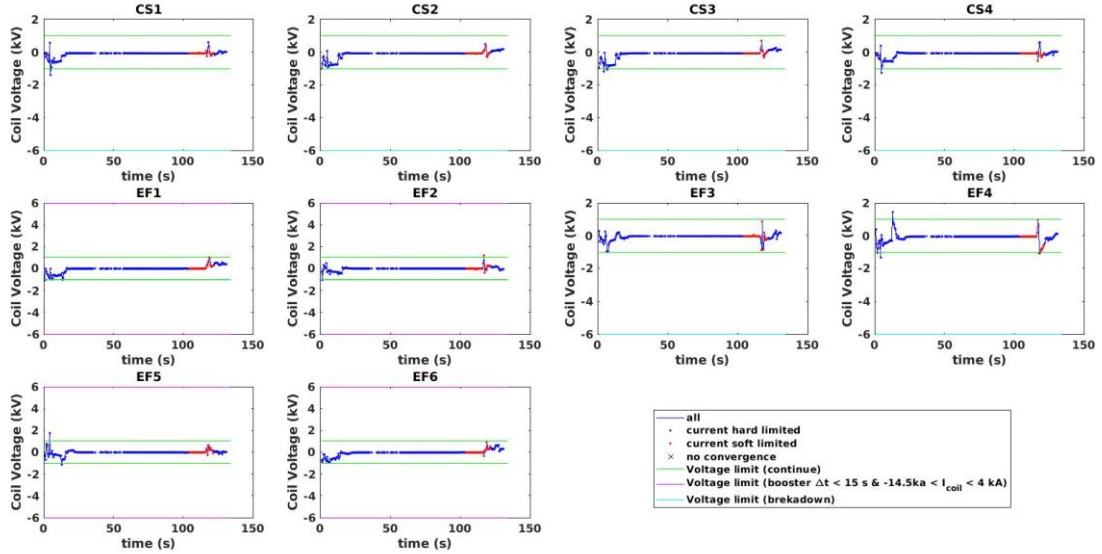
Feedforward coil currents (Visualization)



- Plot graphs for inverse modes:
  - Select file with name ending by “inverse4metis_results_fast” for fixed boundary fast inverse mode
  - Select file with name ending by “inverse4metis_results” for static inverse mode
  - Select file with name ending by “inverse4metis_results_evo” for evolutive inverse mode

Example of graphs plotted for inverse equilibrium





21. Proton-Boron in METIS

Possible fusion fuels and reactions

- Reactions are considered neutron-free if neutron are not involved directly in the reaction
- Most of the reaction involve neutrons or can generate indirectly neutrons
- Only He3 + He3 reactions do not involve some radioactivity
- D+Li6 come with D+D reactions

Alternative Fusion Fuels and Systems,
S. V. Ryzhkov and A. Yu. Chirkov, CRC Press, Taylor & Francis Group
DOI: 10.1201/9780429399398

https://en.wikipedia.org/wiki/Aneutronic_fusion

Table 1. Basic and neutron-free (non-radioactive) thermonuclear reactions

Reaction (energy of products, MeV)	Energy release, keV	Direct radioactivity	Induced radioactivity
D + T → *n* (14.07) + ^{4}He (3.52)	17589	*n*	*n*, T
D + D → *n* (2.45) + ^{3}He (0.82)	3269	*n*	
D + D → *p* (3.02) + T (1.01)	4033	T	*n*
D + ^{3}He → *p* (14.68) + ^{4}He (3.67)	18353	–	*n*, T
D + ^{6}Li → 2 ^{4}He + 22.37	22371	–	*n*, T
D + ^{6}Li → *p* + T + ^{4}He + 2.257	2257	T	*n*
D + ^{6}Li → *p* (4.397) + ^{7}Li (0.628)	5025	–	
D + ^{6}Li → *n* (2.958) + ^{7}Be (0.423)	3381	*n*, ^{7}Be	
D + ^{6}Li → *n* + ^{4}He + ^{3}He + 1.794	1794	*n*	
D + ^{7}Li → *n* + ^{4}He + ^{4}He + 15,121	15121	*n*	
D + ^{7}Be → *p* + ^{4}He + ^{4}He + 16.766	16766	^{7}Be	
p + ^{6}Li → ^{4}He (1.7) + ^{3}He (2.3)	4018	–	*n*, T, ^{7}Be, 11 C
p + ^{11}B → 3 ^{4}He + 8.68	8681	–	*n*, ^{14}C
^{3}He+^{3}He → 2*p* (8.573) + ^{4}He (4.287)	12860	–	–

“Parasitic” reactions

- The main reaction come with some secondary reactions:
  - $p + {}^{11}B \rightarrow {}^{12}C + \gamma + 16\,MeV$ (reactivity ratio to main reaction ~ 10^{-4}, $T_i < 300keV$)
  - $p + {}^{11}B \rightarrow {}^{11}C + n - 2.76\,MeV$ (reactivity ratio to main reaction < 3 10^{-3}, $T_i < 300keV$, 20.4 mn)
  - For reaction involving fast proton or boron, above 675 keV resonance, cross section increase and become equivalent to the cross section of main reaction at 4 MeV.

- α particles can react with other nucleus:
  - ${}^{4}He + {}^{11}B \rightarrow {}^{14}C + p + 0.783\,MeV$ (reactivity ratio to main reaction ~ 5 10^{-4}, $T_i < 300keV$, ~5700 y)
  - ${}^{4}He + {}^{11}B \rightarrow {}^{14}N + n + 0.157\,MeV$ (reactivity ratio to main reaction ~ 5 10^{-4}, $T_i < 300keV$)
  - For fast α particles, cross section for these reactions are close the cross section of main reaction
  - Helium ashes density should be kept low in the discharge

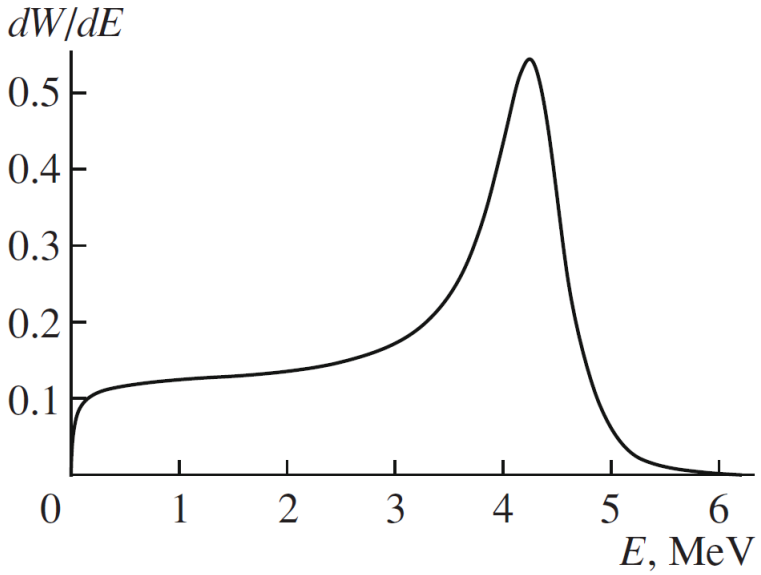
- Less than 0.1% of the reactions in a thermal p–11B plasma produce neutrons (less than 0,2% of released energy)

- Some radioactivity is produced by proton-boron plasma even if it is far less than DT plasma.

Proton-boron fusion reaction

$$p + {}^{11}B \rightarrow \alpha + {}^{8}Be^{*} \rightarrow 3\alpha$$

- ${}^{8}Be$ half life time ~ $10^{-16}\ s$
- On averaged 2.9 MeV per α particle but:
  $\langle \alpha_1 \rangle \sim 4\ MeV$ and $\langle \alpha_{2,3} \rangle \sim 2.3\ MeV$
- But due to two steps reaction, the energy spectrum of α particles is quite broad.

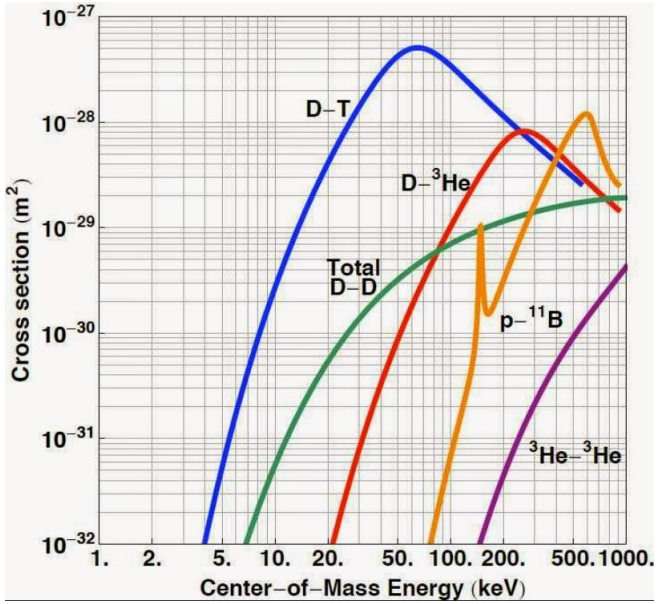


Radiation Limit for the Energy Gain of the p–11B Reaction, A.i Yu. Chirkov and K. D. Kazakov Plasma 2023, 6(3), 379-392; https://doi.org/10.3390/plasma6030026

α-Particle Spectrum in the Reaction p+11B→ α+ 8Be ∗ → 3α*, V. F. Dmitriev, Physics of Atomic Nuclei, 2009, Vol. 72, No. 7, pp. 1165–1167, DOI: 10.1134/S1063778809070084

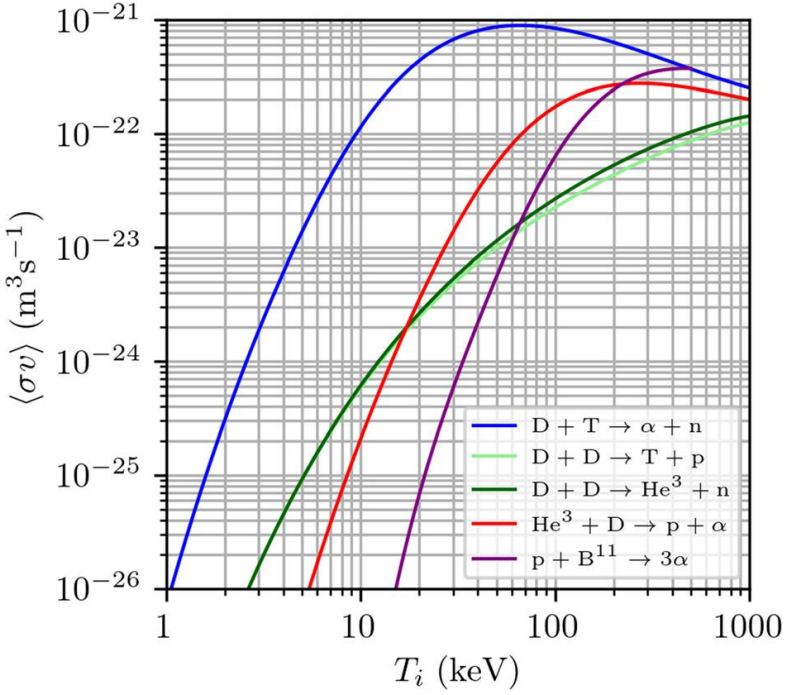
Nuclear Reaction Cross Sections and Reactivity Parameter; R. Feldbacher, IAEA: Vienna, Austria, 1987. https://inis.iaea.org/collection/NCLCollectionStore/_Public/22/030/22030714.pdf

Cross sections and reactivities comparison



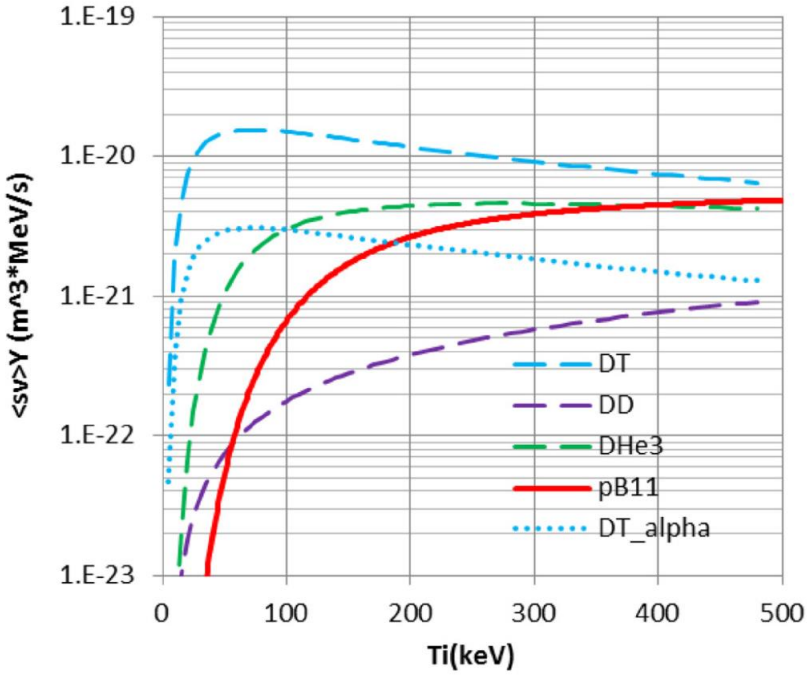
Cross section

https://doi.org/10.1038/s41598-022-08433-4



Reactivity

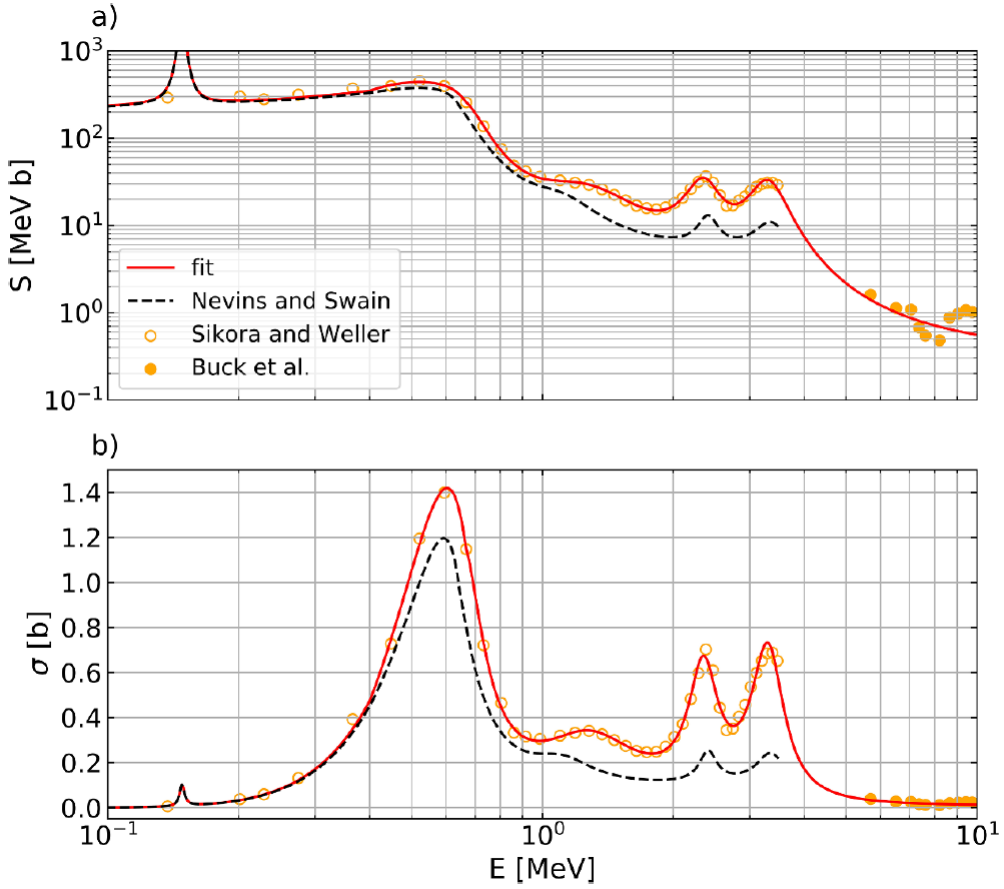
https://doi.org/10.1063/5.0083990



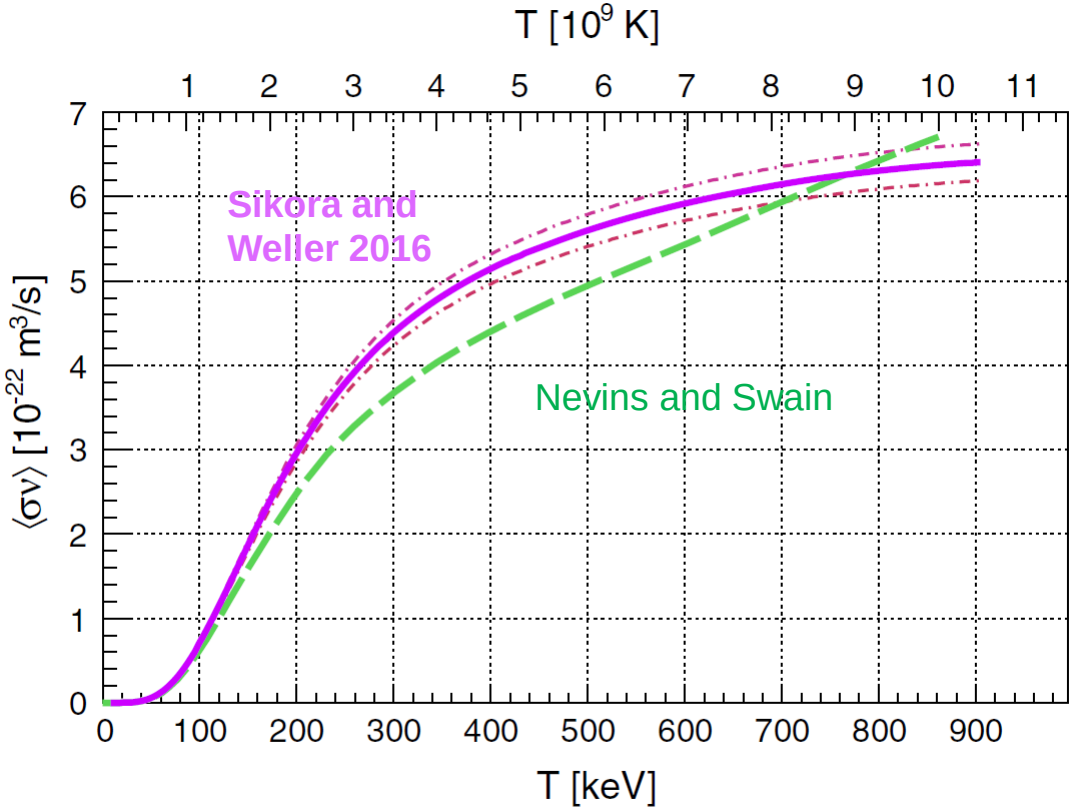
Reactivity * released energy

https://doi.org/10.1088/1741-4326/ab1a60

Cross section and reactivity revisited (2016)



Cross section

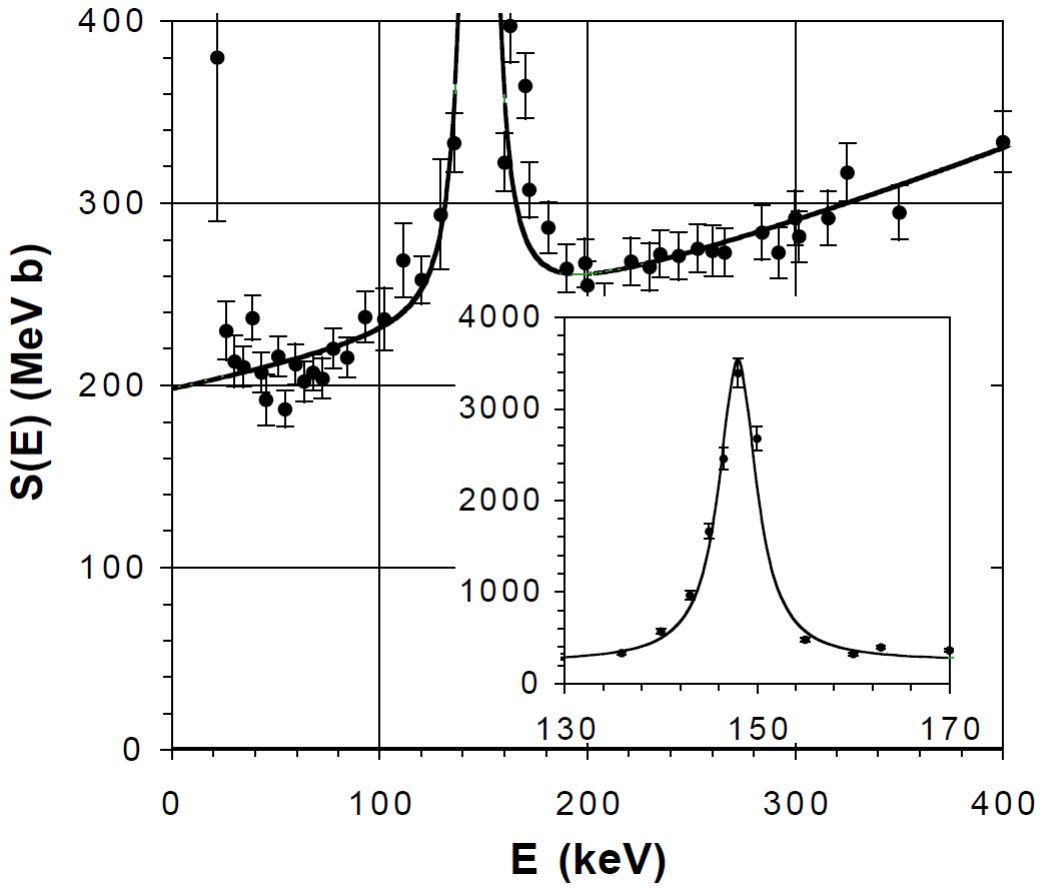


Reactivity

Revisiting p-11B fusion cross section and reactivity, and their analytic approximations, A. Tentori and F. Belloni, 2023 Nucl. Fusion 63 086001, DOI 10.1088/1741-4326/acda4b

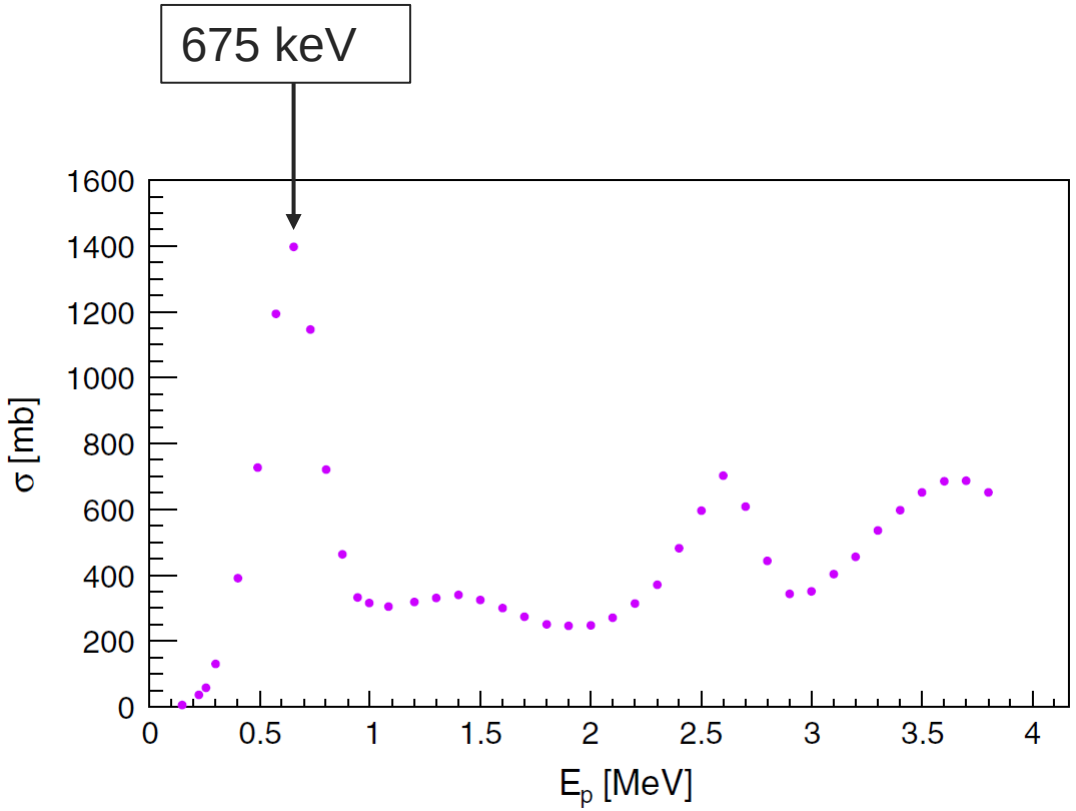
New Evaluation of the 11B-p Reaction Rates, M. H. Sikora and H. R. Weller, J Fusion Energ (2016) 35:538–543, DOI 10.1007/s10894-016-0069-y

Cross section resonances: for non thermal fusion



149 keV resonance

The thermonuclear fusion rate coefficient for p-11B reactions, W.M. Nevins and R. Swain 2000 Nucl. Fusion 40 865, DOI 10.1088/0029-5515/40/4/310



675 keV resonance

New Evaluation of the 11B-p Reaction Rates, M. H. Sikora and H. R. Weller, J Fusion Energ (2016) 35:538–543, DOI 10.1007/s10894-016-0069-y

Fuel polarisation (reactivity*1.6)

- As for DT fuel, proton and boron nucleus spin can be polarized:
  - Increase of cross section and reactivity up to a factor 1.6
  - Solid fuel injection look mandatory to use it (hydrogen pellet and solid boron)
  - Need of a facility to prepare polarized fuel

Polarized fusion and potential in situ tests of fuel polarization survival in a tokamak plasma, L. Baylor et al 2023 Nucl. Fusion 63 076009, DOI 10.1088/1741-4326/acc3ae

Nuclear Spin-Polarized Proton and 11B Fuel for Fusion Reactors: Advantages of Double Polarization in the 11B(p, a)8Be* Fusion Reaction, M. W. Ahmed • H. R. Weller, J Fusion Energ (2014) 33:103–107 DOI 10.1007/s10894-013-9643-8

Effect of Polarization on the Cross Section for the Reaction 11B(p,α)8Be∗ and Angular Distributions of Its Products, V. F. Dmitriev, ISSN 1063-7788, Physics of Atomic Nuclei, 2006, Vol. 69, No. 9, pp. 1461–1462, Pleiades Publishing, Inc., 2006. DOI: 10.1134/S1063778806090043

How to produce energy with p-11B fusion

- p-11B fusion parameters (thermal plasma):
  - $T_i \geq 200\ keV$
  - $n_i \tau_E \geq 6\ 10^{21} m^{-3} s$ ($\tau_E \geq 10\ s$ for a tokamak $= 3 * \tau_{E,ITER}$)
- High electron temperature plasma come with:
  - High bremsstrahlung radiative losses

$$P_{\mathrm{Brem}}/W_0 = 8.511 t^{1/2} \left\{ Z_{\mathrm{eff}} \left(1 + 1.78 t^{1.34}\right) + 2.12 t \left(1 + 1.1 t + t^2 - 1.25 t^{2.5}\right) \right\}$$

$$t = T_{\mathrm{e}}/mc^2, \quad W_0 = e^6 n_{\mathrm{e}}^2 / mc^2 \hbar$$

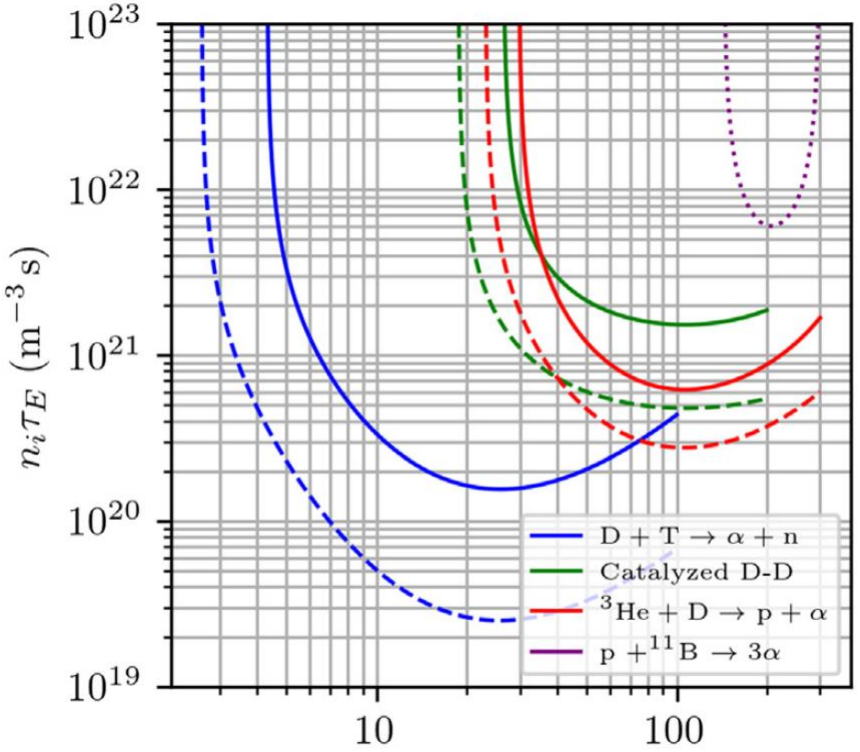
  - High synchroton raditiation (must be reflected by the wall and reabsorbed by the plasma). In tokamaks:

$$P_{\mathrm{cyclo}}(W) = 3.84\ 10^{-2} (1 - r_{\mathrm{w}})^{1/2} Ra^{1.38} \kappa^{0.79} B_{\mathrm{t}}^{2.62} n_{\mathrm{e,c}}^{0.38} \times T_{\mathrm{e,c}} (16 + T_{\mathrm{e,c}})^{2.61} \left(1 + 0.12 \frac{T_{\mathrm{e,c}}}{P_{a0}^{0.14}}\right)^{-1.51} K(\alpha_n, \alpha_{\mathrm{T}}, \beta_{\mathrm{T}})\, G\left(\frac{R}{a}\right)$$

$$K(\alpha_{\mathrm{n}}, \alpha_{\mathrm{T}}, \beta_{\mathrm{T}}) = (\alpha_{\mathrm{n}} + 3.87\ \alpha_{\mathrm{T}} + 1.46)^{-0.79} \times (1.98 + \alpha_{\mathrm{T}})^{1.38} \beta_{\mathrm{T}}^{2.14} \left(\beta_{\mathrm{T}}^{1.53} + 1.87\ \alpha_{\mathrm{T}} - 0.16\right)^{-1.33}$$

$$G(A) = 0.93\ \left(1 + 0.85 e^{-0.82A}\right)$$

$$P_{\mathrm{a,c}} = 6.04 \times 10^3 \frac{a n_{\mathrm{e,c}}}{B_{\mathrm{t}}}$$



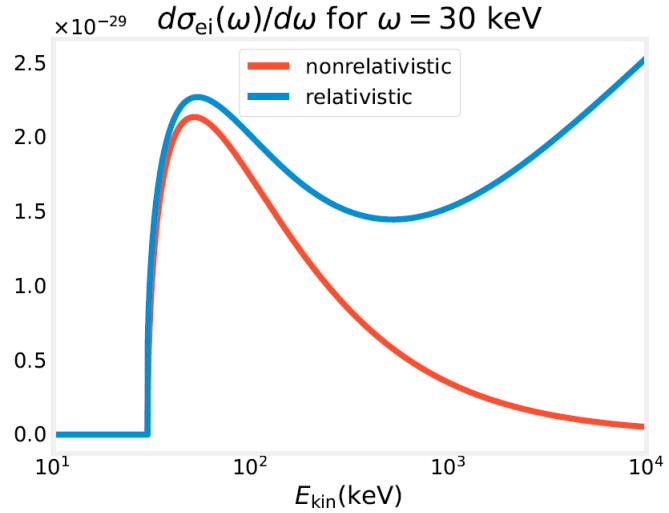
- Radiative losses are the main limiting factor for p-11B thermal fusion

Fusion reactivity of the pB11 plasma revisited, S.V. Putvinski et al 2019 Nucl. Fusion 59 076018, DOI 10.1088/1741-4326/ab1a60

Improved calculation of synchrotron radiation losses in realistic tokamak plasmas, F. Albajar et al, 2001 Nucl. Fusion 41 665, DOI 10.1088/0029-5515/41/6/301

Remarks on bremsstrahlung radiation

- Formulation from Putvinski paper should be accurate at 1%
- Relativistic correction can become inaccurate at high temperature (but larger than the one needed for proton-boron fusion)
  - Overestimation of radiation power by a factor of 2~3 possible in some case
  - Fully relativistic bremsstrahlung computation is needed for high temperature
- At very high magnetic field or at very high density, bremsstrahlung radiation is reduced due to quantum effects:
  - Mainly relevant for inertial fusion



Suppression of bremsstrahlung losses from relativistic plasma with energy cutoff, V.R. Munirov and N.J. Fisch, Phys. Rev. E 107, 065205, DOI:https://doi.org/10.1103/PhysRevE.107.065205

High field suppression of bremsstrahlung emission in high intensity laser–plasma interactions, M. Habibi et al, Phys. Plasmas 30, 103108 (2023) https://doi.org/10.1063/5.0167288

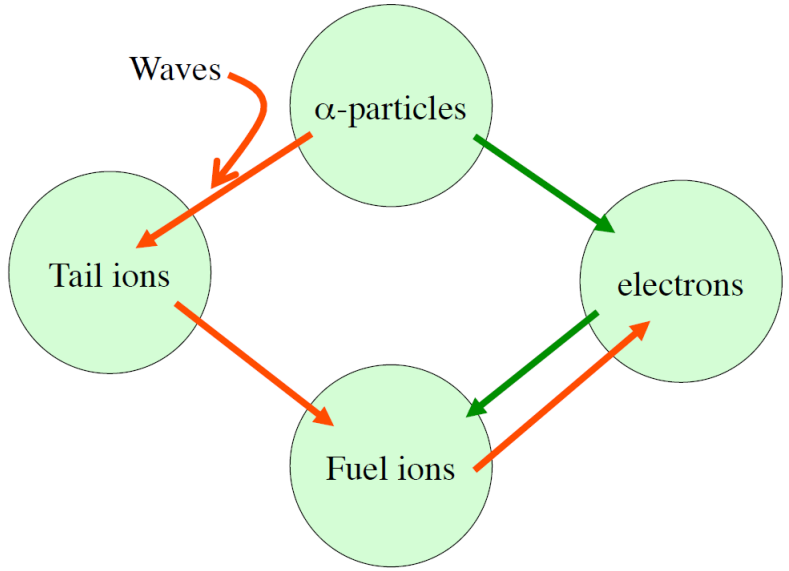
Focus Fusion: Overview of Progress Towards p-B11 Fusion with the Dense Plasma Focus, E. J. Lerner et al, Journal of Fusion Energy (2023) 42:7 https://doi.org/10.1007/s10894-023-00345-z

Aneutronic fusion in a degenerate plasma, S. Son and N.J. Fisch, Physics Letters A 329 (2004) 76–82, https://doi.org/10.1016/j.physleta.2004.06.054

Controlled fusion with hot-ion mode in a degenerate plasma, S. Son and N.J. Fisch, Physics Letters A 356 (2006) 65–71, https://doi.org/10.1016/j.physleta.2006.05.008

Waves assisted non thermal fusion

- Use one or two waves heating system to sustain fast ions population near cross section resonance
  - IC or IBW
  - LH ion heating
- Amplifying wave energy using alpha channeling mechanism (like in a MASER, see Fisch publications)
  - Transfer of power from fast α to the wave that later heat ions
  - Tested in TFTR
  - Configuration depending on device geometry



Wave-supported hybrid fast-thermal p-11B fusion, E. J. Kolmes et al, Phys. Plasmas 29, 110701 (2022), https://doi.org/10.1063/5.0119434

Elementary processes underlying alpha channeling in tokamaks , N. J. Fisch, AIP Conf. Proc. 1478, 80–90 (2012), https://doi.org/10.1063/1.4751641

On the optimal conditions for alpha channelling in tokamaks, Francesco Cianfrani and Francesco Romanelli 2019 Nucl. Fusion 59 106005, https://doi.org/10.1088/1741-4326/ab3018

Ignition threshold for non-Maxwellian plasmas, M. J. Hay; N.J. Fisch, Phys. Plasmas 22, 112116 (2015), https://doi.org/10.1063/1.4936346

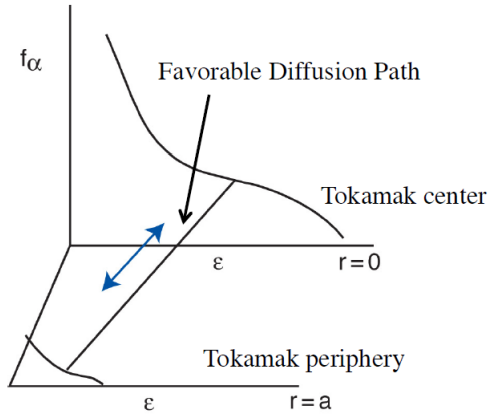
Interaction of energetic alpha particles with intense lower hybrid waves, N.J. Fisch and J-M. Rax, Phys. Rev. Lett. 69, 612, https://doi.org/10.1103/PhysRevLett.69.612

Alpha power channelling with two waves, N.J. Fisch and M.C. Herrmann 1995 Nucl. Fusion 35 1753, https://doi.org/10.1088/0029-5515/35/12/I40

Stochastic ion heating by a lower hybrid wave, C. F. F. Karney, Phys. Fluids 21, 1584–1599 (1978), https://doi.org/10.1063/1.862406

Fusion plasma experiments on TFTR: A 20 year retrospective, R. J. Hawryluk et al, Phys. Plasmas 5, 1577–1589 (1998), https://doi.org/10.1063/1.872825

Alpha particle losses from Tokamak Fusion Test Reactor deuterium–tritium plasmas, D. S. Darrow et al, Phys. Plasmas 3, 1875–1880 (1996), https://doi.org/10.1063/1.871983



Remarks on collisional heating

- For high electron temperature (T_e >= 100 keV) or high ion temperature over electron temperature, the correction become non negligible:

$$P_{ie} = 7.61 \times 10^{-28} n_e \left(1 + \frac{0.3 T_e}{m_e c^2}\right) \sum_i \frac{Z_i^2 n_i m_p}{m_i T_e^{3/2}} \ln \Lambda$$

$$\times \exp\left[-\left(3.5 \sum_i \frac{Z_i^2 n_i}{n_e} \frac{m_e}{m_i} \frac{T_i}{T_e}\right)^{2/3}\right] (T_i - T_e) \frac{\mathrm{W}}{\mathrm{cm}^3}$$

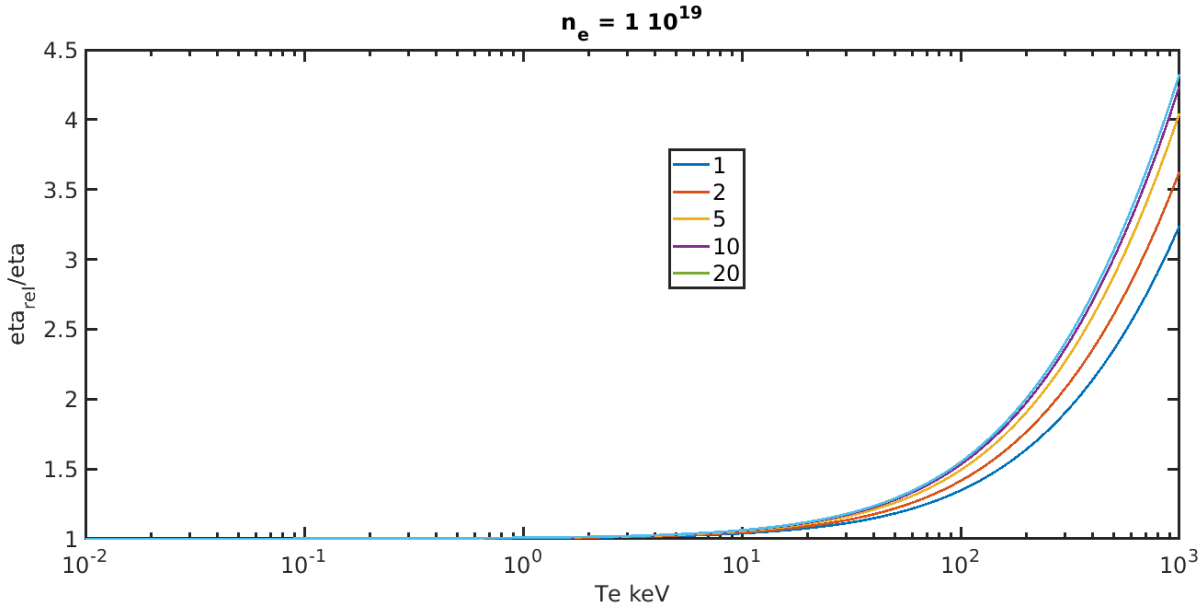
- METIS included also a relativistic effect correction

Modification of classical Spitzer ion–electron energy transfer rate for large ratios of ion to electron temperatures, T.H. Rider and P.J. Catto, Phys. Plasmas 2, 1873–1885 (1995), https://doi.org/10.1063/1.871274

Relativistic thermal plasmas, Susan Stepney, Institute of Astronomy and Newnham College University of Cambridge, PhD, 1983, https://www-users.york.ac.uk/~ss44/bib/ss/ast/thesis.pdf

Resistivity and bootstrap

- Correction of resistivity due to relativistic effect →
- No correction known for the boostrap current
  - Limit of Redl fit ~ 50 keV ?
  - Used of relativitistic DKE solver can provide some insigth (can be tested with LUKE code)
  - Maybe inacurrate, can be corrected by hand in METIS



Transport coefficients of a relativistic plasma ,O. J. Pike and S. J. Rose, Phys. Rev. E 93,2016, https://doi.org/10.1103/PhysRevE.93.053208

Relativistic transport theory for a two-temperature magnetized plasma, T. Metens , R. Balescu,

Phys. Fluids B 2, 2076–2090 (1990), https://doi.org/10.1063/1.859428

Synergism of Radio-Frequency Current Drive with the Bootstrap Current, Joan Decker, https://dspace.mit.edu/bitstream/handle/1721.1/87209/51176004-MIT.pdf?sequence=2

Specific settings for proton boron plasma (1/2)

- Plasma composition: change from DT to pB11:
  - Gas parameter = 11 for pB11
  - nT/nD reference become nB/nH: used to tune the fraction of boron in hydrogen plasma
  - FTNBI should be set to 0 or NBI should be forced to H in parameter settings
  - Residual fraction of D is by default set to the natural value (0.000115)
  - Reactivity can be enhanced using pellet injection and fuel polarisation
  - Remark: boron density information are stored in tritium fields with gas option = 11

- Maximum temperature, equipartition and transport
  - The maximum temperature in METIS should be increase (by default it is limited to 100 keV) up to 1 MeV
  - Equipartition formulation should be changed to take into account large Ti/Te and higher temperature.
  - The ratio between ion heat transport coefficient and electron heat transport cofficient can, optionnaly, be reduced to promote decoupling between ion and electron temperature. This tuning should be later support by evidence from first principle transport model.

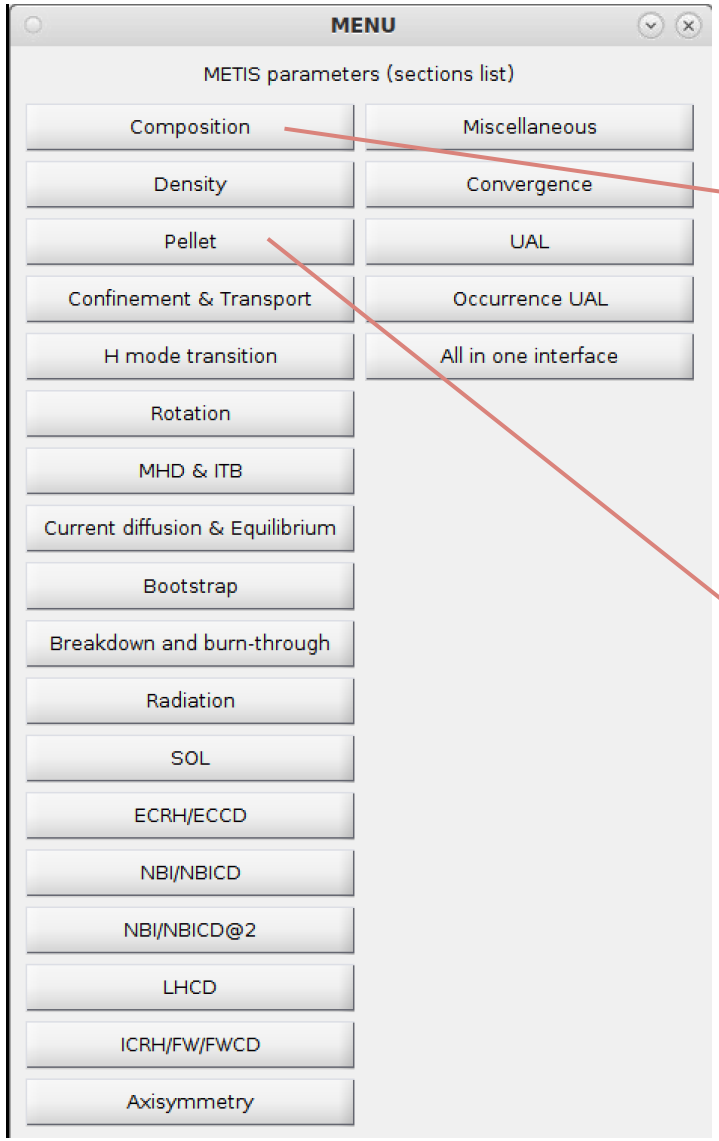
Specific settings for proton boron plasma (2/2)

- Radiation
  - Bremsstralhung formulation should be switched to relativistic one and the complete formulation for the Gaunt factor should be used
  - Synchrotron radiation formulation is already suitable for high temperature, but is larger at high remperature, so now the wall reflection coefficient (rw) plays a major role in performances of the discharge and should be carefully tuned.

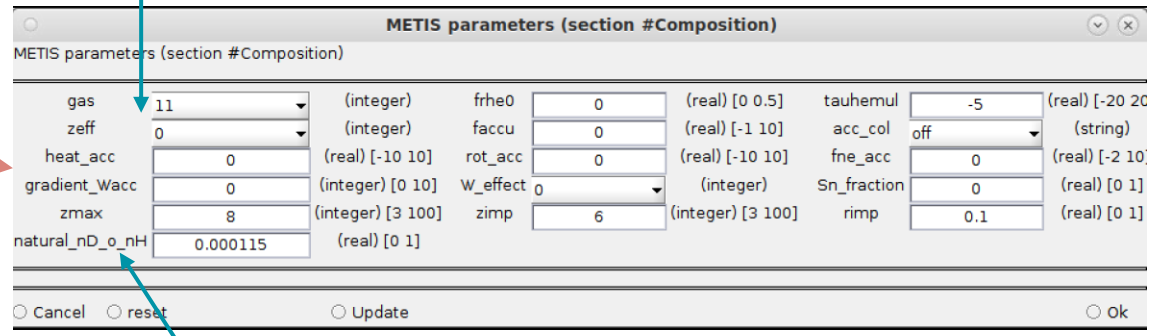
- Plasma resistivity and boostrap current
  - Relativistic plasma resistivity correction should switched on
  - Boostrap current factor can be adjusted if some information are available from external code

- Alpha channeling
  - If the rigth conditions are fullfilled, energy from fast alphas can be transfered to wave and then to plasma ions. A simple parameter can be set in METIS to increase the energy transfered from alpha to bulk plasma ions and enhanced the ion temperature. The effect should be check with external wave and F-P codes.

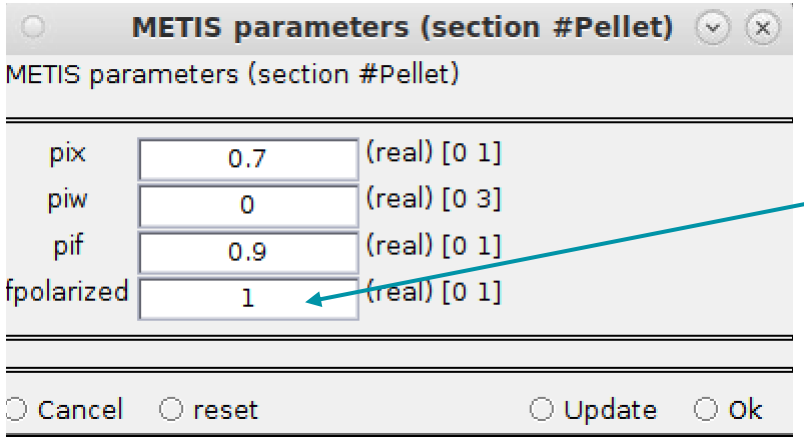
Where are these parameters in METIS GUI (1/5)



Gas = 11 → proton-boron

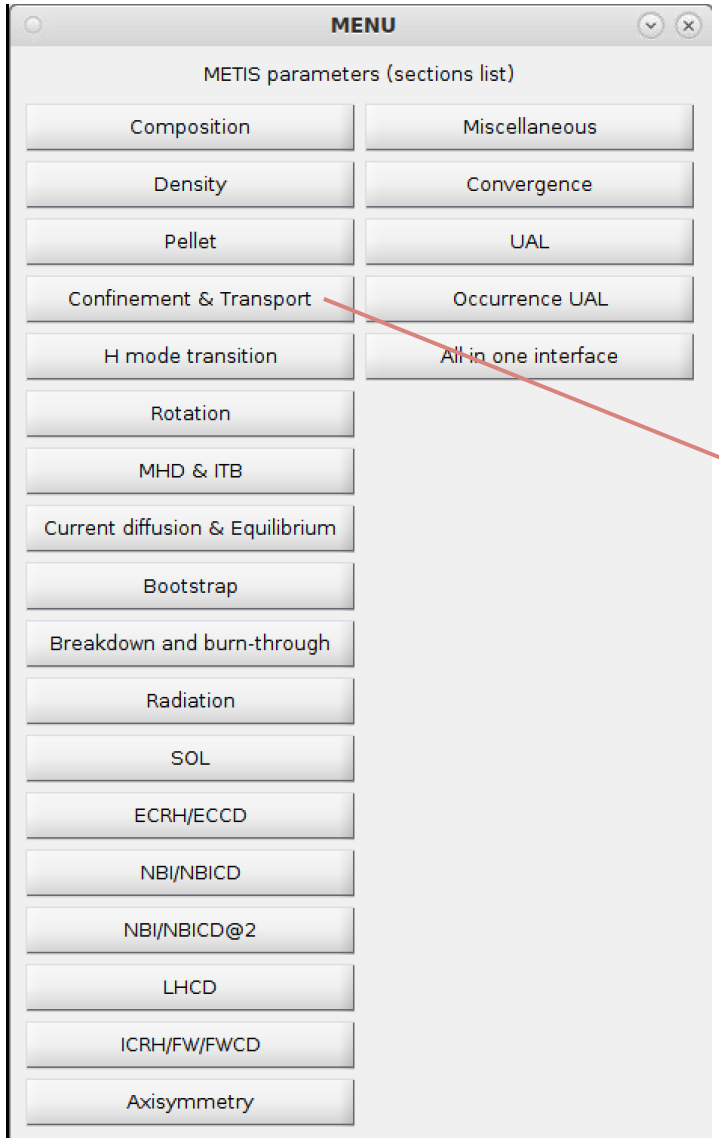


Fraction of D in H

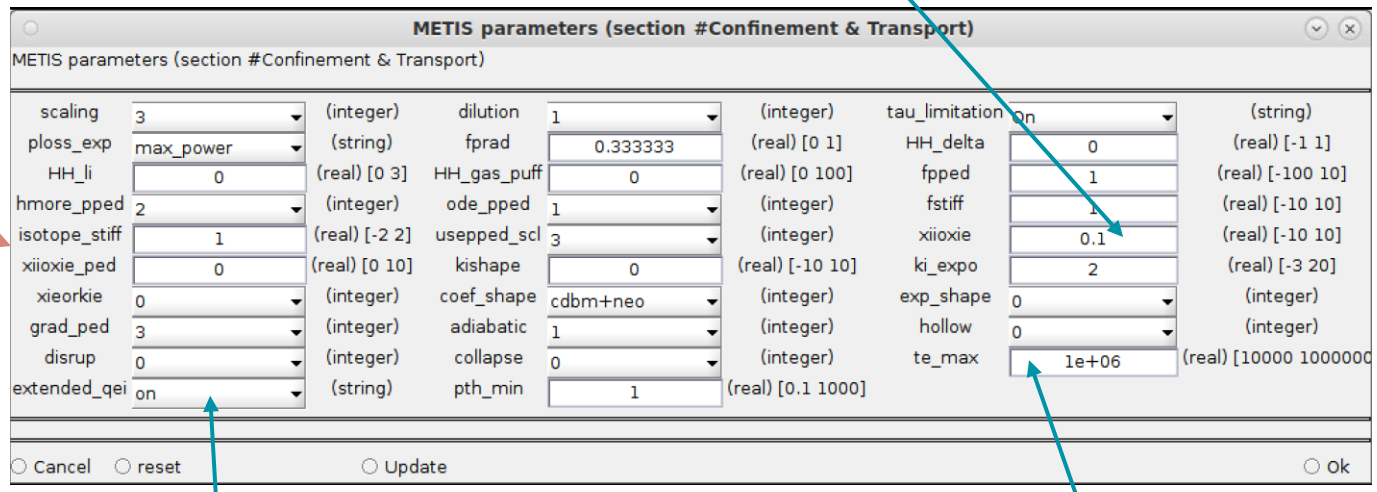


Polarized fuel requires fueling by pellet injection (pif >0)

Where are these parameters in METIS GUI (2/5)



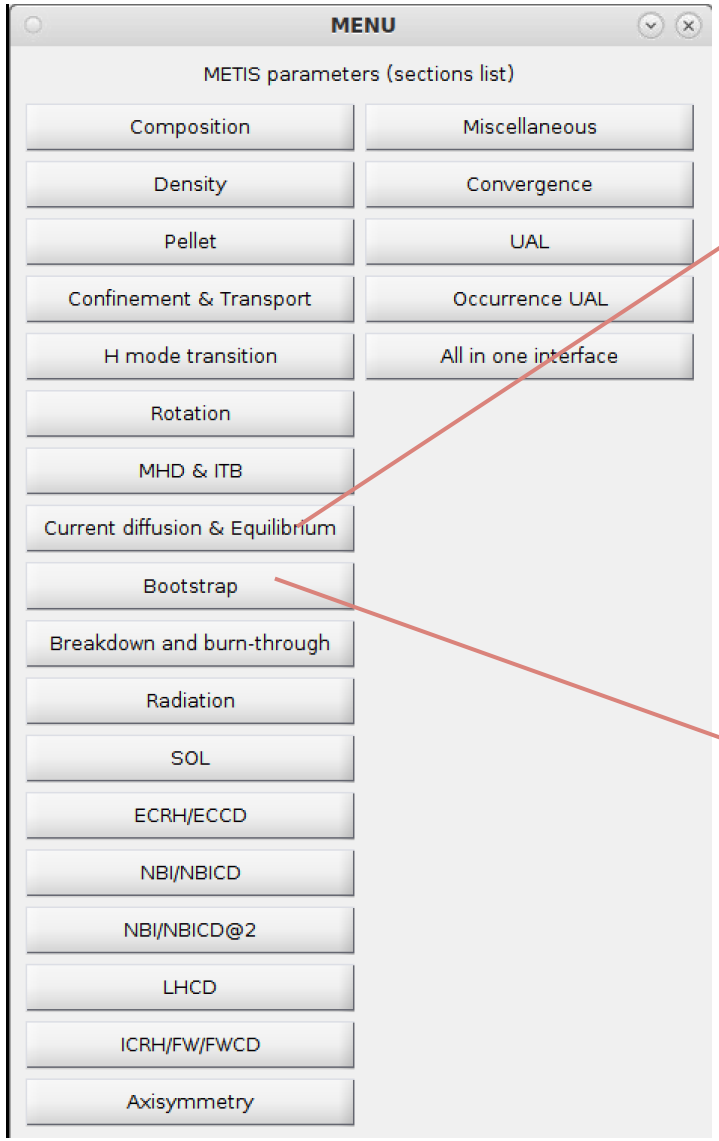
Ratio ion transport coefficient over electron transport coefficient

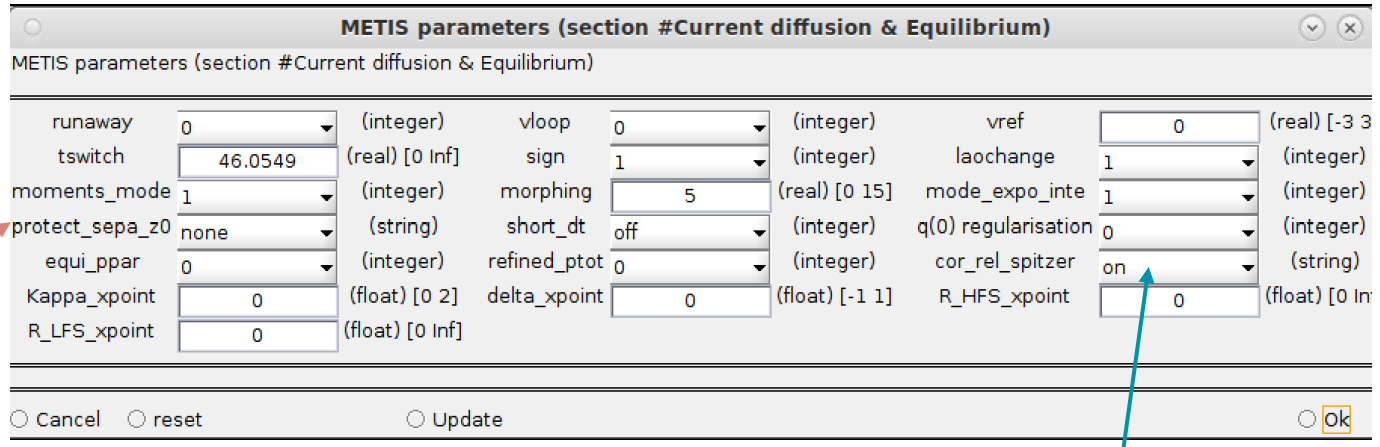


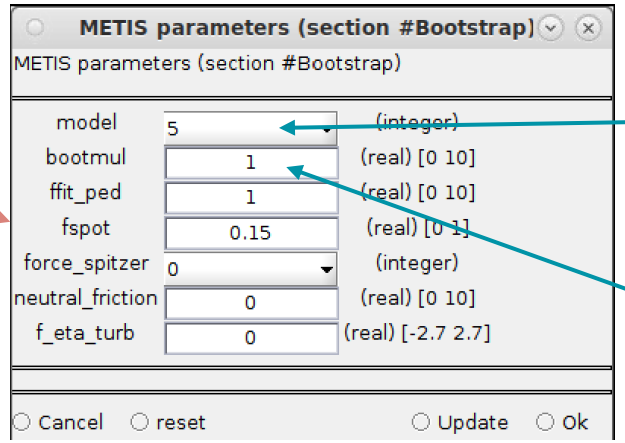
High Te, high Ti/Te and relativistic correction to equipartition term

Maximum Te increased from 1e5 to 1e6 eV

Where are these parameters in METIS GUI (3/5)





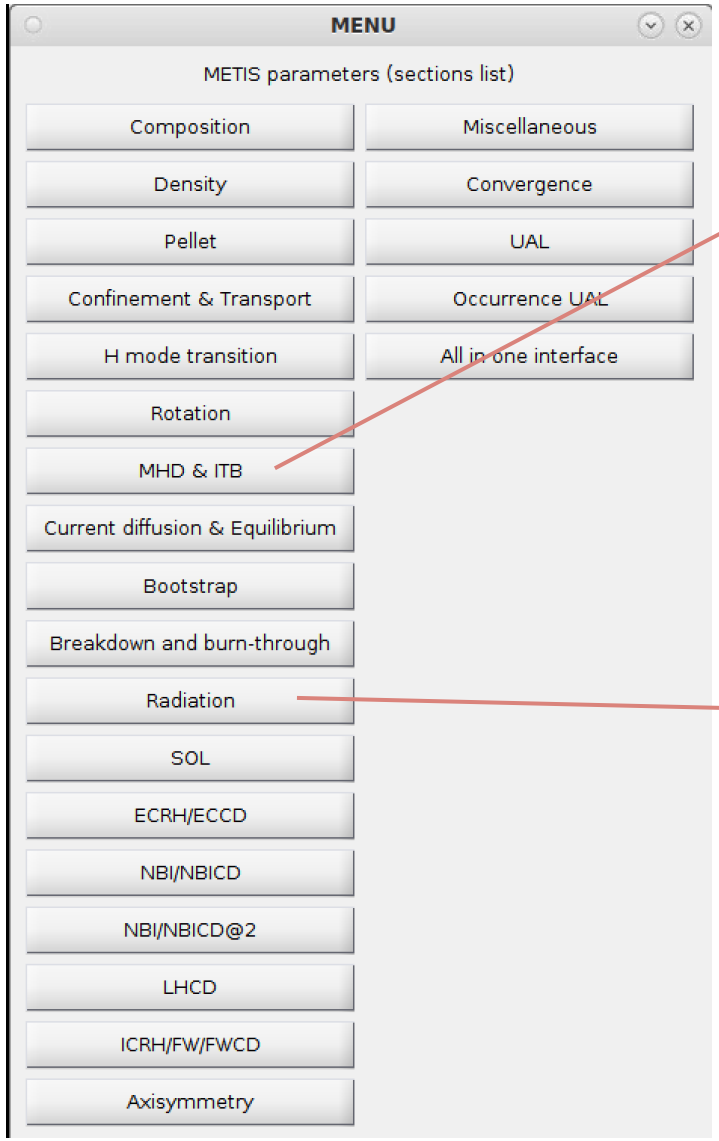


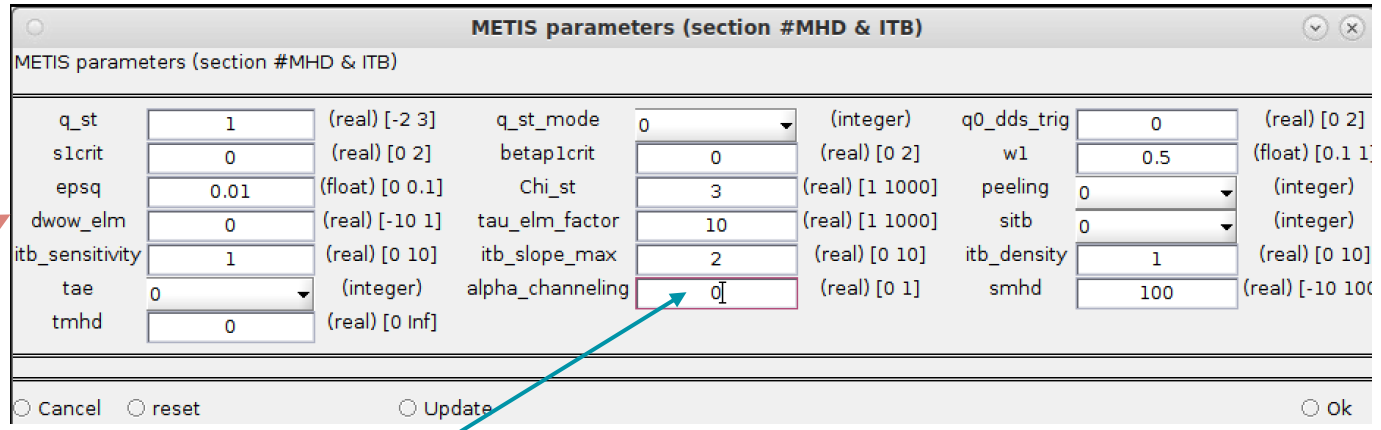
Relativistic correction for the resistivity

Redl fit for bootstrap current → 5

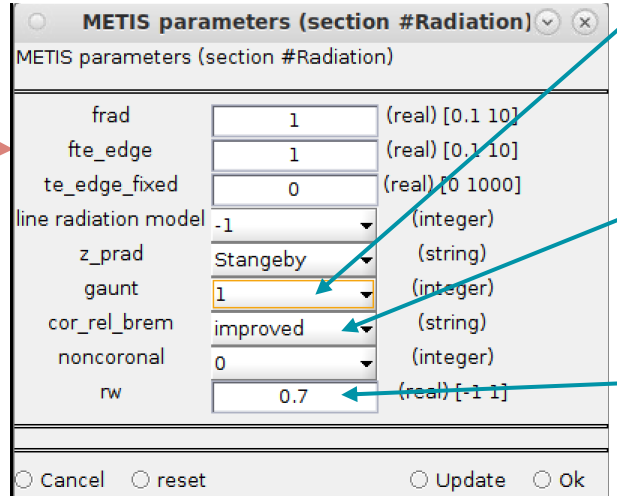
Factor applied to bootstrap current

Where are these parameters in METIS GUI (4/5)





Alpha channeling factor (arbitrary !)

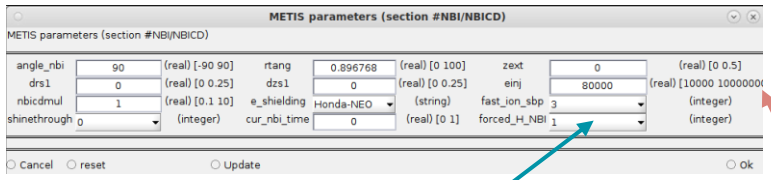


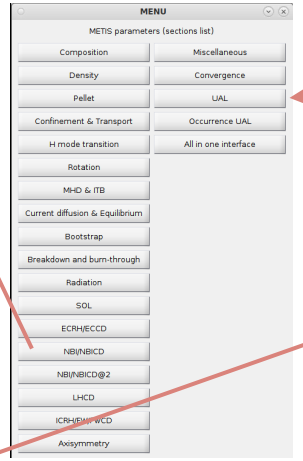
Complete formulation of the Gaunt factor (instead of a constant)

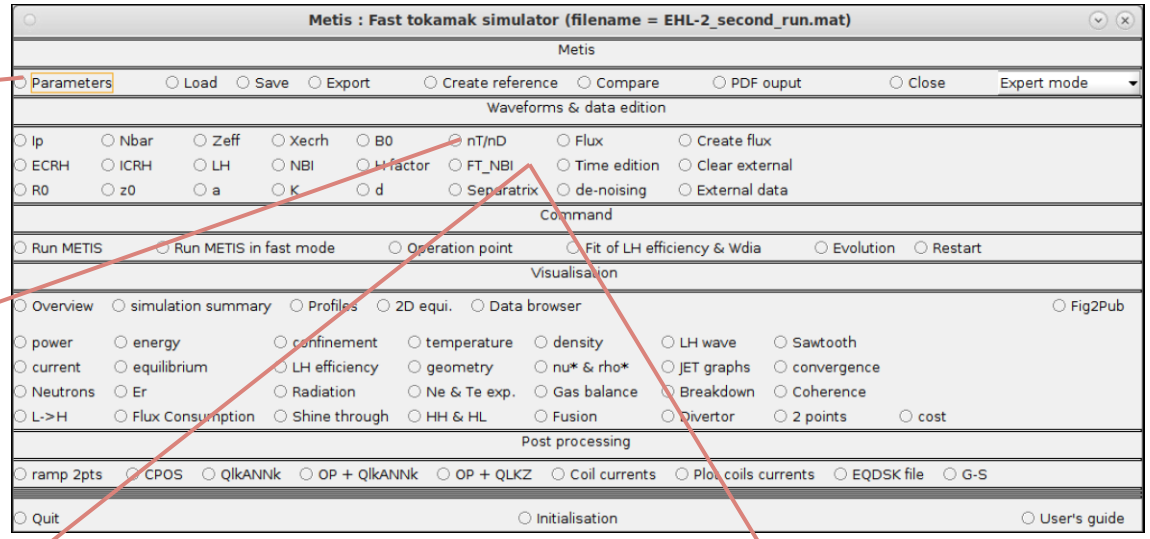
Precise relativistic correction for the Bremsstrahlung

Fraction of synchrotron radiation reflected by the wall

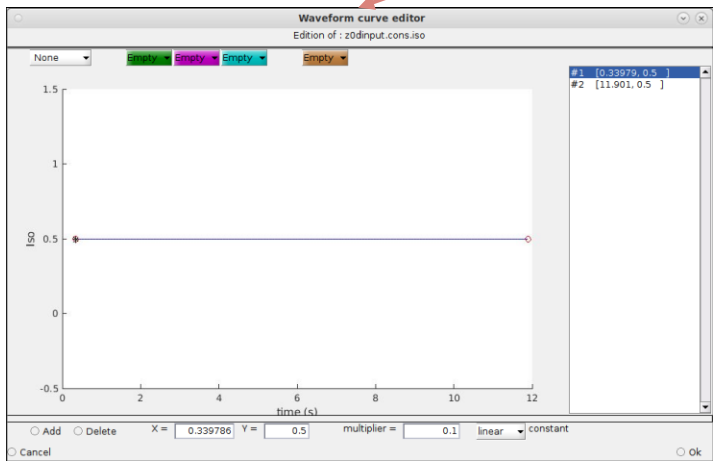
Where are these parameters in METIS GUI (5/5)



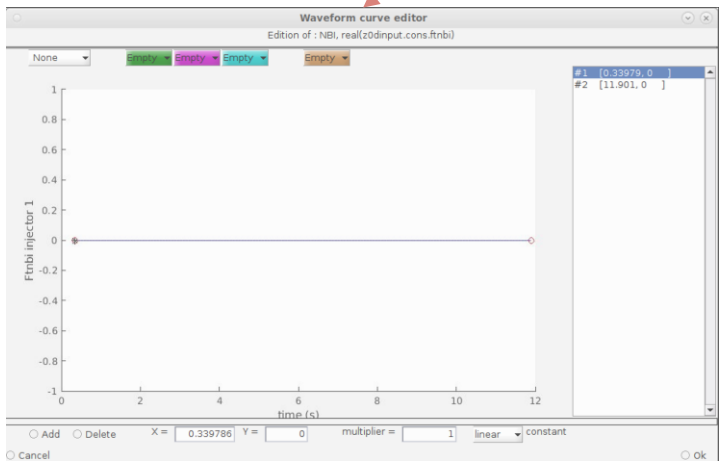




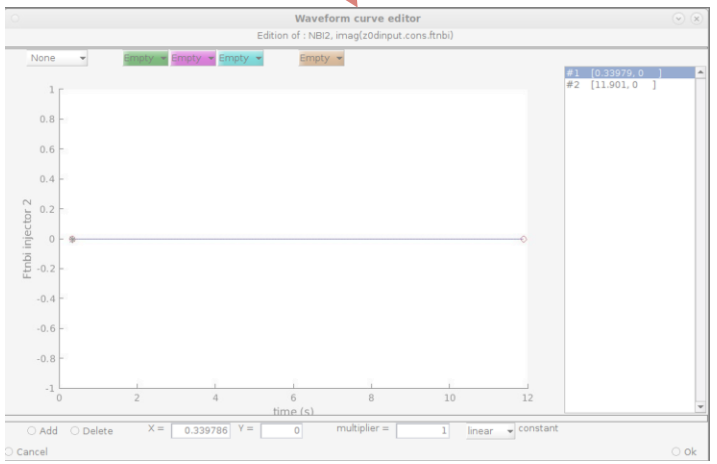
Force pure hydrogen NBI or used FT_NBI and set it to 0



5% boron in H plasma



Pure H NBI (NBI 1)



Pure H NBI (NBI 2, optional)

Thermal fusion relative difficulty (1/2)

fuel	$\langle\sigma v\rangle/T^2$	penalty/bonus	inverse reactivity	Lawson criterion	power density [W/m^3/kPa^2]	inverse ratio of power density
$^2_1$D–$^3_1$T	1.24×10^{-24}	1	1	1	34	1
$^2_1$D–$^2_1$D	1.28×10^{-26}	2	48	30	0.5	68
$^2_1$D–$^3_2$He	2.24×10^{-26}	2/3	83	16	0.43	80
p$^+$–$^6_3$Li	1.46×10^{-27}	1/2	1700		0.005	6800
p$^+$–$^{11}_5$B	3.01×10^{-27}	1/3	1240	500	0.014	2500

The maximum value of $\langle\sigma v\rangle/T^2$ is taken from a previous table. The "penalty/bonus" factor is that related to a non-hydrogenic reactant or a single-species reaction. The values in the column "inverse reactivity" are found by dividing 1.24×10^{-24} by the product of the second and third columns. It indicates the factor by which the other reactions occur more slowly than the $^2_1$D–$^3_1$T reaction under comparable conditions. The column "Lawson criterion" weights these results with E_{ch} and gives an indication of how much more difficult it is to achieve ignition with these reactions, relative to the difficulty for the $^2_1$D–$^3_1$T reaction. The next-to-last column is labeled "power density" and weights the practical reactivity by E_{fus}. The final column indicates how much lower the fusion power density of the other reactions is compared to the $^2_1$D–$^3_1$T reaction and can be considered a measure of the economic potential.

https://en.wikipedia.org/wiki/Nuclear_fusion

Based of old data for reactivity

Thermal fusion relative difficulty (2/2)

fuel	T_i [keV]	$P_{fusion}/P_{Bremsstrahlung}$
${}^{2}_{1}D-{}^{3}_{1}T$	50	140
${}^{2}_{1}D-{}^{2}_{1}D$	500	2.9
${}^{2}_{1}D-{}^{3}_{2}He$	100	5.3
${}^{3}_{2}He-{}^{3}_{2}He$	1000	0.72
$p^{+}-{}^{6}_{3}Li$	800	0.21
$p^{+}-{}^{11}_{5}B$	300	0.57

$$T_e = T_i$$

- ✓ Bremsstrahlung relativist correction incomplete
- ✓ Based on old data for reactivity

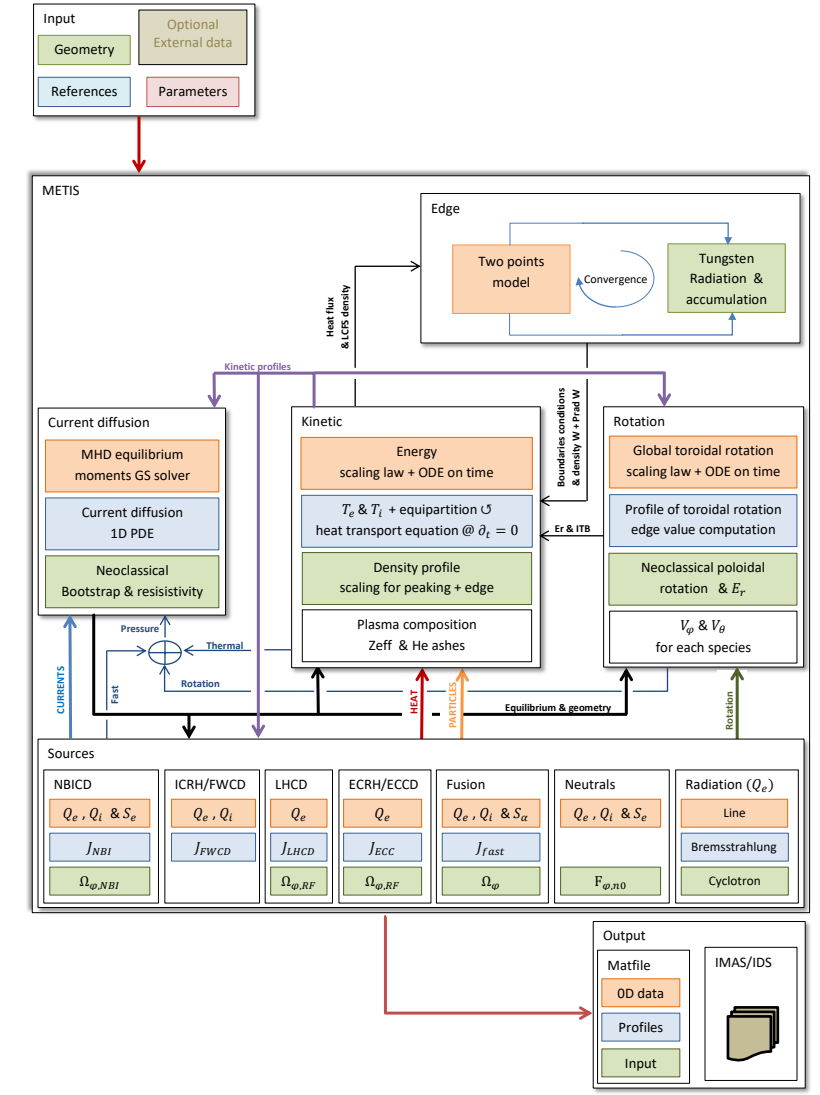
https://en.wikipedia.org/wiki/Nuclear_fusion

12. Summary

METIS: a fast integrated tokamak modeling tool for scenario design

- METIS is a numerical code aiming at fast full tokamak plasma analyses and predictions:
  - It combines 0D scaling-law normalised heat and particle transport with 1D current diffusion modelling and 2D equilibria.
  - It contains several heat, particle and impurities transport models, as well as heat, particle, current and momentum sources
  - It allows faster than real time scenario simulations for reactor simulations.

- METIS applications and use cases:
  - Tokamaks discharges analysis (WEST, Tore Supra, JET, TCV, DIII-D, EAST, HL2A, COMPAS, …)
  - Scenario predictions (JT-60SA, DTT, HL2M)
  - Reactor studies (ITER, DEMO, CFETR, …)
  - Data provider for other codes (as CRONOS)
  - Plasma plant simulator in control workflow
  - Students teaching and training



J.F. Artaud et al 2018 Nucl. Fusion 58 105001 (http://iopscience.iop.org/article/10.1088/1741-4326/aad5b1)